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Large Language Models

Foundations, Systems, Alignment, and
Applications

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Preface

This book is written as a modern, research-grounded introduction to large language models. The prose, organization, notation, examples, and arguments are original. Each chapter is intended to connect three views that are too often taught separately: the mathematical object being optimized, the system that makes the optimization feasible, and the evaluation procedure that determines whether the result is useful.

The book deliberately avoids an implementation-by-implementation narrative. Instead, it follows the lifecycle of a model: data and tokenization, architecture, pretraining, distributed systems, serving, adaptation, alignment, retrieval, reasoning, multimodality, and safety. Historical systems such as GPT, Alpaca, LoRA, RLHF, and LLaMA are treated as conceptual anchors rather than as recipes to copy without context.

The exposition is evidence-first. Claims about algorithms, scaling behavior, evaluation, or deployment are tied to primary papers or technical reports whenever possible. No third-party slide text, media transcript, dataset record, or code listing is reproduced as manuscript prose.

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The book discusses model training, deployment, alignment, data collection, evaluation, retrieval, and agentic systems. These topics carry risks involving privacy leakage, copyright infringement, unsafe generation, benchmark contamination, dual-use automation, and misleading evaluation. Chapters therefore include explicit safety, licensing, and evaluation cautions where those risks are material.

No claim in the manuscript should be interpreted as legal advice, medical advice, security advice, or a recommendation to deploy an unreviewed model in a high-stakes environment.

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Acronyms

API	Application programming interface.
BPE	Byte-pair encoding.
CoT	Chain-of-thought prompting.
DPO	Direct preference optimization.
FFN	Feed-forward network.
FSDP	Fully sharded data parallelism.
GQA	Grouped-query attention.
KV	Key-value.
LLM	Large language model.
LoRA	Low-rank adaptation.
MoE	Mixture of experts.
MQA	Multi-query attention.
PEFT	Parameter-efficient fine-tuning.
PPO	Proximal policy optimization.
RAG	Retrieval-augmented generation.
RLHF	Reinforcement learning from human feedback.
RLVR	Reinforcement learning with verifiable rewards.
RoPE	Rotary position embedding.
SFT	Supervised fine-tuning.
VLM	Vision-language model.

Part I

Foundations

Chapter 1

What Makes a Language Model Large

Abstract Large language models are not defined by parameter count alone. They are coupled artifacts whose behavior emerges from data selection, tokenization, architecture, optimization, post-training, inference systems, and evaluation. This chapter establishes that lifecycle view. It explains why causal prediction became the dominant training interface, why scaling laws changed the engineering economics of language modeling, why instruction following and preference learning turned pretrained models into assistant systems, and why reasoning models made inference-time computation part of the modeling problem.

1.1 The Object of Study

A language model assigns probabilities to token sequences, but a modern large language model is better understood as a trained conditional computation system whose distribution is shaped by corpus design, optimizer dynamics, architectural constraints, and deployment-time decoding. For a tokenized sequence $x_{1:T}$, the causal language-model factorization is

$$p_{\theta}(x_{1:T}) = \prod_{t=1}^T p_{\theta}(x_t \mid x_{<t}), \quad (1.1)$$

and the usual training objective minimizes the average negative log-likelihood,

$$\mathcal{L}(\theta) = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(x_t \mid x_{<t}), \quad (1.2)$$

with perplexity defined as $\exp(\mathcal{L})$ when the loss is measured in natural-log units. The Transformer made this system practical at scale because self-attention exposes long-range token interactions while remaining highly parallel on accelerators [108]. GPT-style causal models then showed that a single left-to-right objective can support broad transfer when the model is trained on enough heterogeneous text [90, 10]. BERT-style

Table 1.1 Axes of largeness in modern language-model systems.

Axis	Symbol	What Scales	Typical Failure Mode
Parameters	N	Dense weights or total MoE weights	Capacity improves while memory, bandwidth, and overfitting risks grow
Active compute	N_{active}	Parameters used per token in sparse models	Routing imbalance or expert under-use hides nominal scale
Training data	D	Tokens, documents, domains, and languages	More tokens amplify duplication, leakage, and licensing problems
Training compute	C	FLOPs and accelerator time	Undertraining or wasted compute from poor data/model balance
Context	T	Prompt and generated sequence length	KV-cache memory and attention cost dominate serving
Inference budget	B	Samples, search, retrieval, tools, verifiers	Higher pass rates can hide weak calibration or invalid verifiers

Data and provenance → tokenizer → architecture → objective and optimizer → distributed training → post-training → inference system → evaluation and governance

Fig. 1.1 The lifecycle stack used throughout the book. Each arrow represents a contract: the next layer inherits the assumptions, artifacts, and failure modes of the previous layer.

masked models remain important for representation learning and retrieval, and dense retrieval systems later made representation quality central to open-domain question answering and RAG pipelines [22, 42, 53].

The word “large” therefore refers to several coupled axes, summarized in Table 1.1. Parameter count changes capacity, but active parameters matter separately in sparse mixture-of-experts systems; token count changes the statistical evidence seen during training, but token provenance determines whether the evidence is useful or publishable; context length changes what can be conditioned on, but KV-cache memory and attention cost determine whether that context can be served economically [41, 35, 20]. Inference budget is now also part of model behavior, because retrieval, reranking, sampling, verification, tool calls, and explicit reasoning traces can spend additional computation after pretraining has ended [9, 53, 13, 118].

This book treats an large language model as a stack rather than as a single neural network. The stack begins with raw data and a tokenizer; passes through architecture, loss, optimizer, and distributed training; continues through supervised instruction tuning and preference optimization; and ends in an inference service that must decide how much context, retrieval, search, and safety filtering to apply. A mistake in any layer can be misread as a model limitation: a weak tokenizer can harm multilingual coverage, a contaminated benchmark can exaggerate capability, an unstable optimizer can waste compute, and a poorly designed serving path can make a strong model unusable under latency constraints [100, 115].

1.2 Why Prediction Became a General Interface

The central technical bargain of language modeling is that next-token prediction converts unstructured text into a dense self-supervised training signal. Every token in a document can contribute a target, so web-scale corpora become usable without manual labels [10]. The bargain is imperfect: the objective rewards distributional continuation, not truth, helpfulness, calibration, or harmlessness [84]. Yet it is powerful because language co-occurs with facts, code, dialogue, procedures, arguments, and instructions, so a model trained to predict text can internalize many latent tasks before any task-specific fine-tuning [10].

The Transformer architecture amplified this bargain. Self-attention gives each position a data-dependent path to previous positions, so the model can condition on both local syntax and distant context [108]. Multi-head attention gives the model multiple learned projection subspaces for comparing and aggregating tokens, while feed-forward sublayers supply high-capacity tokenwise transformations. Residual pathways and normalization make very deep stacks trainable, and modern decoder families refine these ingredients with rotary position embeddings, RMS normalization, gated feed-forward networks, grouped-query attention, and mixture-of-experts routing [104, 106, 1, 20].

Prediction also gives a clean implementation path. The model receives token ids, maps them to embeddings, repeatedly mixes information through attention and feed-forward blocks, and projects the final hidden states to vocabulary logits. Training minimizes cross-entropy between predicted logits and the next token, while generation samples or selects tokens from the resulting distribution. Later chapters derive this computation from tensor shapes and turn the derivations into original implementation exercises.

1.3 Scaling Changed the Research Program

Before the modern scaling era, natural language processing was organized around task-specific datasets, architectures, and losses. Scaling-law work reframed the field by showing that loss follows empirical power laws over model size, dataset size, and compute across broad ranges [41]. Compute-optimal training then shifted attention from simply increasing parameters to balancing model size and token count, arguing that many early large models were undertrained relative to their compute budgets [35]. The consequence is practical: a serious training plan must budget parameters, tokens, sequence length, hardware throughput, and data quality together.

Open model reports made this systems view explicit, although they should be read as technical reports rather than as a substitute for independent replication. LLaMA demonstrated that carefully trained smaller models can be competitive when token budgets and data mixtures are chosen well [106]. Llama 2 expanded the public discussion of pretraining, supervised fine-tuning, preference data, and safety evaluation [107]. Llama 3 pushed the recipe toward longer context, multilingual coverage, coding, safety models, and extensive post-training [29]. DeepSeek-V3 and Kimi K2 indicate that

frontier-scale open-weight systems increasingly combine mixture-of-experts routing, specialized attention mechanisms, and agentic or coding-oriented post-training, but their claims still require careful benchmark and deployment-context interpretation [20, 43].

Scaling also exposed failure modes. Training loss can improve while benchmark validity degrades through leakage, contamination, or overfitting to public leaderboards [100, 115]. Larger context windows can increase cost faster than they increase robust long-context reasoning. More samples at inference time can improve pass rates while hiding weak calibration or brittle verification. This book therefore treats evaluation as a modeling component, not as an afterthought.

1.4 From Base Models to Assistants

A pretrained base model is not yet an assistant. It has learned broad continuation behavior, but user-facing behavior requires additional training signals that specify what counts as a helpful response. Supervised instruction tuning supplies demonstrations of desired behavior, and RLHF adds preference rankings so the model can be optimized toward responses humans prefer under a specific annotation protocol [84]. Direct preference optimization and related methods simplify this stage by fitting preferences without an explicit online reinforcement-learning loop, although they still depend on the quality and scope of the preference data [91].

Parameter-efficient adaptation changed who can perform this second stage. LoRA freezes the base model and learns low-rank updates inside selected linear layers, which makes adaptation far cheaper than full fine-tuning for many settings [38]. QLoRA combines low-rank adaptation with quantized base weights, making instruction tuning feasible on much smaller hardware budgets [21]. These methods are not magic compression; they change the trainable subspace, memory footprint, optimizer state, and failure modes. Chapter 10 treats rank, target modules, quantization error, and evaluation as design decisions rather than as fixed defaults.

Assistants also need retrieval and tools when the answer depends on fresh, private, or long-tail knowledge. Retrieval-augmented generation inserts external evidence into the context, while tool-use patterns let the model call search, calculators, databases, or code execution environments [53, 120]. These systems trade parametric memory for runtime grounding, but they introduce their own failure modes: retriever recall, reranker precision, context ordering, citation faithfulness, and prompt-injection risk.

1.5 Reasoning Makes Inference Part of Training

Reasoning in large language models is not a single mechanism. Chain-of-thought prompting elicits intermediate natural-language steps, self-consistency samples multiple paths, tree search branches over candidate partial solutions, and verifier-guided methods score or filter reasoning traces [112, 119]. These traces can improve task performance without being faithful causal explanations of the model’s internal computation,

so they must be evaluated as generated artifacts rather than accepted as transparent reasoning records [71]. Reasoning-oriented reinforcement learning adds another layer by rewarding correct final answers, verified intermediate steps, or task-specific executable outcomes [19, 113]. A recurring post-2024 trend is that inference-time compute and training-time incentives must be studied together: some gains come from better policies, while others come from searching, sampling, verifying, or allocating more computation at test time [13, 102, 118].

This perspective changes how we teach algorithms such as PPO, DPO, GRPO-style training, process reward models, and Monte Carlo tree search. The book does not present reinforcement learning as a detached prerequisite followed by an unrelated RLHF recipe. Instead, it follows the sequence-level control problem: what is being rewarded, what distribution is being constrained, what verifier is trusted, and how much extra computation is spent at inference time.

1.6 A Roadmap for the Book

Part I defines the foundation: what is being modeled, how text becomes tokens, how corpora become training signals, and why provenance matters. Part II builds the architecture and systems spine, moving from attention and GPT-style decoders to LLaMA-class blocks, optimization, distributed training, and inference serving. Part III studies adaptation through instruction tuning, synthetic data, LoRA, QLoRA, domain adaptation, and multilingual specialization. Part IV covers retrieval, tools, agents, preference learning, reasoning, multimodality, evaluation, safety, and governance.

The intended reading discipline is simple. Each chapter starts from a concrete problem, states the mathematical object being optimized, connects the object to implementation, and then asks how the claim would be evaluated. That discipline is the main defense against treating slides, benchmark tables, or model cards as self-evident truth. The goal is not to memorize the names of model families; it is to understand why a modeling choice works, when it fails, and how to test it in a reproducible system.

1.7 Key Terms

Base model	A pretrained model before instruction tuning, preference optimization, or task-specific adaptation.
Context length	The number of tokens available to condition the model during a forward pass.
Inference budget	The computation spent after a prompt is received, including decoding, retrieval, search, verification, and tool use.
Perplexity	The exponential of average negative log-likelihood under a language model, useful for comparing models only when tokenization and evaluation data are controlled.

Post-training	Training after base pretraining, including SFT, preference optimization, safety tuning, and domain adaptation.
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1.8 Exercises

1. Write the causal factorization and cross-entropy loss for a batch with shape $B \times T$. State which tokens should be masked for a chat-style SFT example.
2. Pick two rows from Table 1.1. For each row, describe one benchmark result that could improve while the real system becomes worse.
3. Explain why a model with fewer total parameters can be more expensive to serve than a larger model under a different context length or KV-cache layout.
4. Design a provenance checklist for one dataset, one code repository, one figure, and one model checkpoint before they can be used in this book.
5. Give an example where a chain-of-thought answer is correct but the explanation is not a faithful record of the underlying computation.

Chapter 2

Tokens, Corpora, and Training Signals

Abstract This chapter turns raw text into a defensible training signal. It covers tokenization, corpus construction, deduplication, filtering, contamination control, packing, licensing, and the difference between pretraining, instruction, preference, retrieval, and evaluation data. The central message is that data is not an inert input. It fixes the model’s vocabulary, compute budget, benchmark validity, legal exposure, and much of its downstream behavior.

2.1 Data as Specification

The first trainable layer of a language model sees integers, not text. Everything before that point is a data contract: which documents are admitted, how they are normalized, how they are split into tokens, how examples are packed into sequences, which targets are masked, and which records are reserved for evaluation. A model trained on noisy, duplicated, or legally unusable data can still reduce pretraining loss, but the resulting artifact may be less useful, less trustworthy, and less publishable than a smaller model trained on a better specified corpus.

Let $\mathcal{D}$ be a collection of documents and let τ be a tokenizer mapping text strings into token ids. A causal pretraining example is usually a token sequence

$$x_{1:T} = \tau(d)_{1:T}, \quad (2.1)$$

and the base training signal is the next-token loss

$$\mathcal{L}_{\text{pre}}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{(x,t) \in \mathcal{B}} \log p_{\theta}(x_t \mid x_{<t}). \quad (2.2)$$

The equation hides many decisions. The same document can become a different number of training targets under a different tokenizer; a repeated document can receive many times its intended weight; a benchmark item accidentally included in $\mathcal{D}$ can make a model look more capable than it is; a chat example can train the model to imitate the user

Table 2.1 Data artifacts in an large language model project.

Artifact	Main Contract	Common Failure Mode
Tokenizer corpus	Defines reusable units and token efficiency	Rare scripts, code, or domain terms become long token sequences
Pretraining corpus	Supplies dense self-supervised targets	Duplication, low-quality text, private data, or benchmark leakage
Instruction data	Demonstrates desired input-output behavior	Labels train on the prompt, formatting leaks, or style dominates task skill
Preference data	Ranks competing responses under a policy	Annotator preferences are underspecified or reward spurious verbosity
Retrieval corpus	Provides evidence at inference time	Chunks are stale, poorly licensed, badly segmented, or not actually retrieved
Evaluation set	Estimates generalization or safety	Contamination, prompt overfitting, weak rubrics, or hidden distribution shift

if its labels are not masked correctly. Data engineering is therefore part of modeling, not a preprocessing chore.

Table 2.1 separates common data artifacts by the contract they impose. The same raw string should not be reused casually across these roles. Training data teaches behavior, retrieval data grounds behavior at inference time, preference data changes the model’s response distribution, and evaluation data measures behavior. Mixing these roles without tracking provenance creates leakage and invalid comparisons.

2.2 Tokenization as a Modeling Choice

A tokenizer is a lossy modeling interface even when it is exactly reversible at the byte or character level. It chooses the units over which the model predicts probabilities. Word-level tokenizers have intuitive units but fail on rare words, names, spelling variants, code, and multilingual text. Character-level tokenizers avoid unknown words but make sequences much longer. Subword tokenizers occupy the middle: frequent strings become single tokens, while rare strings decompose into smaller pieces.

Byte-pair encoding (BPE) adapts a compression idea to subword segmentation by repeatedly merging frequent adjacent symbols [97]. In a simple word-internal version, the initial vocabulary contains characters, and each merge replaces the most frequent pair with a new symbol. After M merges, the final vocabulary contains the original alphabet plus learned merges. Modern GPT-style tokenizers often operate close to bytes so they can represent arbitrary Unicode text without an unknown-token failure. SentencePiece made another important design choice: it trains subword models directly from raw text rather than assuming a pre-tokenized word sequence, and it supports both BPE and unigram language-model tokenization [45]. These details matter for languages without whitespace word boundaries, code, mathematical notation, and mixed-script documents.

For a corpus C and tokenizer τ , a useful diagnostic is token efficiency. For language or domain group g , define

$$\rho_g = \frac{\sum_{d \in C_g} \text{bytes}(d)}{\sum_{d \in C_g} |\tau(d)|}. \quad (2.3)$$

Higher ρ_g means more bytes fit into the same context window. This is not automatically better, because bytes are not semantic units, but a very low value often indicates that the model must spend many decoding steps on text that humans would regard as short. Token efficiency is also an inference cost: autoregressive generation pays one forward pass per output token, so a tokenizer that expands a language by 40% relative to another tokenizer imposes a direct latency and cost penalty for that language.

The tokenizer and the model checkpoint are coupled. The embedding matrix has shape $V \times d_{\text{model}}$, where V is vocabulary size. Changing the tokenizer after pretraining changes token ids, special-token conventions, and the distribution of sequence lengths. Vocabulary extension can be useful for domain adaptation, but it is not a harmless metadata edit: new rows in the embedding and output matrices must be initialized and trained, old tokens may split the same string differently, and tied input-output embeddings make the change affect both reading and generation. A production system should therefore version the tokenizer, the chat template, special tokens, normalization rules, and model weights together.

Implementation note.

A tokenizer pipeline should be tested as a reversible function before it is used for training. The tests should include empty strings, whitespace-only strings, non-ASCII names, code indentation, emoji, malformed Unicode, markup, and the exact special tokens used by the model. If a tokenizer has a beginning-of-sequence token, end-of-sequence token, padding token, or chat role markers, the training code should state which component inserts each token. Silent double insertion of BOS or EOS tokens is a common source of off-by-one labels and surprising loss curves.

2.3 Corpus Construction and Mixtures

A corpus record should be more than a text field. A defensible record includes at least a stable document id, source, acquisition date, license or terms category, language, content type, filtering decisions, and enough lineage to reproduce exclusions. In large projects the raw corpus, filtered corpus, tokenized corpus, and packed training stream should have separate manifests. Without those manifests, it is hard to explain why a model learned a behavior, whether a benchmark was contaminated, or whether a source can be redistributed.

Corpus mixtures are usually heterogeneous. Web pages provide scale and diversity; books provide long-form prose; code provides formal syntax and executable patterns; math and scientific text provide notation and derivations; dialogue and instruction data

provide interaction patterns; domain corpora provide specialized vocabulary and tasks. If mixture component i contains D_i available tokens and is sampled with probability p_i , then a training run of S tokens will expose the model to approximately

$$E_i = Sp_i \quad (2.4)$$

tokens from that component, possibly over multiple epochs when $E_i > D_i$. A small high-quality component can therefore be intentionally upsampled, but repeated exposure increases memorization and overfitting risk. Mixture weights are modeling assumptions and should be reported with enough precision to interpret downstream behavior.

Filtering is a sequence of recall-precision tradeoffs. Language identification can remove off-target text but harm code-switching and low-resource languages. Quality classifiers can remove boilerplate but also encode the biases of their training labels. Toxicity filters can reduce harmful content but remove documents that discuss abuse, medical issues, or newsworthy events in necessary context. Exact rules should be logged because a model trained on the absence of certain topics can fail just as seriously as a model trained on their worst examples.

Deduplication is both a quality intervention and an evaluation intervention. Exact deduplication removes byte-identical or normalized-identical records. Near deduplication uses shingles, hashes, or embedding-like signatures to remove documents with large overlapping spans. Lee et al. show that duplication in language-model datasets can increase memorized generations and distort validation estimates, and that deduplication can reduce memorization while improving training efficiency [49]. The important practical distinction is scope. Deduplicating within the training split reduces overweighted examples; deduplicating between training and evaluation protects benchmark validity; deduplicating across licenses may be required to remove a prohibited source that was mirrored elsewhere.

Implementation note.

Store filtering and deduplication decisions as data, not only as logs. A common pattern is a document table with boolean columns such as `is_language_ok`, `is_quality_ok`, `is_dedup_kept`, and `is_eval_overlap`. That table lets later investigators reconstruct why a document is absent from the final training stream without rerunning every expensive filter.

2.4 Packing Text into Training Sequences

Accelerators prefer dense rectangular tensors, but documents have variable length. A simple batching scheme pads each document to the maximum length in the batch. Padding is easy to reason about, but it wastes compute whenever lengths vary. If a batch has lengths $\ell_1, \dots, \ell_B$ and maximum length T , the useful-token fraction is

$$\eta_{\text{pad}} = \frac{\sum_{b=1}^B \ell_b}{BT}. \quad (2.5)$$

A value of $\eta_{\text{pad}} = 0.55$ means 45% of attention and feed-forward work is spent on padding positions unless kernels explicitly skip them. Pretraining systems therefore usually concatenate tokenized documents into long streams, insert separator or EOS tokens, and slice fixed-length blocks. This approach keeps utilization high but creates another contract: the model may see unrelated documents adjacent in a context window. EOS tokens, document-boundary masks, or careful packing policies determine whether that adjacency becomes a training signal.

For causal language modeling, a packed block of length $T + 1$ often yields inputs $x_{1:T}$ and labels $x_{2:T+1}$. This one-token offset is simple, but it is also the source of many bugs. Padding labels should be set to an ignore index, not to the padding token id, unless the model is intentionally trained to predict padding. In instruction tuning, prompt tokens are commonly masked so that the model is optimized only on the assistant response. If the labels include user tokens, the model is trained partly to imitate the user rather than to answer the user.

Packed datasets also have systems costs. Token streams are large enough that random Python string processing becomes a bottleneck. Practical training pipelines use memory-mapped arrays, binary shards, precomputed indexes, pinned host memory, and asynchronous transfer to the accelerator. Sequence length is a compute multiplier: attention score tensors scale as BHT^2 for batch size B , heads H , and sequence length T . Even when efficient attention kernels reduce memory traffic, longer packed blocks still increase arithmetic and KV-cache size.

2.5 Training Signals Beyond Pretraining

Pretraining supplies broad continuation behavior, but assistant systems need additional signals. Supervised instruction tuning (SFT) trains on demonstrations. A typical record has a prompt, optional context, and a target response. The loss is still cross-entropy, but the label mask changes:

$$\mathcal{L}_{\text{sft}}(\theta) = -\frac{1}{|\mathcal{A}|} \sum_{t \in \mathcal{A}} \log p_{\theta}(x_t \mid x_{<t}), \quad (2.6)$$

where $\mathcal{A}$ indexes assistant-response tokens. The formatting template is part of the data. Role tags, stop tokens, and tool-call delimiters must match the inference template, or the model is trained under one protocol and served under another.

Preference data changes the signal from a single target to comparisons. A record may contain a prompt, a chosen response, and a rejected response. RLHF trains a reward model and then optimizes a policy against that reward under a regularization constraint [84]; direct preference optimization fits a policy more directly from paired preferences [91]. In both cases, the data does not reveal a universal human preference. It

reveals preferences under a rubric, annotator pool, user population, and model sample distribution. Changing any of those can change the meaning of the same pair.

Verifier data and tool traces are yet another signal. A math verifier may label only final answers, intermediate steps, or executable programs. A tool-use trace may include hidden retrieval calls, database queries, or code execution. These traces should be separated from ordinary assistant text because they teach different behavior and may include private credentials, proprietary schemas, or brittle tool outputs.

Retrieval corpora are not training labels, but they are still data. Dense retrieval systems use embedding models and nearest-neighbor indexes to provide evidence at inference time [42, 53]. Their chunking, metadata, freshness, and access controls affect the generated answer as much as the base model does. A retrieval corpus should be evaluated for recall and attribution, not only for downstream answer quality, because a generator can produce plausible answers even when the retriever failed.

2.6 Contamination, Evaluation, and Provenance

Evaluation requires separation between the evidence used for training and the evidence used for measurement. Contamination can be exact, near duplicate, paraphrastic, or procedural. A benchmark solution may appear in a GitHub repository; a multiple-choice question may appear in a web page with reordered options; a code benchmark may be present in a tutorial; a synthetic instruction dataset may have been generated from benchmark prompts. Public benchmarks are especially vulnerable because model developers can optimize prompts, filters, and post-training data against them. Recent work on benchmark leakage and contamination detection should be read as a warning that a high score is not self-validating [100, 115].

A practical contamination audit should run at several levels: normalized string matching, token n-gram overlap, document-source overlap, and semantic or template-level review for high-value benchmarks. The audit should be asymmetric. It is not enough to ask whether the exact benchmark item appears in training data; one should also ask whether explanations, answer keys, generated variants, or translated versions appear. When a benchmark is used for model selection, its prompts and labels should be access-controlled like test data in any other empirical science.

Provenance and licensing are publication constraints. Notes, slides, notebooks, code repositories, datasets, figures, and model outputs can be useful research artifacts, but they should not be copied into a textbook unless their license and authorship permit it. The safer pattern is to use such material to identify topics and then write original exposition from primary sources. For datasets, provenance review should record the source license, whether redistribution is allowed, whether personal data may be present, whether robots or platform terms restrict collection, and whether generated derivatives can be published.

2.7 Key Terms

Byte fallback	A tokenizer design that can represent arbitrary text by falling back to byte-level units instead of producing unknown tokens.
Contamination	Overlap between training or development data and evaluation data that invalidates a generalization estimate.
Deduplication	Removal or downweighting of exact or near-duplicate records within or across data splits.
Mixture weight	The probability with which a data component is sampled during training.
Sequence packing	Concatenating tokenized records into fixed-length training blocks to reduce padding waste.
Tokenizer-model coupling	The dependency between a model checkpoint and the tokenizer, vocabulary, special tokens, and formatting rules used to train it.

2.8 Exercises

1. Train a tiny BPE tokenizer on a small corpus with English prose, Python code, and Chinese text. Report token counts for the same five strings before and after 50, 500, and 5000 merge operations.
2. Given three data components with available sizes $D = (20, 200, 2000)$ million tokens and mixture weights $p = (0.4, 0.3, 0.3)$, compute how many epochs of each component are seen during a 1 billion token run. Which component has the highest memorization risk?
3. Design a label mask for a chat SFT example with system, user, assistant, and tool messages. State exactly which tokens contribute to the loss and why.
4. Implement an exact decontamination check between a toy training set and a toy benchmark using normalized 13-gram overlap. List two cases that this method will miss.
5. For a retrieval corpus, propose metadata fields that would allow answers to cite source, date, license, and access permissions. Explain how missing metadata can create a safety or publication problem.

Part II
Architectures, Optimization, and Systems

Chapter 3

Transformer Mechanics

Abstract This chapter derives the Transformer from tensor shapes. It explains attention, masking, residual streams, normalization, feed-forward layers, positional information, and the implementation invariants needed to debug a model from first principles. The goal is not to memorize a diagram, but to understand the contracts that make deep attention stacks trainable and implementable.

3.1 The Tensor Contract

A Transformer layer is a map from a sequence of hidden vectors to another sequence of hidden vectors. Let a batch of token ids have shape $B \times T$, where B is batch size and T is sequence length. An embedding table maps ids into vectors of width d_{model} , producing

$$X \in \mathbb{R}^{B \times T \times d_{\text{model}}}. \quad (3.1)$$

Every standard Transformer block preserves this outer shape. Attention mixes information across the T positions, and the feed-forward network transforms each position independently, but the residual stream remains $B \times T \times d_{\text{model}}$ from the first block to the final normalization.

The original Transformer combined self-attention, feed-forward sublayers, residual connections, and layer normalization into an architecture that parallelizes over sequence positions far more effectively than recurrent networks [108]. Modern language models adjust many details, but they keep the same central invariant: the model maintains a residual stream and repeatedly applies content-dependent mixing followed by tokenwise nonlinear transformation.

tokens $\rightarrow$ embeddings and positions $\rightarrow$ [norm $\rightarrow$ attention $\rightarrow$ residual $\rightarrow$ norm $\rightarrow$ MLP $\rightarrow$ residual] $\times L \rightarrow$ logits or task head

Fig. 3.1 A shape-preserving view of the Transformer block. The attention sublayer changes which positions can exchange information; the MLP changes each position’s representation; residual addition keeps a stable path through depth.

3.2 Embeddings and Position

Token embeddings supply content but not order. If two sequences contain the same token ids in a different order, a permutation-invariant attention layer without position information cannot distinguish them. Position must enter the computation somehow. There are three broad choices.

Learned absolute position embeddings add a trainable vector P_t to each token embedding:

$$X_{b,t} = E[x_{b,t}] + P_t. \quad (3.2)$$

This is simple and was used in GPT-style models, but it fixes a maximum trained context length and gives no built-in rule for unseen positions. Sinusoidal encodings use deterministic sine and cosine functions at multiple frequencies, allowing extrapolation in principle and giving attention a smooth position signal [108]. Rotary position embeddings (RoPE) instead rotate query and key coordinates by a position-dependent angle before the dot product, so relative offsets influence attention scores directly [104].

Position can also enter as an attention bias. ALiBi, for example, adds a head-specific linear penalty as a function of distance, encouraging recency and enabling some train-short, test-long behavior [86]. Bias methods do not require adding position vectors to the residual stream, but they alter the attention score matrix and therefore interact directly with masking and numerical stability.

Implementation note.

Position ids are not always the same as array indices. Packed sequences, left padding, cached decoding, and sliding-window generation can all make the logical position differ from the tensor column. A robust implementation treats position ids as an explicit tensor or constructs them from a well-tested cache update rule.

3.3 Scaled Dot-Product Attention

Self-attention creates three projections of the same residual stream:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V. \quad (3.3)$$

For a single head with head dimension d_h , the attention output is

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}} + M\right)V, \quad (3.4)$$

where M is an additive mask or bias broadcastable to the score matrix. The scale factor $\sqrt{d_h}$ keeps dot products in a numerically useful range as the head dimension grows. Without it, the softmax can become overly sharp early in training, reducing gradient signal.

In batched multi-head attention, the projections are reshaped into

$$Q, K, V \in \mathbb{R}^{B \times H \times T \times d_h}, \quad (3.5)$$

where $Hd_h = d_{\text{model}}$. The score tensor has shape $B \times H \times T_q \times T_k$. After softmax and multiplication by V , the head outputs are transposed and concatenated back to $B \times T \times d_{\text{model}}$, then multiplied by an output projection. Multi-head attention is not merely parallel copies of the same computation. Each head has its own learned projections and therefore its own similarity space, value space, mask view, and often its own positional bias.

The computational cost of exact full attention is dominated by the score and value products:

$$O(BHT^2d_h) = O(BT^2d_{\text{model}}). \quad (3.6)$$

The memory footprint of materialized attention weights is $O(BHT^2)$. This quadratic term is why context length is a systems decision, not just a modeling hyperparameter.

Numerical stability.

The softmax in Eq. 3.4 should be computed after subtracting the row maximum, and masked positions should be represented by a sufficiently negative additive value before softmax. In mixed precision, many implementations compute score reductions in a higher precision than the stored activations. A model can appear to have an architecture bug when the real problem is overflow, underflow, or a mask with the wrong dtype.

3.4 Masks and Causality

Masks specify which positions may exchange information. A padding mask prevents real tokens from attending to padded tokens. A causal mask prevents position t from attending to positions $> t$. Encoder-decoder cross-attention uses a different pattern: decoder positions attend to all allowed encoder positions, while decoder self-attention remains causal.

The common causal mask is lower triangular:

$$M_{ij} = \begin{cases} 0, & j \leq i, \\ -\infty, & j > i. \end{cases} \quad (3.7)$$

The sign convention matters. An additive mask uses $-\infty$ for disallowed positions; a multiplicative boolean mask may use `True` for allowed positions or for blocked positions depending on the library. Confusing these conventions can train a model that sees the future, attends only to padding, or produces all-`nan` attention rows.

In decoder-only language models, causality is part of the statistical objective. If token x_t can attend to x_{t+1} during training, the model can reduce loss by copying information that will not exist at inference time. This creates a false loss curve and poor generation. Causal masking is therefore not an optional implementation detail; it is the mechanism that makes teacher-forced next-token training match autoregressive decoding.

3.5 Residual Streams and Normalization

Deep Transformers are trainable because each block modifies a residual stream rather than replacing it. In a pre-normalized decoder block,

$$Y = X + \text{Attention}(\text{Norm}_1(X)), \quad (3.8)$$

$$Z = Y + \text{MLP}(\text{Norm}_2(Y)). \quad (3.9)$$

The residual path gives gradients a direct route through depth, while normalization controls the scale of the inputs to attention and MLP sublayers. Post-normalized blocks instead apply normalization after residual addition. Post-norm was used in the original Transformer, but pre-norm variants are often easier to optimize at substantial depth because the sublayer input distribution is more controlled.

Layer normalization centers and rescales a vector using its mean and variance. For hidden vector $h \in \mathbb{R}^d$,

$$\text{LayerNorm}(h)_i = \gamma_i \frac{h_i - \mu}{\sqrt{\sigma^2 + \epsilon}} + \beta_i. \quad (3.10)$$

RMSNorm removes the mean-centering step and rescales by the root mean square [124]:

$$\text{RMSNorm}(h)_i = g_i \frac{h_i}{\sqrt{\frac{1}{d} \sum_{j=1}^d h_j^2 + \epsilon}}. \quad (3.11)$$

The difference is small in notation but significant in implementation: RMSNorm uses fewer reductions and has become a common default in LLaMA-class decoders.

Normalization does not make scale irrelevant. Residual branches, initialization, optimizer settings, activation functions, and precision still determine whether activations remain in a useful range. A useful debugging habit is to log activation norms per layer during early training. Sudden layer-local norm explosions often reveal a mask, initialization, learning-rate, or mixed-precision error before the loss becomes unrecoverable.

Table 3.1 Implementation invariants for a Transformer block.

Check	Expected Property	Symptom When Broken
Residual shape	Every block returns $B \times T \times d_{\text{model}}$	Later projection shape errors or silent broadcasting
Head split	d_{model} divisible by number of heads	Incorrect reshape or mixed head dimensions
Causal mask	Token t attends only to positions $\leq t$	Unrealistically low train loss and bad generation
Padding labels	Padding does not contribute to loss	Model learns to emit padding or loss depends on batch padding
Position ids	Cached and uncached paths use the same logical positions	Prefill works but token-by-token decoding drifts
Softmax rows	Every unmasked query has at least one allowed key	nan probabilities after masking

3.6 Feed-Forward Blocks

The feed-forward sublayer is positionwise: it applies the same MLP to each token independently. The original Transformer uses a two-layer network with a larger hidden width:

$$\text{FFN}(h) = W_2 \phi(W_1 h + b_1) + b_2. \quad (3.12)$$

The expansion width is often around $4d_{\text{model}}$ in GPT-style blocks. This sublayer supplies a large fraction of the model’s parameters and compute because it is applied at every position in every layer.

Gated variants split the hidden projection into a value branch and a gate branch. A SwiGLU-style layer can be written

$$\text{SwiGLU}(h) = W_o (\text{SiLU}(W_g h) \odot W_u h), \quad (3.13)$$

where $\odot$ denotes elementwise multiplication. GLU variants have been found effective in Transformer feed-forward sublayers [99]. Modern implementations often shrink the intermediate width relative to the simple $4d$ MLP so the parameter count remains comparable after adding the gate projection.

3.7 Debugging Invariants

Most Transformer bugs are shape bugs, mask bugs, or convention bugs. Table 3.1 lists invariants that should hold before training at scale. These checks are cheap compared with a wasted run.

Attention weights are useful diagnostics but weak explanations. A head that attends to a token does not prove that the token caused the output, because later layers, value projections, residual paths, and MLPs can transform or erase that information. Evalua-

tion of a Transformer should therefore combine mechanistic inspection with controlled interventions, held-out loss, downstream tasks, and generation behavior.

3.8 Key Terms

Attention head	One learned query-key-value projection subspace inside multi-head attention.
Causal mask	A mask that prevents a position from attending to future positions.
Residual stream	The shape-preserving hidden state that passes through all blocks by residual addition.
RMSNorm	A normalization layer that rescales by root mean square without subtracting the mean.
Scaled dot-product attention	Attention computed by softmax-normalized query-key dot products divided by $\sqrt{d_h}$.
Tokenwise MLP	A feed-forward network applied independently to each sequence position.

3.9 Exercises

1. For $B = 8$, $T = 1024$, $d_{\text{model}} = 768$, and $H = 12$, write the shapes of Q , K , V , the attention score tensor, the attention output before concatenation, and the final output projection input.
2. Implement single-head causal attention for a batch tensor and verify with an assertion that changing token x_{t+1} cannot change the output at position t .
3. Compare learned absolute positions, sinusoidal positions, RoPE, and ALiBi. For each, state where position enters the computation and one extrapolation risk.
4. Derive the number of parameters in a two-layer MLP with expansion ratio 4 and compare it with a gated MLP that uses two input projections of width m and one output projection.
5. Create a mask convention test for a deep learning library of your choice. Show how a boolean mask and an additive mask represent the same causal pattern.

Chapter 4

From Transformer to GPT

Abstract This chapter specializes the Transformer into a causal decoder. It covers next-token loss, GPT-style blocks, sampling, repetition control, loss curves, minimal training loops, and efficient attention as used in practical GPT implementations. GPT is best understood as an interface: train by teacher-forced next-token prediction, then use the same conditional distribution for autoregressive generation.

4.1 The Decoder-Only Contract

A GPT-style model is a stack of masked self-attention blocks trained to predict the next token. The architecture is decoder-only: there is no separate encoder memory, no bidirectional self-attention, and no masked-token objective. For a token sequence $x_{1:T}$, the model defines

$$p_{\theta}(x_{1:T}) = \prod_{t=1}^T p_{\theta}(x_t \mid x_{<t}). \quad (4.1)$$

This factorization turns every position into a supervised target and makes arbitrary text usable as self-supervised data. GPT-2 and GPT-3 demonstrated that the same interface can support broad zero-shot and few-shot behavior when scaled over heterogeneous corpora [90, 10].

The training and inference contracts must match. During training, teacher forcing supplies the true prefix $x_{<t}$ for all positions in parallel. During generation, the model receives its own sampled prefix one token at a time. Causal masking is the bridge between the two: it prevents the training computation at position t from seeing targets that will be unavailable during generation.

4.2 Causal Language Modeling

In implementation, a packed block normally contains $T + 1$ token ids. The model input is the first T tokens and the label sequence is shifted by one:

$$\text{input} = x_{1:T}, \quad \text{label} = x_{2:T+1}. \quad (4.2)$$

If logits have shape $B \times T \times V$, where V is vocabulary size, the token-level cross-entropy loss is

$$\mathcal{L}(\theta) = -\frac{1}{N_{\text{valid}}} \sum_{b=1}^B \sum_{t=1}^T m_{b,t} \log \frac{\exp z_{b,t,y_{b,t}}}{\sum_{v=1}^V \exp z_{b,t,v}}, \quad (4.3)$$

where $m_{b,t}$ is 1 for valid labels and 0 for ignored labels. Ignored labels are needed for padding, prompt masking in instruction tuning, and occasionally for boundary tokens when documents are packed.

Perplexity is $\exp(\mathcal{L})$ when natural logs are used. It is a useful loss-scale diagnostic, but it is not a universal capability metric. Perplexity depends on tokenizer, corpus, context length, and evaluation preprocessing. A character-level model and a BPE model can assign different numbers of targets to the same string; comparing their perplexities without normalization is misleading.

Implementation note.

The most common GPT data bug is an incorrect shift. A quick test is to feed a known sequence such as [10, 20, 30, 40] and inspect one batch. The labels should be [20, 30, 40, next] for a stream sample or [20, 30, 40, ignore] for a bounded sample. If labels equal inputs, the model is being trained as an autoencoder, not a causal language model.

4.3 A Minimal GPT Block Stack

A minimal GPT configuration fixes vocabulary size V , maximum block size T_{max} , number of layers L , number of heads H , hidden width d_{model} , dropout, and whether linear and normalization layers use bias terms. The forward pass has five stages:

1. Map token ids to token embeddings and add or apply position information.
2. Pass the residual stream through L causal decoder blocks.
3. Normalize the final hidden states.
4. Project hidden states to vocabulary logits.
5. Compute cross-entropy loss if labels are supplied.

GPT-style blocks commonly use pre-normalization:

$$h^{(\ell+\frac{1}{2})} = h^{(\ell)} + \text{MHA}\left(\text{Norm}(h^{(\ell)})\right), \quad (4.4)$$

$$h^{(\ell+1)} = h^{(\ell+\frac{1}{2})} + \text{MLP}\left(\text{Norm}(h^{(\ell+\frac{1}{2})})\right). \quad (4.5)$$

The attention sublayer uses a causal mask; the MLP sublayer is tokenwise; residual additions preserve shape. Weight tying often reuses the token embedding matrix as the output projection matrix, reducing parameters and making input and output token spaces share representations.

Optimizer grouping is part of the implementation recipe. Weight decay is usually applied to large matrix weights but not to biases or normalization scale parameters. AdamW, learning-rate warmup, cosine decay, gradient clipping, mixed precision, checkpointing, and distributed data parallelism belong to the training system rather than the mathematical block, but they determine whether the block can be trained reliably. Chapter 6 returns to those details.

Checkpoint metadata should include the model configuration, tokenizer identity, optimizer state, training step, random number state when resumability matters, and enough data-stream state to avoid accidentally repeating or skipping large parts of a corpus. A checkpoint that contains only weights is often sufficient for inference, but it is not sufficient for reproducible training.

4.4 Batching, Packing, and Loss Curves

The number of tokens per optimizer step is

$$N_{\text{tok/step}} = B_{\text{micro}} \times T \times G \times W, \quad (4.6)$$

where B_{micro} is micro-batch size, T is sequence length, G is gradient accumulation steps, and W is data-parallel world size. This quantity matters more than batch size alone because it controls gradient noise scale, throughput reporting, and how quickly the model consumes the corpus.

A healthy pretraining loss curve is smooth only after averaging. Individual steps can spike because a batch contains unusual data, long code spans, repeated tokens, or numerical instability. Validation loss should be estimated over enough batches to reduce sampling noise, and the validation set should be deduplicated from training data. Compute-optimal planning connects model size, token budget, and loss, but a run can still be wasteful if the data stream is low quality or the tokenizer expands important domains excessively [35].

Loss curves must be interpreted with evaluation context. A falling validation loss on web text does not guarantee better instruction following, factuality, long-context use, or safety. Conversely, a small domain model can have worse broad perplexity and better performance on a narrow, well-specified domain. The correct question is not whether loss improved, but whether loss improved on a measurement distribution that matches the intended claim.

Table 4.1 Common decoding controls and their main risks.

Control	Intended Effect	Risk
Greedy	Deterministic highest-probability continuation	Repetition and poor exploration
Temperature	Adjust distribution sharpness	Can amplify low-quality tails or collapse diversity
Top- k	Remove low-rank candidates	Fixed k ignores uncertainty variation across steps
Top- p	Keep a dynamic probability nucleus	Sensitive to calibration and tokenization
Repetition penalty	Discourage repeated tokens or spans	Can damage legitimate repetition in code, poetry, and lists
Stop condition	End generation under protocol rules	String-level stops can fail on token-boundary variants

4.5 From Logits to Text

At generation step t , the model produces logits $z_t \in \mathbb{R}^V$ for the next token. Greedy decoding selects $\arg \max_v z_{t,v}$. Greedy decoding is deterministic and useful for some constrained tasks, but it can produce repetitive or brittle text. Sampling draws from a probability distribution, usually

$$p(v \mid x_{<t}) = \text{softmax}\left(\frac{z_{t,v}}{\tau}\right), \quad (4.7)$$

where $\tau > 0$ is temperature. Lower temperature sharpens the distribution; higher temperature flattens it. Temperature is not a creativity knob in isolation. Its effect depends on model calibration, prompt, vocabulary, and any truncation rule.

Top- k sampling restricts the distribution to the k highest-probability tokens before sampling. Nucleus, or top- p , sampling chooses the smallest set of tokens whose cumulative probability exceeds p and samples within that set. Holtzman et al. introduced nucleus sampling as a response to degeneration in open-ended neural text generation [36]. Repetition penalties, presence penalties, frequency penalties, and no-repeat n -gram constraints further modify logits or candidate sets to discourage loops.

Stopping is also part of decoding. Generation may stop at an EOS token, a maximum length, a user-defined delimiter, a tool-call boundary, or a safety policy. A stop string applied after detokenization is not equivalent to a stop token applied during decoding. Stop logic should be tested on partial tokens, whitespace variants, multilingual punctuation, and adversarial prompts that try to hide delimiters.

4.6 Efficient Attention and Generation

Training a GPT block over a length- T sequence computes all positions in parallel. Generation has two phases. In prefill, the prompt is processed as a full sequence and the model creates key-value tensors for every layer. In decode, one new token is processed

at a time while reusing the previous keys and values. Without a KV cache, generating n tokens after a prompt would repeatedly recompute the entire prefix. With a KV cache, each step computes new queries, keys, and values for the latest token and attends to cached keys and values.

The cache size for a dense multi-head decoder is approximately

$$\text{bytes}_{\text{KV}} = 2 \times B \times L \times T \times H \times d_h \times \text{bytes_per_element}. \quad (4.8)$$

The factor of 2 stores keys and values. Long contexts and large batches can make KV memory the limiting serving resource even when model weights fit comfortably on the device.

FlashAttention improves exact attention by changing how the computation is tiled through the memory hierarchy [17]. It does not approximate the softmax attention formula; it avoids materializing large intermediate attention matrices in high-bandwidth memory and recomputes or streams quantities in an IO-aware way. This distinction is important. Some efficient attention methods change model semantics; FlashAttention is an implementation of the same attention operation under suitable masks and precisions.

Efficient generation also depends on batching policy. Continuous batching improves device utilization by admitting new requests as others finish, but it complicates cache layout and stop handling. Prefix caching can reuse KV states for shared prompts, but it requires exact tokenizer and prompt-template equality. Quantized KV caches reduce memory but can affect long-context quality. These are serving-system choices, yet they feed back into model design through context length, number of KV heads, and tokenizer efficiency.

4.7 Evaluation Cautions

GPT evaluation should separate loss, decoding behavior, and task accuracy. Loss is measured under teacher forcing; generation is measured under the model’s own sampled prefixes. A model can have lower loss and worse open-ended samples if decoding is poorly chosen. A model can score well on a benchmark and fail in deployment if the benchmark is contaminated, too narrow, or unlike user prompts. A decoding configuration can improve a demo while reducing factuality or calibration.

Report the tokenizer, context length, prompt format, decoding settings, number of samples, random seeds when relevant, and exact metric. For stochastic decoding, a single sample is a weak estimate. For pass-at- k style coding or reasoning metrics, increasing k spends more inference compute and should not be compared directly with a single-sample result unless that is the intended product budget.

4.8 Key Terms

Autoregressive generation	Producing tokens one at a time while conditioning each step on previously generated tokens.
Causal language modeling	Training a model to predict each token from only earlier tokens.
KV cache	Stored key and value tensors reused during decoding to avoid recomputing the prefix.
Nucleus sampling	Sampling from the smallest token set whose cumulative probability exceeds a threshold p .
Teacher forcing	Training with true previous tokens supplied as context for every target position.
Weight tying	Sharing the token embedding matrix with the output projection matrix.

4.9 Exercises

1. For a packed token array of length 17 and block size 8, construct two training examples using the $T + 1$ shift convention. Mark inputs, labels, and ignored labels.
2. Implement a tiny GPT block with shape assertions after embedding, attention, MLP, final normalization, and logits. Test that logits have shape $B \times T \times V$ during training and $B \times 1 \times V$ during cached single-token decoding.
3. Train a character-level causal model on a small public-domain text. Plot training loss and validation loss, then explain one sign of overfitting and one sign of a data or label bug.
4. On the same prompt and model, compare greedy, temperature sampling, top- k , and top- p decoding. Hold maximum length and stop conditions fixed, and describe how conclusions change across at least five random seeds.
5. Compute KV-cache memory for a 24-layer model with $B = 8$, $T = 4096$, $H = 16$, $d_h = 64$, and bfloat16 cache tensors. How does the answer change if grouped-query attention uses 4 KV heads?

Chapter 5

LLaMA-Class Architectures

Abstract This chapter studies the architectural conventions used by modern open decoder models: RMSNorm, RoPE, SwiGLU-style feed-forward networks, grouped-query attention, KV cache design, long-context extensions, and mixture-of-experts variants. LLaMA-class models are not a different species from GPT; they are a refined decoder-only recipe whose details change stability, memory, throughput, and context behavior.

5.1 What Makes a Decoder LLaMA-Class

Modern open large language models are mostly decoder-only Transformers, but the defaults have changed since early GPT implementations. LLaMA and its successors popularized a compact recipe: pre-normalized decoder blocks, RMSNorm, rotary position embeddings, bias-free linear layers, SwiGLU-style feed-forward networks, careful tokenizer and data choices, and post-training pipelines that make the base model usable as an assistant [106, 107, 29]. Later families add grouped-query attention, longer context, stronger multilingual and code mixtures, tool-oriented post-training, and sparse mixture-of-experts variants [1, 20, 43].

A LLaMA-class block can be written as

$$y^{(\ell)} = x^{(\ell)} + \text{Attn}\left(\text{RMSNorm}(x^{(\ell)})\right), \quad (5.1)$$

$$x^{(\ell+1)} = y^{(\ell)} + \text{SwiGLU}\left(\text{RMSNorm}(y^{(\ell)})\right). \quad (5.2)$$

This is still the residual-stream structure from Chapter 3. The significance is in the defaults. RMSNorm changes normalization cost and behavior; RoPE changes where positions enter attention; SwiGLU changes MLP expressivity and parameter allocation; grouped-query attention changes the KV-cache budget. These are not cosmetic differences when training or serving billions of parameters.

5.2 RMSNorm and Pre-Normalization

RMSNorm rescales a hidden vector without subtracting its mean [124]. For $h \in \mathbb{R}^d$,

$$\text{RMSNorm}(h) = g \odot \frac{h}{\sqrt{\frac{1}{d} \sum_{j=1}^d h_j^2 + \epsilon}}, \quad (5.3)$$

where g is a learned scale vector. Compared with LayerNorm, RMSNorm removes the mean reduction and bias term in common implementations. The saved work is small per vector but large across every token, layer, and generation request. More importantly, it is part of a block recipe that has been empirically stable for deep decoder-only models.

Pre-normalization means the attention and MLP sublayers see normalized inputs, while residual additions carry unnormalized accumulated state forward. This tends to improve optimization at depth because each sublayer receives a controlled scale even as the residual stream evolves. The final RMSNorm before the language-model head is also important: it makes logits depend on a normalized representation rather than on raw residual scale.

Implementation note.

RMSNorm is easy to implement incorrectly in mixed precision. The mean of squares should usually be accumulated in a sufficiently stable dtype, and ϵ should match the checkpoint family. Changing ϵ or whether the scale is applied before or after casting can produce small logit differences that become visible during long autoregressive generation.

5.3 SwiGLU Feed-Forward Networks

The simple GPT MLP expands from d to $4d$, applies GELU, and projects back to d . LLaMA-class models often use a gated feed-forward network inspired by GLU variants [99]:

$$\text{FFN}(h) = W_o (\text{SiLU}(W_g h) \odot W_u h). \quad (5.4)$$

There are two input projections, W_g and W_u , and one output projection, W_o . To keep parameter count near the ungated $4d$ MLP, the hidden width is often set near $\frac{8}{3}d$ and rounded to a hardware-friendly multiple. The gate lets the model modulate candidate features elementwise, while the up projection supplies the candidate values.

The feed-forward block is often the largest parameter component of a dense decoder layer. With hidden width m , the gated MLP has approximately $3dm$ weights, ignoring biases. A standard $4d$ MLP has approximately $8d^2$ weights. Setting $m \approx \frac{8}{3}d$ makes these comparable. Rounding m upward can improve tensor-core utilization but changes parameter count, activation memory, and checkpoint size.

5.4 Rotary Position Embeddings

RoPE applies a position-dependent rotation to query and key vectors before the attention dot product [104]. Pair adjacent dimensions and rotate each pair by angle $m\theta_i$ at position m :

$$R_{m,i} = \begin{bmatrix} \cos(m\theta_i) & -\sin(m\theta_i) \\ \sin(m\theta_i) & \cos(m\theta_i) \end{bmatrix}. \quad (5.5)$$

If a query at position m and a key at position n are rotated, their dot product contains

$$(R_m q)^\top (R_n k) = q^\top R_m^\top R_n k, \quad (5.6)$$

and $R_m^\top R_n$ depends on the relative offset $n - m$. This is the geometric reason RoPE gives attention access to relative position while remaining compatible with standard dot-product attention.

RoPE is not a free long-context solution. A model trained up to one context length has learned to use a particular range of rotation phases. Extrapolating to much longer contexts can create phase patterns that were rare or absent during training. Long-context extensions use methods such as position interpolation, frequency scaling, continued training, sliding attention, or attention biases. ALiBi is a contrasting approach that adds linear distance biases rather than rotating query and key vectors [86]. The evaluation question is not whether a model accepts a longer input tensor, but whether it uses distant evidence reliably without damaging shorter-context behavior.

Implementation note.

RoPE caches must align with the decode cache. In prefill, positions are usually $0, \dots, T-1$. In cached decoding, the new token’s query and key must use the next logical position, not position 0. A one-token error can be almost invisible on short prompts and severe on long generations.

5.5 KV Cache and Attention Variants

During autoregressive decoding, every layer stores keys and values from previous tokens. For a dense model with L layers, batch B , context length T , head dimension d_h , and H_{kv} key-value heads, the approximate cache memory is

$$\text{bytes}_{kv} = 2BLTH_{kv}d_h s, \quad (5.7)$$

where s is bytes per element. Multi-head attention has $H_{kv} = H_q$, one KV head per query head. Multi-query attention shares a single KV head across all query heads, reducing cache memory and memory bandwidth during decoding [98]. Grouped-query attention uses an intermediate number of KV heads, grouping several query heads per

Table 5.1 Attention variants by KV-cache structure.

Variant	KV Heads	Main Benefit	Main Risk
Multi-head attention	H_q	Maximum per-head KV flexibility	Largest decode cache and bandwidth
Multi-query attention	1	Very small KV cache	Quality loss on some tasks or scales
Grouped-query attention	$1 < H_{kv} < H_q$	Good memory-quality trade-off	Requires compatible checkpoint design
Latent or compressed attention	Model-specific	Smaller or transformed memory state	Harder to compare semantics and kernels

KV head, and often preserves more quality than full multi-query attention at similar serving benefits [1].

Equation 5.7 explains why grouped-query attention became a practical default. Weight memory is paid once per loaded model replica, but KV memory grows with active batch and context. In a long-context serving system, KV cache can dominate capacity. Reducing H_{kv} improves throughput not only by saving memory, but also by reducing memory bandwidth in token-by-token decoding.

Some newer architectures go beyond GQA with latent or compressed attention states, specialized cache layouts, or sparse attention patterns [20]. These variants should be evaluated under the exact serving regime of interest. A method that improves peak throughput for long prompts may add overhead for short prompts; a cache compression that preserves average benchmark scores may fail on retrieval-heavy tasks requiring exact distant evidence.

5.6 Tokenizer and Vocabulary Contracts

LLaMA-class checkpoints inherit the tokenizer-model coupling from Chapter 2. The vocabulary determines embedding and output projection shapes, special-token ids, chat-template conventions, and token efficiency. Open model families often change tokenizer vocabularies across generations to improve multilingual and code coverage. Such changes are architectural from the checkpoint’s point of view: the embedding table, output head, and all learned token statistics are tied to the tokenizer.

Vocabulary size is also a systems parameter. Embedding and output matrices scale as Vd_{model} . During training, the final softmax over V can be expensive; during inference, logits processors and sampling operate over V each step. Implementations often pad vocabulary size to a multiple that improves matrix-kernel efficiency. Padding rows must not become ordinary sampled tokens.

5.7 Long Context

Long context has three separate meanings. The first is acceptance: the model can run a forward pass at length T . The second is retrieval: the model can identify relevant information somewhere in that context. The third is reasoning or synthesis: the model can combine distant evidence correctly. Many failures occur because a system demonstrates the first and claims the third.

Training for long context increases attention cost, activation memory, data requirements, and evaluation burden. Continuing training at longer sequence lengths can adapt positional behavior, but it may also change short-context performance. Synthetic needle-in-a-haystack tests are useful sanity checks but incomplete. A model can find a unique key phrase and still fail when evidence is paraphrased, conflicting, tabular, or distributed across sections. Long-context evaluation should include retrieval, aggregation, contradiction handling, citation faithfulness, and latency/cost reporting.

Serving long context is dominated by prefill and cache pressure. Prefill cost grows with the prompt length and full attention pattern; decode cost grows with cache length. Sliding-window attention, retrieval-augmented generation, summary memory, and cache eviction policies are alternatives to simply increasing $T_{\max}$. These alternatives change the task semantics and should be exposed in evaluation.

5.8 Mixture-of-Experts Variants

Mixture-of-experts (MoE) models replace some dense feed-forward computation with a set of expert networks and a router. For token hidden state h_t , a router produces scores over experts. A top- k router selects a small subset S_t and computes

$$\text{MoE}(h_t) = \sum_{e \in S_t} \alpha_{t,e} E_e(h_t), \quad (5.8)$$

where E_e is expert e and $\alpha_{t,e}$ is a routing weight. The model can have many total parameters while activating only a fraction per token. This changes the meaning of model size: total parameters, active parameters, routing overhead, and communication cost must be reported separately.

Sparse computation has failure modes that dense models do not. Routers can collapse onto a few experts, leaving others undertrained. Load-balancing losses can improve hardware utilization while interfering with specialization. Expert parallelism introduces all-to-all communication across devices. Capacity limits can drop or reroute tokens. Evaluation can hide these problems if it reports only aggregate accuracy rather than per-domain behavior, routing distribution, latency percentiles, and failure cases.

MoE models such as DeepSeek-V3 and Kimi K2-style systems show how sparse scaling can support strong open-weight performance claims, but the systems contract is more complex than a dense LLaMA-class model [20, 43]. A publishable comparison should specify active parameters, total parameters, context length, data mixture where known, post-training recipe, inference budget, and serving hardware.

5.9 Beyond the LLaMA-Class Decoder

Decoder-only Transformers remain the reference architecture for most deployed large language models, but they are not the only plausible sequence backbone. Retentive networks, selective state-space models, and hybrid memory architectures try to change the long-context cost model by replacing all-pairs attention with recurrent or chunkwise state updates [105, 30, 18, 7]. The attraction is clear: if a model can summarize the prefix into a compact recurrent state, decoding no longer needs a KV cache that grows linearly with every past token in every layer.

These architectures should be compared by contract, not by slogan. Attention exposes direct content-addressed access to previous token representations; recurrent or state-space layers compress history into a state whose update rule determines what can be remembered, forgotten, or retrieved. Mamba-style selective state spaces make the dynamics input-dependent, improving content selectivity relative to older SSMs [30]. Mamba-2 and related state-space duality work further connect attention-like computation and structured recurrent algorithms [18]. RetNet proposes parallel training with recurrent inference, while memory-augmented designs such as Titans make test-time memory a learned component rather than a fixed KV table [105, 7].

For a book about large language models, the practical conclusion is conservative. These models are essential frontier material for long-context and streaming systems, but a replacement claim needs evidence at comparable data scale, tokenizer, training budget, context distribution, and serving workload. Many promising sequence models win on asymptotic or medium-scale efficiency but must still prove broad instruction following, retrieval fidelity, tool use, and post-training compatibility at frontier scale.

5.10 Evaluation Cautions

Architecture claims should be tested under matched training and serving budgets. Replacing LayerNorm with RMSNorm, GELU with SwiGLU, MHA with GQA, or absolute positions with RoPE changes parameter count, activation memory, cache memory, and often the optimal learning rate. A fair ablation controls for tokens, compute, data order, tokenizer, optimizer, and evaluation prompts as much as possible.

Open model reports are valuable primary sources, but they are not interchangeable with independent replication. They often combine architecture changes with data improvements, post-training changes, filtering, safety tuning, and benchmark-specific evaluation choices. When a report attributes gains to an architecture, ask whether the evidence isolates that factor. When a deployment chooses a LLaMA-class model, evaluate the actual checkpoint and decoding path rather than relying on the family name.

5.11 Key Terms

Activated parameters	Parameters used for a particular token or forward pass, especially in sparse MoE models.
Grouped-query attention	Attention in which multiple query heads share each key-value head.
KV-cache memory	Memory required to store past keys and values for autoregressive decoding.
RoPE	Rotary position embeddings, which rotate query and key dimensions by position-dependent angles.
RMSNorm	Normalization that rescales activations by their root mean square.
SwiGLU	A gated feed-forward layer using a SiLU gate and element-wise multiplication.

5.12 Exercises

1. Modify a GPT-style block on paper into a LLaMA-style block. List every changed component: normalization, bias terms, position handling, MLP activation, and attention cache assumptions.
2. For $d = 4096$, compare the approximate MLP parameter counts for a GELU MLP with hidden width $4d$ and a SwiGLU MLP with hidden width $\frac{8}{3}d$ rounded up to the nearest multiple of 256.
3. Derive the relative-position property of RoPE by multiplying $R_m^\top R_n$ for a two-dimensional coordinate pair.
4. Compute KV-cache memory for MHA, MQA, and GQA with $L = 32$, $B = 16$, $T = 8192$, $H_q = 32$, $H_{kv} \in \{32, 8, 1\}$, $d_h = 128$, and bfloat16 elements.
5. Design an ablation plan to test whether GQA changes downstream quality independently of total parameter count and training tokens.
6. For an MoE layer with 64 experts, top-2 routing, and capacity limits, name three metrics besides benchmark accuracy that should be reported to make the architecture claim credible.

Chapter 6

Optimization and Pretraining

Abstract This chapter connects the mathematical objective of language-model pretraining to the engineering recipe used to make large runs stable. It covers autoregressive loss accounting, AdamW, parameter groups, learning-rate schedules, warmup, gradient clipping, mixed precision, checkpoint recovery, data order, loss prediction, compute-optimal planning, and evaluation cautions.

6.1 The Pretraining Problem

Pretraining is the stage in which a model learns a broad distribution before any assistant-specific behavior is imposed. For a causal decoder, the training example is a token sequence $x_{1:T}$ and the objective is next-token prediction:

$$\mathcal{L}(\theta) = -\frac{1}{M} \sum_{(b,t) \in \mathcal{M}} \log p_{\theta}(x_{b,t} \mid x_{b,<t}), \quad (6.1)$$

where $\mathcal{M}$ is the set of positions that contribute to loss. In simple packed pretraining, $\mathcal{M}$ contains almost every non-padding position. In instruction tuning, many prompt tokens are masked, but base pretraining usually treats the stream itself as the supervision signal. This difference matters: a pretraining loss curve is a measurement of distributional compression over the corpus mixture, not a direct measurement of helpfulness, truthfulness, or safety.

The recipe is a coupled choice of architecture, token budget, context length, data mixture, optimizer, precision, hardware layout, and stopping rule. Scaling-law studies made this coupling explicit by showing predictable loss trends with model size, data size, and compute over broad ranges [41]. Compute-optimal work then showed that many early large models were too parameter-heavy for the number of tokens they saw, shifting modern practice toward training smaller or similarly sized models on far more data [35]. Open model reports reinforce the point: LLaMA-class models are not only architectural artifacts; they are also data and optimization artifacts [106, 107, 29].

A practical pretraining plan begins with a units table. Let N be trainable parameters, D be training tokens, T be sequence length, B_μ be micro-batch size per device, G be the number of gradient-accumulation micro-steps, and P be the data-parallel world size. The global token batch is

$$B_{\text{tok}} = B_\mu T G P. \quad (6.2)$$

The number of optimizer updates is approximately D/B_{tok} . For dense decoder-only Transformers, a common planning approximation is that training costs about $6ND$ floating-point operations, excluding attention details, embedding/output costs, recomputation, data loading, and system inefficiency. The approximation is useful because it gives the right first-order economics: doubling either parameters or tokens roughly doubles training compute, while increasing context length changes both model-side attention cost and systems-side activation memory.

6.2 Objective Accounting and Data Order

The loss is only meaningful when tokenization, masking, and data provenance are controlled. Two models can report the same numerical loss while using different tokenizers, different document separators, different treatment of repeated text, or different validation mixtures. Perplexity is therefore most useful within a fixed experimental family. Across model families, it should be interpreted as a diagnostic rather than as a universal capability score.

Packed sequence training converts documents into fixed-length blocks. This improves accelerator utilization because every position participates in a dense matrix computation. It also creates subtle boundary choices. If document boundaries are ignored, the model may learn unrealistic transitions across unrelated documents. If every document starts with a boundary token, the model receives a distributional signal about resets, but the vocabulary and tokenizer must reserve and consistently use that marker. If samples are packed with loss masking across boundaries, the data loader must keep masks synchronized with token ids after shuffling and distributed partitioning.

Data order is an optimization variable. A uniform shuffle over all tokens is simple, but large corpora are mixtures: web pages, books, code, papers, multilingual text, math, dialogue, and synthetic data. Mixture weights determine both what the model learns and how stable the loss becomes. Code and math can produce different gradient statistics from casual web text; low-quality duplicates can lower training loss while harming downstream robustness; benchmark-like documents can inflate evaluation through contamination [100, 115]. A serious run keeps a manifest of mixture weights, token counts, deduplication policy, filtering policy, and validation splits.

The validation set should be fixed before the run and should be separated by source, not merely by random token slice. Random slices from a large web corpus may share near-duplicate documents with training. A useful validation dashboard reports loss by mixture component: general web, code, math, books, academic text, multilingual buckets, and any domain-specific data. When a global loss spike occurs, per-component

Table 6.1 Pretraining bookkeeping that should be fixed before a run starts.

Item	Why It Matters	Failure Mode
Tokenizer and boundary tokens	Defines the loss units and document tran- sition signal	Perplexity cannot be compared across runs
Loss mask	Specifies which positions train the model	Padding, separators, or prompt to- kens accidentally dominate gradi- ents
Mixture weights	Controls the distribution being learned	A global loss hides regressions in code, math, or low-resource lan- guages
Shard order and seed	Determines reproducibility and restart be- havior	A resumed run silently repeats or skips data
Validation sources	Separates optimization diagnostics from benchmark claims	Near-duplicate validation makes the run look healthier than it is
Contamination checks	Protects downstream evaluation	Public benchmark text is learned during pretraining

losses often reveal whether the problem is optimizer instability, data corruption, a single bad shard, or a distribution shift introduced by a curriculum boundary.

6.3 AdamW and Parameter Groups

Most modern large language model pretraining uses AdamW or a close variant. Adam maintains exponential moving averages of first and second moments of the stochastic gradient [44]. For gradient $g_t = \nabla_{\theta} \mathcal{L}_t(\theta_t)$,

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad (6.3)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \quad (6.4)$$

with bias-corrected estimates $\hat{m}_t = m_t / (1 - \beta_1^t)$ and $\hat{v}_t = v_t / (1 - \beta_2^t)$. A simplified AdamW update is

$$\theta_{t+1} = \theta_t - \eta_t \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} - \eta_t \lambda \theta_t, \quad (6.5)$$

where η_t is the learning rate and λ is decoupled weight decay [68]. The word “decoupled” is important. Classical L_2 regularization adds $\lambda \theta$ to the gradient, so the adaptive denominator changes the effective regularization. AdamW applies decay as a separate multiplicative shrinkage term, which makes the meaning of weight decay less entangled with gradient scale.

Parameter groups are part of the algorithm, not cosmetic code. Common practice applies weight decay to large weight matrices but excludes biases, normalization parameters, and sometimes embeddings. The reason is structural: a normalization scale or bias does not behave like a high-dimensional weight matrix whose norm should be regularized. A typical implementation builds two groups:

$$\Theta_{\text{decay}} = \{W : \dim(W) \geq 2\}, \quad \Theta_{\text{nodecay}} = \Theta \setminus \Theta_{\text{decay}}. \quad (6.6)$$

The exact rule should be logged, because small changes can affect reproducibility across codebases.

The Adam hyperparameters also encode assumptions about gradient noise. Many large language model recipes use β_1 near 0.9 and β_2 lower than the original Adam default of 0.999, often around 0.95 or 0.98, so the second-moment estimate adapts faster during long runs with changing data mixtures. The ϵ value is not merely a numerical afterthought in low precision: it affects the denominator when second moments are small. Any comparison of optimizer variants should state $(\beta_1, \beta_2, \epsilon)$, weight decay, clipping rule, schedule, batch size, and precision.

6.4 Schedules, Warmup, and Batch Size

The learning-rate schedule controls how aggressively the run spends optimization budget. A common schedule warms up linearly, decays with a cosine curve, and ends at a nonzero minimum:

$$\eta_t = \begin{cases} \eta_{\max} \frac{t}{t_w}, & t < t_w, \\ \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\pi \frac{t-t_w}{t_d-t_w}\right)\right), & t_w \leq t \leq t_d, \\ \eta_{\min}, & t > t_d. \end{cases} \quad (6.7)$$

Warmup prevents the first few thousand steps from taking full-size adaptive updates before moment estimates and activation scales have settled. Cosine decay gradually reduces update size without a sharp discontinuity. Some production runs reserve a final annealing phase on a changed mixture, but such phases should be treated as new experiments: changing the data distribution near the end can improve targeted evaluations while obscuring what the base model learned from the main corpus.

Batch size is best measured in tokens, not examples. A batch of eight sequences at context length 8192 contains as many token positions as sixty-four sequences at context length 1024. Larger token batches reduce gradient noise and improve device utilization, but they also reduce the number of optimizer updates for a fixed token budget. Gradient accumulation simulates a larger batch when the micro-batch that fits in memory is small. With data parallelism, the effective batch is the product of micro-batch, sequence length, accumulation steps, and replicas. If that product changes, the learning-rate schedule and warmup should be reconsidered rather than inherited blindly.

A useful diagnostic is the tokens-per-update curve together with loss. If a run changes sequence length, packing efficiency, data-parallel size, or accumulation steps after resumption, the apparent update count no longer corresponds to the same number of tokens. For large runs, logs should use token count as the primary training-time axis and optimizer step as a secondary axis.

6.5 Gradient Clipping and Stability

Large pretraining runs fail in recognizable ways: loss spikes, NaNs, diverging gradient norms, stuck data loaders, corrupted checkpoints, and sudden throughput drops. Gradient clipping is a simple guardrail. Global norm clipping rescales gradients when

$$\|g\|_2 = \sqrt{\sum_i \|g_i\|_2^2} \quad (6.8)$$

exceeds a threshold c , replacing g with $g \cdot c / (\|g\|_2 + \epsilon)$. Clipping does not fix a bad schedule or bad data, but it can keep a rare large batch from destroying the optimizer state. The clipped fraction should be logged; constant clipping means the threshold or learning rate is probably wrong, while rare clipping during outlier batches may be acceptable.

Stability also depends on architecture. Pre-normalization improves gradient flow in deep decoders. RMSNorm, residual scaling, careful initialization, and gated feed-forward layers change activation distributions. Long context increases attention score ranges and KV-cache sizes. Mixture-of-experts models add router logits and load-balancing losses, so a stable dense recipe may not transfer directly to a sparse model [20]. Optimizer settings should therefore be tied to the architecture version and not recorded as generic “LLM defaults.”

Mixed precision is a second source of instability. FP16 has limited exponent range, so loss scaling became a standard technique for avoiding gradient underflow [72]. BF16 has fewer mantissa bits but a wider exponent range, which makes it attractive for large Transformer training because it often avoids explicit loss scaling. FP8 and other lower-precision formats can improve throughput but require careful scaling, accumulation, and validation against higher-precision baselines. A publishable training report states which tensors are stored, multiplied, reduced, and accumulated in which precision.

Loss spikes need triage rather than superstition. The first question is whether the spike is local to one worker, one data shard, one mixture component, or the whole job. The second is whether weights, optimizer moments, or both have been corrupted. If only a bad batch caused a transient spike, skipping or clipping may be enough. If optimizer state contains infinities, resuming from a checkpoint before the spike is safer. If the validation loss changes permanently, the run has entered a different region of parameter space and should be treated as a new branch.

6.6 Checkpointing and Reproducibility

A checkpoint is more than model weights. For reproducible pretraining it must contain model parameters, optimizer state, scheduler state, random-number generator states, data-loader position, tokenizer identity, configuration, and enough sharding metadata to reconstruct the model. AdamW stores two moment tensors per parameter, so optimizer state can be larger than the weights themselves. Distributed systems may save sharded

checkpoints to avoid gathering a full model on one host; the checkpoint format then becomes part of the system contract.

Checkpoint cadence is a cost tradeoff. Frequent checkpoints reduce lost work after failure but consume I/O bandwidth and storage. In a large cluster, writing checkpoints can stall training if all workers flush simultaneously to a shared filesystem. A practical plan staggers writes when possible, keeps a small rolling window of recent recovery checkpoints, and separately exports less frequent archival checkpoints for evaluation. The archival checkpoint should include enough metadata to identify exactly which data and code produced it.

Resumption should be tested before the expensive run. A small dry run should train for a few updates, save, resume, and verify that loss and learning rate continue on the expected trajectory. This catches common errors: forgetting optimizer state, resetting the scheduler, repeating data shards, changing dropout seeds, or loading weights under a slightly different architecture. The dry run is not administrative overhead; it is a systems test of the scientific claim.

6.7 Compute-Optimal Planning

Scaling laws are planning tools, not laws of nature. They are empirical fits over specific architectures, tokenizers, corpora, and training procedures. The useful lesson is not that a single formula determines the next model, but that compute should be allocated deliberately between parameters and tokens. If a model is too large for the available token budget, it may have impressive capacity but underdeveloped weights. If a model is too small for the token budget, extra data may produce diminishing returns because capacity becomes the bottleneck [35].

A planning worksheet usually includes:

1. the target model family and parameter count N ;
2. the token budget D by mixture component;
3. the sequence length and expected packing efficiency;
4. the estimated FLOPs, $6ND$ as a baseline plus attention and overhead adjustments;
5. expected model FLOPs utilization (MFU) or hardware utilization;
6. wall-clock time under realistic failure and checkpoint overhead;
7. validation checkpoints and go/no-go criteria.

The worksheet should be updated with measured throughput from a scaled-down run. Small-batch microbenchmarks often overestimate full-run throughput because they omit data loading, checkpointing, network contention, stragglers, validation, and restarts.

Undertraining is a common interpretation trap. A larger model trained on too few tokens can beat a smaller model on some few-shot benchmarks while having worse cost-performance. Conversely, a smaller model trained longer can be easier to serve and adapt. Model reports such as LLaMA, Llama 2, and Llama 3 should be read with this tradeoff in mind: parameter count alone is a poor proxy for training investment or deployment cost [106, 107, 29].

Table 6.2 First-order compute and memory quantities used in pretraining planning. Constants vary by architecture and implementation, so these formulas are planning approximations.

Quantity	Approximation	Interpretation
Training FLOPs	$6ND$	Dense decoder baseline for forward and backward passes
Token batch	$B_\mu TGP$	Tokens consumed per optimizer update
Optimizer steps	$D/(B_\mu TGP)$	Schedule length should track tokens as well as steps
Parameter memory	$N \times \text{bytes per parameter}$	Weight storage before gradients and optimizer states
Adam state memory	$2N \times \text{state bytes}$	First and second moments dominate training memory
Activation memory	Proportional to layers, B_μ , T , hidden size	Often the reason for checkpointing or smaller micro-batches

6.8 Monitoring, Evaluation, and Stop Rules

Training dashboards should separate optimization health from model quality. Optimization health includes training loss, validation loss by mixture, gradient norm, update norm, learning rate, clipping rate, tokens processed, throughput, memory reserved, data-loader wait time, all-reduce time, checkpoint time, and error counts. Model quality includes held-out tasks, generation samples, code tests, math tests, multilingual probes, safety probes, and contamination-aware benchmark suites. The first set answers “is the run functioning?” The second answers “is the artifact useful?”

Loss prediction can detect problems early. If prior scaling fits or smaller pilot runs predict a smooth loss curve, deviations deserve explanation. A loss above prediction may indicate poor data quality, an overly aggressive learning rate, insufficient warmup, a bug in masking, or lower-than-expected batch quality. A loss below prediction may be good news, but it can also signal validation leakage, duplicate-heavy data, or a tokenizer difference. The direction of surprise is less important than whether the team can explain it.

Stop rules should be written before the run. Examples include a maximum token budget, a target validation loss, a budget cap, or a decision checkpoint where marginal loss improvement no longer justifies cost. Without a stop rule, teams are tempted to continue because the curve still decreases, even when the deployment target would be better served by spending compute on data cleaning, post-training, evaluation, or inference optimization.

6.9 Implementation Notes

The local implementation exercises that motivate this chapter use a progression from a small Transformer training loop to GPT-style and LLaMA-style pretraining scripts. The textbook lesson is not the literal code but the invariants those scripts expose: zero

gradients before backward, scale loss during accumulation, clip after unscaling and before the optimizer step, step the scheduler consistently, save optimizer state, and log tokens per second. In a distributed run, only some workers should perform human-readable logging, but all workers must agree on seeds, data partitions, and checkpoint barriers.

Several bugs are easy to miss in review. If the target tensor is shifted incorrectly, the model may learn to copy the current token rather than predict the next. If padding is not masked, the model can achieve low loss by learning padding statistics. If loss is divided by accumulation steps after the backward pass instead of before, gradient scale changes with accumulation. If validation uses training dropout or an unfrozen data order, the loss becomes noisy for reasons unrelated to generalization. If the final partial accumulation step is mishandled, the last updates of an epoch can have a different effective learning rate.

6.10 Key Terms

- AdamW** Adam with weight decay applied as a separate parameter shrinkage term rather than as an adaptive gradient penalty.
- Gradient accumulation** Multiple forward/backward micro-steps whose gradients are summed before one optimizer step.
- Global norm clipping** Rescaling all gradients together when their combined norm exceeds a threshold.
- MFU** Model FLOPs utilization, the ratio of achieved model-relevant FLOPs to hardware theoretical peak.
- Packed sequence** A fixed-length token block formed by concatenating or slicing documents for efficient training.
- Warmup** An initial schedule phase that increases learning rate gradually while optimizer moments and activations settle.

6.11 Exercises

1. A model has $N = 7$ billion parameters and is trained on $D = 1.4$ trillion tokens. Estimate dense training FLOPs with the $6ND$ rule. If the cluster sustains 1.2×10^{18} FLOPs/s on model work, estimate ideal wall-clock days before checkpointing and failures.
2. Suppose $B_\mu = 4$, $T = 4096$, $G = 16$, and $P = 128$. Compute tokens per optimizer update and the number of optimizer updates for 300 billion tokens. Explain how the answer changes if the context length doubles while the token budget stays fixed.
3. Write pseudocode for one accumulation cycle using mixed precision, loss scaling if needed, global norm clipping, AdamW, and a cosine schedule. State exactly where gradients are divided by the accumulation factor.

4. Design a validation dashboard with at least six mixture-specific losses. For each loss, name one data or optimization failure it could reveal before global validation loss does.
5. A run resumes from checkpoint and immediately reports a lower training loss but a higher validation loss. List four possible causes and the evidence you would inspect for each.
6. Compare two plans with the same compute budget: a larger model on fewer tokens and a smaller model on more tokens. Describe one deployment scenario where each plan could be the better choice.

Chapter 7

Distributed Training Systems

Abstract This chapter explains the systems layer that makes large-scale training feasible. It develops memory accounting for parameters, gradients, optimizer states, activations, and communication buffers; then covers data parallelism, fully sharded training, ZeRO-style optimizer partitioning, tensor parallelism, pipeline parallelism, expert parallelism, precision, collectives, throughput accounting, checkpointing, and operational reliability.

7.1 Why Training Systems Are Part of the Model

Large-scale pretraining is not obtained by placing a textbook optimizer inside a larger loop. The distributed system changes what batch size is affordable, how sequence length is chosen, which precision is safe, how often checkpoints can be written, and how quickly failures can be recovered. A model recipe that is mathematically clear but cannot fit in memory or communicate efficiently is not a runnable recipe.

The central design problem is to map a computation graph onto accelerators with limited memory and finite interconnect bandwidth. If the model fits on one GPU but the desired batch does not, data parallelism may be enough. If optimizer state is too large, sharding is needed. If a single layer's matrix multiplications do not fit or do not run efficiently, tensor parallelism becomes relevant. If depth or activation memory dominates, pipeline parallelism or activation checkpointing may be the right lever. If the model is sparse, expert parallelism adds a routing and load-balancing problem. Modern frontier training commonly composes several of these strategies rather than choosing only one [77, 20].

The discipline of this chapter is accounting. Each strategy moves bytes, computation, or idle time from one place to another. It is not enough to say that a method “saves memory.” A useful systems design says which bytes are saved, which collectives are added, how overlap is achieved, what happens during checkpointing, and which failure mode becomes more likely.

Table 7.1 Major memory categories during dense large language model training.

Category	Scaling Driver	Typical Mitigation
Parameters	N and parameter precision	Tensor parallelism, FSDP, ZeRO-3, CPU or NVMe offload
Gradients	N and gradient precision	ZeRO-2/3, reduce-scatter, gradient accumulation
Optimizer states	$2N$ or more for Adam-style methods	ZeRO optimizer partitioning, lower-precision states, optimizer choice
Activations	$LB_\mu TH$ plus attention intermediates	Smaller micro-batch, activation checkpointing, sequence parallelism
Communication buffers	Collective size and overlap strategy	Bucket sizing, topology-aware process groups, overlap tuning
Temporary workspaces	Kernel and allocator behavior	Memory headroom, fused kernels, static shapes where possible

7.2 Memory Accounting

For dense training with AdamW, the persistent state per parameter usually includes the parameter value, a gradient, and two optimizer moments. If parameters and gradients are stored in BF16 and Adam moments are stored in FP32, a rough per-parameter budget is

$$2 \text{ bytes} + 2 \text{ bytes} + 4 \text{ bytes} + 4 \text{ bytes} = 12 \text{ bytes}. \quad (7.1)$$

Some implementations also keep FP32 master weights, which raises the total. A 70-billion-parameter dense model can therefore require hundreds of gigabytes before activations, temporary buffers, CUDA allocator fragmentation, or communication workspaces are counted. This is why single-device reasoning about parameter count is misleading.

Activation memory is often the limiting term for long context or large micro-batches. During backpropagation, the system needs enough intermediate values to compute gradients. For a decoder with L layers, hidden size H , micro-batch B_μ , and sequence length T , residual activations alone scale like $O(LB_\mu TH)$, while attention internals can add terms that depend on heads and sequence length. Memory-efficient attention kernels reduce some temporary storage, but they do not remove the need to retain or recompute the values required for backward.

Temporary memory is the least visible category and one of the most common sources of out-of-memory errors. Matrix multiplication libraries allocate workspaces; attention kernels may allocate scratch buffers; distributed collectives may require staging buffers; model code may create transient copies through layout conversions; and memory fragmentation can prevent a large allocation even when total free memory looks sufficient. Production training systems leave headroom rather than filling the device to the last megabyte.

7.3 Data Parallelism

Data parallelism replicates the model on each worker, gives each replica a different micro-batch, and averages gradients before the optimizer step. In synchronous data parallel training, the mathematical update is close to a larger batch on one device, assuming identical numerics and deterministic reduction. The main collective is an all-reduce over gradients:

$$g = \frac{1}{P} \sum_{r=1}^P g^{(r)}, \quad (7.2)$$

where P is the data-parallel world size. Distributed data parallel implementations usually bucket gradients so communication for early layers can overlap with backward computation for later layers.

The strength of data parallelism is simplicity. It preserves the local model code and increases token throughput as long as communication is not dominant. The weakness is memory replication: every worker stores all parameters, gradients, and optimizer states. For small and medium models this is acceptable. For very large models, data parallelism alone runs out of memory before it runs out of compute.

Large data-parallel worlds also change optimization. If the global token batch grows with the number of workers, the number of optimizer updates for a fixed token budget falls. Learning-rate schedules, warmup, and gradient noise properties must be retuned. Scaling out a run is therefore both a systems change and an optimization change.

7.4 ZeRO and Fully Sharded Data Parallel

ZeRO-style training partitions redundant data-parallel state across workers [92]. The usual stages are easiest to remember by what they shard:

- Stage 1 Shard optimizer states.
- Stage 2 Shard optimizer states and gradients.
- Stage 3 Shard optimizer states, gradients, and parameters.

The memory savings can be dramatic because Adam moments and gradients no longer exist in full on every data-parallel worker. The cost is communication: workers must gather parameter shards before computation, reduce or reduce-scatter gradients after backward, and coordinate optimizer updates.

Fully Sharded Data Parallel (FSDP) is a widely used PyTorch implementation of this idea, especially close to ZeRO-3 in spirit [125]. A module wrapped with FSDP stores only local parameter shards most of the time. Before the module's forward or backward computation, workers all-gather the full parameters needed for that module; after use, full parameters can be freed. Gradients are reduce-scattered so each rank keeps only its shard. Nested wrapping controls the granularity of gathering and freeing.

The important design question is granularity. If wrapped units are too large, peak memory rises because full parameters for a large region are materialized at once. If units

are too small, collective overhead and scheduling overhead increase. Transformer block wrapping is a common compromise. Auto-wrapping policies should be inspected, not trusted blindly, because shared embeddings, tied output heads, adapters, and mixture-of-experts layers can violate simple assumptions.

FSDP also affects checkpointing. A full state dict is convenient for evaluation and sharing but expensive to materialize. A sharded state dict is efficient for resuming training but tied to metadata about world size, parameter flattening, and sharding layout. A robust training pipeline tests both resumption and export. It should be possible to recover the job after a node failure and separately export a canonical checkpoint for inference or post-training.

7.5 Tensor Parallelism

Tensor parallelism splits individual layer computations across devices. Megatron-LM popularized practical tensor parallel layouts for Transformer language models [101]. For a linear layer $Y = XW$, column parallelism splits W by output columns, so each device computes a shard of Y . Row parallelism splits W by input rows, so devices compute partial outputs that must be summed. MLPs and attention projections can be arranged so some collectives are delayed or paired with the next operation.

The advantage is that very large layers can run when a single device cannot store or multiply the full matrices efficiently. Tensor parallelism also increases arithmetic capacity for one model replica. The cost is frequent communication inside the forward and backward pass. These collectives are latency-sensitive and topology-sensitive, so tensor-parallel groups are usually kept within a high-bandwidth island such as GPUs connected by NVLink or within one server. Spanning tensor parallelism across slower inter-node links can destroy utilization.

Attention introduces additional choices. Query, key, and value projections can be partitioned by head because different heads are mostly independent until output projection. Sequence parallelism splits some activation tensors along the sequence dimension to reduce replicated activation memory, but it adds collectives around operations that need full sequence or hidden dimensions. The implementation must align partitioning with kernel expectations; otherwise layout conversions can erase the intended savings.

7.6 Pipeline Parallelism

Pipeline parallelism partitions layers into stages. Stage 0 processes early layers, sends activations to stage 1, and so on. Backward sends gradients in the reverse direction. GPipe introduced a clean micro-batch pipeline approach for training very deep networks [39]. In language-model training, pipeline parallelism is often combined with tensor and data parallelism for large models [77].

The basic problem is the pipeline bubble. If there are S stages and only one micro-batch, most stages are idle most of the time. Splitting a batch into M micro-batches fills

the pipeline:

$$\text{bubble fraction} \approx \frac{S - 1}{M + S - 1} \quad (7.3)$$

for a simple schedule. More micro-batches reduce idle time but increase activation bookkeeping and may change effective batch-size constraints. Interleaved schedules assign multiple stage chunks to a device to improve load balance, but they increase scheduling complexity.

Stage balance is critical. A pipeline runs at the speed of its slowest stage. Embedding layers, output projections, attention-heavy long-context layers, and MoE layers can make equal layer counts unequal in runtime. A good partitioner measures real execution time and memory rather than assuming identical blocks. It also accounts for activation transfer sizes between stages.

Pipeline parallelism changes failure recovery and debugging. A NaN generated in one stage may only be observed downstream. A shape bug may depend on a boundary between stages. Logging must include stage rank, micro-batch id, and schedule step. Checkpoints must reconstruct partitioned modules correctly, especially if the model is later exported without pipeline partitioning.

7.7 Expert Parallelism

Mixture-of-experts models replace some dense feed-forward computation with a set of experts and a router. Each token is sent to one or more experts, so total parameters can be much larger than active parameters per token. Expert parallelism places different experts on different devices and uses all-to-all communication to move token representations to the devices that own the selected experts. GShard and later MoE systems made this style of conditional computation central to large sparse models [50, 20].

The systems challenge is load balance. If the router sends too many tokens to one expert, that expert becomes a straggler and may overflow its capacity. Load-balancing losses, capacity factors, token dropping policies, and auxiliary routing constraints are therefore part of the training system. They affect both optimization and throughput. A sparse model can have excellent nominal FLOPs per token and still run poorly if routing creates uneven all-to-all traffic.

Expert parallelism composes awkwardly with other strategies. Tensor-parallel experts require communication inside expert computation; data-parallel replicas need consistent router gradients; pipeline stages containing MoE layers may have variable runtime; sharded checkpointing must preserve expert placement and optimizer state. This is a case where the architecture and system are inseparable.

7.8 Activation Checkpointing and Recomputation

Activation checkpointing trades compute for memory. Instead of storing every intermediate activation needed for backward, the system stores selected checkpoints and recomputes missing intermediates during the backward pass. The general technique predates modern large language models and can reduce activation memory substantially at the cost of extra forward computation [12]. In large Transformers, selective recomputation is often applied at the block level or to memory-heavy subcomponents.

The tradeoff is not simply “30 percent more compute for less memory.” Recomputation interacts with pipeline schedules, tensor parallel collectives, dropout determinism, attention kernels, and compiler graph capture. If a recomputed segment includes random operations, random-number states must be replayed correctly. If the segment includes communication, recomputation may add network traffic as well as arithmetic. If the bottleneck is memory bandwidth rather than compute, recomputation can have a different cost than expected.

The practical question is which activations are worth storing. Small cheap activations can be recomputed. Expensive or communication-heavy intermediates may be worth keeping. A measured memory profile is better than a uniform policy. Modern training stacks often expose several checkpointing modes; their names are less important than the actual saved tensors and recomputed graph.

7.9 Precision and Communication

Distributed training depends on both numerical precision and communication precision. FP32 remains common for optimizer moments and some reductions. BF16 is widely used for activations and weights because its exponent range is forgiving. FP16 can be efficient but often requires loss scaling. FP8 and related formats can improve throughput and reduce communication volume, but only when scaling, accumulation, and kernel support are mature enough for the architecture and hardware [72, 20].

Collectives are the language of distributed training:

All-reduce Every rank contributes a tensor and receives the reduced result. Used in classical data-parallel gradient averaging.

Reduce-scatter The reduction result is partitioned across ranks. Used in sharded gradients.

All-gather Shards are gathered so every rank receives the full tensor. Used before FSDP module computation.

All-to-all Each rank sends different slices to different ranks. Used heavily in expert parallelism.

Broadcast One rank sends the same tensor to others. Used for initialization and some checkpoint flows.

The same collective can have different cost depending on message size, topology, contention, and overlap. For example, a small all-reduce may be latency-dominated, while a large reduce-scatter may be bandwidth-dominated.

Overlap is the main performance technique. During backward, gradients for later layers become available before earlier layers finish. A system can launch communication for ready gradient buckets while computation continues. FSDP can prefetch upcoming parameter shards. Pipeline parallelism can overlap different micro-batches. Overlap is fragile: a bucket that is too large starts late; a bucket that is too small has high launch overhead; a dependency inserted by logging, synchronization, or shape conversion can serialize the step.

7.10 Throughput Accounting

Training performance should be reported in tokens per second, time per optimizer step, and model FLOPs utilization. Tokens per second is directly tied to the training budget. Step time is useful for debugging schedules and checkpoint cadence. MFU estimates how much of the hardware’s theoretical compute is used for model-relevant operations:

$$\text{MFU} = \frac{\text{estimated model FLOPs per step}}{\text{step time} \times \text{peak hardware FLOPs}}. \quad (7.4)$$

The numerator is usually approximate, so MFU should be treated as an engineering diagnostic rather than a precise physical measurement.

End-to-end throughput includes data loading, host-to-device transfer, forward, backward, optimizer step, communication, validation, checkpointing, and restarts. A microbenchmark that measures only matrix multiplications is useful for kernel development but not for planning a training run. Conversely, a low end-to-end number does not identify the bottleneck. Profiling should separate data time, compute time, collective time, optimizer time, and I/O time.

Stragglers matter because synchronous training waits. A slow data-loader worker, a thermal-throttled GPU, a congested network path, or a filesystem pause can reduce the throughput of the whole job. Logs should make per-rank timing visible. If only the master rank logs averaged step time, the slow rank can remain hidden until utilization is already poor.

7.11 Operational Reliability

Large training jobs fail often enough that failure recovery is part of the design. A reliable run has a launch manifest, immutable code reference, environment capture, data manifest, checkpoint policy, health checks, and runbook. The runbook should say what to do for a single-rank failure, repeated out-of-memory errors, NaNs, slow filesystem writes, degraded network links, and validation regressions. Waiting until failure occurs is expensive because hundreds or thousands of accelerators may sit idle while the team debates whether a checkpoint is trustworthy.

Table 7.2 Parallelism strategies and their primary tradeoffs.

Strategy	Main Saving	Main Cost	Best First Use
Data parallel	More token throughput	Replicated state, gradient reduce	Model fits per device, all-batch should scale
ZeRO/FSDP	Shards optimizer, gradients, parameters	All-gather and reduce-scatter, checkpoint complexity	Model state exceeds memory
Tensor parallel	Splits large layer compute/state	Frequent intra-layer collectives	Layers too large or too slow on one GPU
Pipeline parallel	Splits depth and activation footprint	Pipeline bubbles, stage balance	Deep model cannot fit as one partition
Expert parallel	Splits sparse experts	All-to-all, load balance	MoE model with many experts
Activation checkpointing	Reduces saved activations	Recomputation determinism constraints	Activation and memory dominates

Checkpointing must match the parallel layout. Data-parallel checkpoints can be simple full replicas. FSDP and ZeRO checkpoints may be sharded. Tensor-parallel checkpoints must preserve partitioned weight layouts. Pipeline-parallel checkpoints must map layers to stages. Expert-parallel checkpoints must preserve expert identities and optimizer shards. When changing world size for resumption, the system may need a conversion step. That conversion should be tested with small models before it is needed in an emergency.

Data-loader reliability is a frequent bottleneck. Tokenized shards should include checksums, lengths, and format versions. The sampler should be able to resume at a token-accurate or shard-accurate position. If workers independently shuffle without a coordinated seed and epoch definition, resumption may repeat or skip data. If remote storage is used, caching policy and backpressure should be measured. A model can only train as fast as tokens arrive.

Experiment tracking should separate high-rate metrics from durable artifacts. High-rate metrics include step time, loss, gradient norm, and memory. Durable artifacts include configs, data manifests, evaluation reports, checkpoints, and environment descriptions. The first group supports live debugging; the second supports scientific interpretation after the run.

7.12 Implementation Notes

A minimal distributed training script hides many contracts. Environment variables define rank, local rank, world size, master address, and communication backend. The data sampler must partition examples so that ranks do not duplicate work unless duplication is intentional. The loss must be scaled correctly under gradient accumulation. Random seeds must differ where stochastic data order should differ and match where initializa-

tion should match. Only designated ranks should write shared artifacts, but all ranks must participate in barriers required by the checkpoint protocol.

Several checks catch serious bugs early:

1. Run one step on one GPU and on multiple GPUs with the same global batch, then compare losses within expected numerical tolerance.
2. Verify that every trainable parameter receives a finite gradient.
3. Measure peak memory by category before increasing model size.
4. Save, resume, and confirm that learning rate, token count, data position, and loss continue as expected.
5. Profile at the intended sequence length, not only at a shorter debug length.
6. Kill one worker in a controlled test if the scheduler and checkpoint system claim fault tolerance.

These tests are cheaper than discovering the same problems during a full run.

7.13 Key Terms

All-gather A collective in which ranks exchange shards so every rank obtains the full tensor.

All-reduce A collective in which ranks receive the reduction of all contributed tensors.

Data parallelism Replicating the model across workers and averaging gradients across different data shards.

Expert parallelism Placing different MoE experts on different devices and routing token representations to them.

FSDP Fully Sharded Data Parallel, a sharded training method that keeps only local shards of model state when full parameters are not needed.

Pipeline bubble Idle time caused by filling and draining pipeline-parallel stages.

Tensor parallelism Splitting individual layer tensors and computations across devices.

7.14 Exercises

1. Estimate persistent training memory for a 13-billion-parameter dense model using BF16 parameters, BF16 gradients, FP32 Adam moments, and FP32 master weights. Then estimate per-rank persistent memory under ZeRO-1, ZeRO-2, and ZeRO-3 for $P = 16$, ignoring activations and buffers.
2. A model trains with 8 pipeline stages and 32 micro-batches. Estimate the simple pipeline bubble fraction. How does the fraction change with 8 micro-batches, and what other costs might increase when using 32?
3. For an MLP block with two linear layers, describe one column-parallel and one row-parallel layout. State where all-reduce or all-gather operations are required.

4. Design a process-group layout for 128 GPUs arranged as 16 nodes with 8 GPUs per node. Choose data, tensor, and pipeline parallel degrees for a dense model and justify which collectives stay within a node.
5. A training job has stable loss but tokens per second drops by 25 percent after two hours. List at least six metrics or logs you would inspect to distinguish data loading, network, checkpointing, thermal, and straggler causes.
6. Explain why a sharded checkpoint that resumes training correctly may still be unsuitable for inference export. What metadata or conversion step is needed?

Chapter 8

Inference and Serving

Abstract This chapter studies large language model inference as a systems problem. It separates prefill from decode, derives KV-cache memory costs, explains batching and paged attention, compares quantization and speculative decoding, and gives practical guidance for streaming APIs, cancellation, load testing, cost accounting, and evaluation under real traffic.

8.1 Serving Is Not Evaluation Mode

Inference is often introduced as “run the trained model with dropout disabled.” That description is correct for a single forward pass and inadequate for a user-facing service. An large language model service must accept variable-length prompts, schedule many concurrent requests, manage KV-cache memory, stream tokens, cancel work when clients disconnect, enforce timeouts, protect shared capacity, and report cost. The same model can feel fast or unusable depending on the serving stack.

Autoregressive generation is the reason. A prompt of length T_p is processed once, but a completion of length T_g is generated one token at a time:

$$p_{\theta}(y_{1:T_g} \mid x_{1:T_p}) = \prod_{t=1}^{T_g} p_{\theta}(y_t \mid x_{1:T_p}, y_{<t}). \quad (8.1)$$

Each generated token depends on all previous tokens, so decode has a serial dependency even when each individual forward pass uses parallel hardware. Serving systems therefore optimize two different phases: prefill, which processes the prompt, and decode, which extends the sequence.

8.2 Prefill and Decode

Prefill computes hidden states for the prompt and writes keys and values for every layer into the KV cache. It resembles training inference over a full sequence: matrix multiplications are large, attention can be compute-heavy, and the system benefits from efficient attention kernels such as FlashAttention [17]. For long prompts, prefill latency can dominate time to first token. Prompt caching and prefix reuse can help when many requests share the same system prompt, retrieved context, or conversation prefix, but the cache must be keyed by exact tokens and model state.

Decode is different. At each step, the model receives the newest token, reads all previous keys and values from the KV cache, computes attention for the new position, and writes one new key/value pair per layer. The matrix multiplications are smaller, and memory bandwidth often becomes the bottleneck because the service repeatedly reads model weights and cache entries. This is why tokens per second during decode depends strongly on batch size, KV-cache layout, quantization, and attention implementation.

The distinction explains a common measurement error. Reporting only generated tokens per second hides prompt cost. Reporting only end-to-end latency hides decode throughput. A useful service dashboard separates:

Time to first token	Queueing plus prefill plus the first decode step.
Inter-token latency	The gap between streamed tokens during decode.
End-to-end latency	Time from request arrival to final token.
Input throughput	Prompt tokens processed per second.
Output throughput	Generated tokens processed per second.

These metrics move differently under load. A scheduler can improve output throughput by batching decode requests while increasing time to first token for short prompts.

8.3 The KV Cache

The KV cache stores attention keys and values for previously processed tokens. For a dense decoder with L layers, H_{kv} key/value heads, head dimension d_h , sequence length T , batch size B , and s bytes per cache element, the approximate cache memory is

$$M_{KV} = 2 B T L H_{kv} d_h s. \quad (8.2)$$

The factor of two counts keys and values. For multi-head attention, H_{kv} equals the number of query heads. For multi-query attention, it may be one. For grouped-query attention, it is between those extremes [98, 1]. This architectural choice directly affects serving capacity.

KV memory grows with active sequence length, not just model size. A smaller model with a very long context window can be harder to serve than a larger model at short context. Multi-turn chat sessions worsen the issue because every generated token extends the cache. Retrieval-augmented systems add retrieved passages to the prompt and therefore consume cache before any answer token is produced. A serving plan

should budget cache memory for realistic prompt and completion length distributions, not only for a maximum context headline.

Cache precision is a systems choice. BF16 or FP16 caches preserve accuracy but consume memory. Quantized KV caches can increase batch capacity and long-context affordability, but they may affect perplexity, retrieval sensitivity, or exact copying. Any KV-cache quantization should be evaluated on tasks that depend on long-range information, not only on short generic benchmarks.

8.4 Batching and Scheduling

Batching improves hardware utilization by combining requests. In prefill, batching prompts of similar length avoids padding waste. In decode, batching many active sequences lets the service perform one larger step instead of many tiny steps. The complication is that requests arrive and finish at different times. Static batching waits to form a batch and then runs it to completion; this wastes capacity when sequences have different output lengths. Continuous batching admits new requests as old ones finish, keeping the decode batch full.

Continuous batching turns serving into a scheduling problem. Each request has prompt length, generated length so far, maximum token budget, priority, deadline, and memory footprint. The scheduler chooses which requests enter prefill, which active sequences decode next, and when to evict or pause work. Policies often balance throughput against latency. A throughput-optimized policy may batch aggressively and hurt short interactive requests. A latency-optimized policy may keep batches small and waste hardware.

PagedAttention, introduced in vLLM, addresses memory management by storing KV cache in fixed-size blocks rather than requiring each sequence to occupy a contiguous allocation [46]. The analogy to virtual memory is useful: logical token positions map to physical cache blocks, and blocks can be shared, allocated, or freed as sequences grow and finish. This reduces fragmentation and makes continuous batching more effective. It does not remove the KV-cache cost; it makes the cost manageable under variable-length traffic.

8.5 Memory Management and Admission Control

An inference server needs an admission policy. If it accepts more active tokens than the hardware can support, it will either run out of memory or degrade latency for every request. A simple policy estimates each request’s maximum cache reservation from prompt length plus maximum new tokens. A more flexible policy admits based on expected length and enforces a hard cap during generation. The latter improves utilization but needs cancellation and partial-progress handling when a request exceeds budget.

Table 8.1 Serving bottlenecks differ across request shapes.

Traffic Pattern	Likely Bottleneck	Useful Metric
Short prompts, short outputs	Queueing and launch overhead	Requests/s, p95 end-to-end latency
Long prompts, short outputs	Prefill compute and attention memory	Time to first token, input tokens/s
Short prompts, long outputs	Decode bandwidth and batch occupancy	Inter-token latency, output tokens/s
Long conversations	KV-cache capacity	Active tokens, cache hit/eviction rates
Shared system prompts	Prefix-cache management	Cache hit rate, first-token latency by prefix
Mixed priorities	Scheduler policy	Latency by class, deadline miss rate

Memory fragmentation is not a theoretical concern. Without paging or careful pooling, variable-length sequences allocate and free differently sized cache regions, leaving holes that cannot satisfy later long requests. Fragmentation can produce out-of-memory errors even when total free memory appears large. Fixed block allocation, request compaction, and preallocated cache arenas are common mitigations.

Eviction policy depends on product semantics. For stateless completion APIs, the cache can be freed when the response ends. For chat sessions, a server may keep conversation prefixes warm for a short time. For retrieval systems, retrieved documents may change between turns, so prefix reuse must respect the exact token sequence and tool outputs. For privacy-sensitive deployments, cache lifetime and tenant isolation are security concerns as well as performance concerns.

8.6 Quantization and Compression

Inference cost is often dominated by memory bandwidth and capacity. Quantization reduces the bytes moved for weights, activations, or KV cache. Weight-only quantization stores weights in low precision while computing with higher-precision activations. This can reduce model memory and improve decode throughput, especially when batch sizes are small and weight loading dominates. GPTQ and AWQ are influential post-training approaches for low-bit LLM weight quantization [25, 61]. SmoothQuant targets weight-and-activation INT8 quantization by moving quantization difficulty from activations to weights through offline scaling [114].

Quantization is not a single knob. Important choices include bit width, per-tensor versus per-channel scaling, group size, calibration data, outlier handling, kernel support, and whether the target hardware accelerates the chosen format. A 4-bit checkpoint without efficient kernels may save memory but fail to improve latency. A calibration set that omits code, math, multilingual text, or long-context examples may hide severe regressions.

Evaluation should include both quality and systems metrics. Quality metrics include held-out perplexity, task accuracy, exact-match behavior for copying, code execution,

long-context retrieval, and safety probes. Systems metrics include model load time, peak memory, time to first token, decode tokens per second, batch capacity, and cost per million tokens. Quantization can improve one metric while hurting another; a publishable claim states the traffic shape and hardware.

8.7 Speculative and Parallel Decoding

The serial dependency of autoregressive decoding makes acceleration difficult. Speculative decoding uses a smaller or cheaper draft model to propose several tokens, then verifies those tokens with the target model in a more parallel computation. The exact algorithm can preserve the target model's output distribution while reducing latency when the draft model is accurate enough and cheap enough [52]. The speedup is not automatic: if draft tokens are frequently rejected, the service pays for both draft and target work.

Speculative decoding introduces new serving decisions. The draft model must be deployed, scheduled, and versioned with the target. The acceptance rate should be monitored by prompt class, language, temperature, and domain. Greedy decoding, nucleus sampling, and temperature sampling have different verification details. A model that is excellent for English chat may be a poor drafter for code or low-resource languages. The operational metric is not only accepted tokens per target pass, but end-to-end latency after adding draft-model overhead.

Other parallel decoding methods add auxiliary heads or predict multiple future tokens directly. These approaches can reduce the number of target passes but may require model modification or extra training. They should be evaluated under the same rule as speculative decoding: measure quality-preserving speedup on the actual request distribution, including queueing and memory pressure.

8.8 Attention Kernels and Long Context

Attention kernels matter in prefill and long-context serving. Standard attention materializes or implicitly handles an attention matrix whose logical size grows with T^2 . FlashAttention improves memory efficiency by tiling computation and avoiding unnecessary reads and writes to high-bandwidth memory while computing exact attention [17]. This is especially important for long prompts, training, and batched prefill. During single-token decode, the attention pattern is a query attending to cached keys and values, so the bottleneck often shifts toward cache bandwidth and layout.

Long context should be evaluated with length-aware tests. A model can accept a long prompt syntactically yet fail to use evidence in the middle. A serving stack can support a maximum context in isolation yet collapse under many concurrent long-context requests. A benchmark that reports only accuracy at maximum length omits the cost curve. For production planning, the relevant graph plots latency, throughput, memory, and accuracy as functions of input length and active batch size.

Chunked prefill is a practical technique for long prompts. Instead of monopolizing the device with one long prefill, the server splits prompt processing into chunks and interleaves other work. This can reduce tail latency for mixed traffic, but it complicates scheduling and may reduce raw throughput. Prefix caching, retrieval caching, and chunked prefill should be measured together because they interact.

8.9 Streaming APIs, Cancellation, and Safety Filters

Streaming changes user experience and server behavior. A streamed response can show the first token quickly while the full answer is still decoding. The server must flush partial text, handle token-to-text boundaries, and decide when to emit tool-call or structured-output fragments. It must also handle cancellation. If a client disconnects, continuing to decode wastes capacity and may hold KV-cache blocks that other requests need.

Timeout policy should distinguish queue timeout, prefill timeout, decode timeout, and total response timeout. A request that waits too long in the queue has not consumed much GPU work; a request that times out during a long decode may already have consumed substantial compute. Retrying blindly can multiply load during incidents. Backpressure, rate limits, and clear error semantics are part of serving correctness.

Safety filters and output validators add latency. Some systems filter prompts before prefill, monitor generated tokens during streaming, and run final classifiers after completion. Tool-using systems may validate JSON schemas or execute sandboxed checks. These components should be in the latency budget and load test, not treated as free post-processing. If a safety classifier uses the same GPU pool as generation, it competes for capacity.

8.10 Load Testing and Cost Accounting

A load test should model traffic, not merely saturate a server with identical prompts. Real traffic has distributions over prompt length, output length, decoding parameters, tenant priority, cache reuse, and cancellation. Synthetic tests are still useful, but they should state the distribution. A minimal load-test report includes average, p50, p90, p95, and p99 latency; time to first token; inter-token latency; input and output tokens per second; requests per second; error rate; cancellation rate; GPU memory; GPU utilization; and cost per million input and output tokens.

Cost per token is not constant. Long prompts spend more prefill compute. Long outputs spend more serial decode time and KV memory. Small batches may waste hardware. Very large batches may improve throughput while hurting latency. Quantization can reduce cost if kernels are efficient. Speculative decoding can reduce cost only if the draft overhead is smaller than the target passes it avoids. A pricing or capacity model should therefore separate input-token and output-token economics.

Tail latency deserves special attention. Users experience the slow request, not the mean. Tail latency can come from queueing, long prompts, cache fragmentation, strag-

Table 8.2 A compact load-test table for an large language model service.

Scenario	TTFT p95	E2E p95	Output toks/s	Cost / 1M toks
Short chat				
Long prompt QA				
Code generation				
RAG with shared prefix				
Mixed production trace				

gler kernels, network backpressure, safety filters, or retries. Averages hide all of these. A good load test captures traces for the slowest requests and attributes time to queue, prefill, decode, post-processing, and streaming flush.

8.11 Implementation Notes

A minimal generation loop is easy to write and hard to serve. The loop tokenizes a prompt, runs prefill, samples a token, appends it, and repeats until an end condition. A service-grade loop separates request state from model execution. Request state includes token ids, sampling parameters, stop sequences, maximum token budget, KV-block table, stream handle, priority, deadline, and cancellation flag. Model execution operates on batches selected by the scheduler.

Sampling code must be deterministic when determinism is promised. Temperature, top- k , top- p , repetition penalties, logit biases, and random seeds should be applied in a defined order. Stop sequences are text-level or token-level depending on the API; confusing the two can leak partial delimiters or stop too late. Structured-output modes need validators that can operate incrementally or repair invalid partial generations.

The service-grade invariants are stable across implementations: separate prefill and decode metrics, budget KV cache explicitly, test variable-length batching, account for cancellation, and evaluate quality after every compression or decoding optimization.

8.12 Key Terms

Continuous batching A scheduling policy that adds and removes requests from active decode batches as they arrive and finish.

Decode The autoregressive phase that generates one or more new tokens using the KV cache.

KV cache Stored attention keys and values for previous tokens, used to avoid recomputing the full prefix at every decode step.

PagedAttention A KV-cache management method that stores cache entries in fixed-size blocks to reduce fragmentation under variable-length requests.

Prefill The phase that processes prompt tokens and initializes the KV cache before generation.

Speculative decoding A method that proposes tokens with a cheaper draft computation and verifies them with the target model.

Time to first token The latency from request arrival until the first output token is available to stream.

8.13 Exercises

1. Derive the KV-cache memory for a model with $L = 32$, $H_{kv} = 8$, $d_h = 128$, BF16 cache elements, batch size $B = 24$, and active sequence length $T = 4096$. How does the answer change under multi-query attention with $H_{kv} = 1$?
2. Design a scheduler for mixed traffic containing short chat, long-document summarization, and code generation. State the admission rule, batching rule, and cancellation rule.
3. Build a load-test table using Table 8.2. Add p50 and p99 columns, input tokens/s, error rate, and cache hit rate. Explain which metric would first reveal KV-cache fragmentation.
4. A 4-bit weight-only quantized model has lower memory use but worse p95 latency than BF16. List five possible causes and one measurement that would distinguish each cause.
5. Explain speculative decoding with a draft model and a target model. Under what prompt distributions would you expect low acceptance rates?
6. Compare two deployments of the same model: one optimized for maximum throughput and one optimized for interactive latency. Specify batch size policy, timeout policy, and the metric each deployment should optimize.

Part III

Adaptation

Chapter 9

Supervised Instruction Tuning

Abstract This chapter explains how pretrained models become instruction-following models through supervised examples. It covers chat templates, task diversity, Alpaca-style self-instruction, synthetic data, refusal data, data quality, and the limits of imitation.

9.1 The Interface Shift

A pretrained decoder is trained to continue text. A deployed assistant is expected to answer a request, respect a role hierarchy, follow a formatting contract, refuse some requests, use tools when permitted, and stop when the answer is complete. Supervised instruction tuning, usually shortened to SFT, is the first post-training stage that teaches this interface. It does not change the causal language-model factorization from Chapter 1; it changes the distribution of sequences on which the model is optimized.

Let an instruction example contain a context c and a target assistant response a . The context may include a system message, one or more user turns, previous assistant turns, retrieved passages, tool observations, or task-specific fields. A chat template τ turns the structured example into a token sequence

$$z_{1:T} = \tau(c, a), \quad (9.1)$$

where some tokens are input-only and other tokens are response targets. SFT minimizes next-token loss only on the target positions,

$$\mathcal{L}_{\text{SFT}}(\theta) = -\frac{1}{\sum_{t=1}^T m_t} \sum_{t=1}^T m_t \log p_{\theta}(z_t \mid z_{<t}), \quad (9.2)$$

where $m_t = 1$ for response tokens to be imitated and $m_t = 0$ for prompt tokens, padding, and usually separator tokens. In many training libraries this mask is implemented by setting ignored labels to `-100`. The small detail matters: if the loss is charged on user instructions or system text, the model is trained to generate prompts rather than only answers.

Table 9.1 Common SFT data types and the contracts they teach.

Data Type	Contract Being Taught	Typical Failure If Under-Specified
Single-turn instructions	Map one user request to one answer	Verbose, generic, or format-breaking answers
Multi-turn dialogue	Maintain state, resolve references, and repair mistakes	Forgetting earlier constraints or over-trusting stale context
Structured tasks	Emit JSON, SQL, code, labels, or other constrained outputs	Invalid syntax despite semantically plausible text
Tool traces	Decide when to call a tool and how to use observations	Hallucinated tool results or repeated invalid calls
Refusal and safety data	Decline disallowed requests while helping with safe alternatives	Over-refusal, under-refusal, or inconsistent tone
Domain demonstrations	Adopt domain vocabulary and workflow conventions	Shallow style imitation without factual grounding

Instruction tuning became central because large base models already possess many latent capabilities, but they do not reliably infer the desired user-facing protocol from a raw prompt. FLAN-style work showed that fine-tuning on tasks expressed as natural-language instructions improves zero-shot generalization to unseen task types [111]. InstructGPT used supervised demonstrations as the first stage before reward modeling and RLHF, making clear that SFT is useful but not a complete alignment method [84]. Modern open model reports treat SFT as one component of a larger post-training pipeline with preference data, safety data, red-teaming, and deployment evaluation [107, 29].

9.2 Instruction Data as a Contract

An instruction dataset is not merely a list of prompts and answers. It is a behavioral contract for the model. A good record states who is speaking, what authority each message has, which external evidence is available, what output format is required, and what should happen when the request is impossible or unsafe. A weak record leaves these details implicit, so the model learns correlations that are convenient in the dataset but fragile in deployment.

Table 9.1 summarizes common SFT example types. A single training run often mixes them, but the mixture should be explicit. The model should not learn that every user request deserves a long explanation, that every code question requires a full file rewrite, or that every safety-sensitive request is either always allowed or always refused. Diversity is useful only when it is organized.

The template is part of the dataset. For a chat model, the same semantic example can train different behavior depending on whether the text is formatted as `system/user/assistant` messages, a plain prompt-completion pair, or a vendor-specific instruction format. The template should make message boundaries unambiguous, include an end-of-turn token or stop condition, and match the inference template

used in serving. A mismatch between training and inference templates is one of the easiest ways to make a capable checkpoint appear broken.

Data quality must be evaluated at several levels. At the example level, the answer should be correct, complete, and consistent with the instruction. At the dataset level, tasks should be balanced across skills, lengths, languages, domains, and refusal categories. At the provenance level, the builder must know which sources can be re-distributed, which sources may contain personally identifiable information, and which benchmarks must be excluded from training. Benchmark contamination is especially serious in instruction tuning because many public evaluation tasks are also phrased as instructions [100, 115].

9.3 Synthetic Instruction Data

Human-written demonstrations are expensive. Synthetic instruction generation tries to expand coverage by asking a capable model to create new tasks, inputs, and answers. Self-Instruct is the canonical recipe: seed the process with a small set of human-written instructions, prompt a model to generate new instructions and demonstrations, filter malformed or near-duplicate examples, and then fine-tune on the resulting set [110]. Alpaca-style recipes popularized the same idea for open models by distilling instruction-following behavior from stronger teacher models.

Synthetic data is attractive because it can cover many formats quickly. It can create paraphrases, edge cases, schema-specific examples, tool traces, or domain-style dialogues long before a human annotation program has reached the same breadth. It is also dangerous because errors scale with the generator. A teacher that fabricates facts, mishandles code, writes insecure SQL, or refuses too aggressively can stamp those mistakes into thousands of training examples. Filtering must therefore test both surface diversity and semantic validity.

Modern synthetic-data pipelines are more than prompt expansion. Evol-Instruct-style methods deliberately rewrite seed tasks into harder variants [116]; Orca-style distillation trains smaller models from richer teacher traces rather than only final answers [76]; STaR-style bootstrapping keeps generated rationales that lead to correct answers and repeats the loop [123]. Rejection sampling is the common control mechanism: sample many candidates, score them with unit tests, exact answers, retrieval support, reward models, or human audits, then train only on accepted examples. This turns data generation into a search problem, and it should be logged with the same care as model training: generator version, prompt template, sampling parameters, acceptance rules, rejection reasons, and overlap checks against benchmarks.

Distillation also changes what is being learned. A student can imitate the teacher’s surface style without acquiring the teacher’s latent competence; it can inherit the teacher’s safety boundary; or it can become brittle because the accepted data contains only easy successes. For reasoning and tool use, the strongest synthetic records usually include the problem, context, tool observations or verifier results, the final answer, and a compact explanation of why the answer is valid. For safety-sensitive domains, teacher-generated

examples require additional review because fluency can hide unsupported medical, legal, financial, or security claims.

A practical synthetic-data pipeline usually has four gates. First, generation prompts should specify the task family, difficulty, allowed sources, answer format, and exclusion rules. Second, automatic filters should remove examples that are too short, too similar, malformed, unsupported by evidence, or outside policy. Third, task-specific validators should execute or check outputs when possible: parse JSON, run unit tests, execute SQL against a sandbox database, or compare against a known answer. Fourth, a sampled human audit should estimate residual error rates and update the generator prompts. The goal is not to make synthetic data look diverse; it is to make it add reliable coverage that the current model lacks.

Synthetic data also changes the ethics and licensing story. A generated answer can still be a derivative of the teacher model’s training distribution, a reproduction of memorized text, or an unsafe medical or legal recommendation. The dataset builder must document the teacher model, prompts, filters, and known failure modes. If those details are absent, downstream users cannot interpret the resulting checkpoint.

9.4 Training Details

The training loop for SFT looks simple: tokenize examples, form batches, run the causal model, compute masked cross-entropy, and update parameters. Most failures come from the details around that loop.

First, label masks should align with message boundaries. In a single-turn example, user tokens normally condition the answer but do not contribute to the loss. In a multi-turn example, previous assistant turns may be treated either as context only or as additional target spans. Both choices are defensible, but they train different behavior. If previous assistant turns are targets, the model receives more supervised tokens per dialogue; if they are context only, the objective focuses on the final answer. The choice should be recorded.

Second, sequence packing improves accelerator utilization but can silently corrupt examples. Packing concatenates multiple short conversations into one fixed-length sequence. The attention mask and position handling must prevent a response in one packed example from attending to the user request in the next example unless the packing format deliberately inserts a boundary that the model can learn to ignore. If the implementation uses ordinary causal attention across packed examples, it may train on artificial cross-example context.

Third, length control is part of the objective. Truncating from the right may remove the target answer; truncating from the left may remove the instruction; truncating retrieved evidence may make a correct answer appear unsupported. A robust preprocessing job reports the fraction of examples truncated, the number of target tokens lost, the distribution of prompt and answer lengths, and the examples most affected.

Fourth, batch composition affects the gradient. A mixture with many long coding answers may dominate token-weighted loss even if those examples are a minority by count. Conversely, many short classification examples may dominate example-weighted

sampling while contributing few target tokens. A useful run log reports both example counts and target-token counts by data source.

The optimizer settings are usually less exotic than pretraining but still important. SFT often uses a lower learning rate than training from scratch, short warmup, weight decay on non-norm parameters, gradient clipping, and early stopping by held-out loss or task metrics. Parameter-efficient variants, especially LoRA and QLoRA, are common when full fine-tuning is too expensive; Chapter 10 studies their tradeoffs. Full fine-tuning can alter the model more deeply, but it also stores a full adapted checkpoint and increases the risk of catastrophic forgetting when the SFT set is narrow.

9.5 Refusal and Safety Data

Instruction tuning can teach refusals, but refusal data must be designed with care. The easy mistake is to train a binary style: answer normal requests, refuse unsafe requests. Real deployments require finer distinctions. A model should refuse requests for wrongdoing, protected private data, or dangerous operational details; it should answer benign nearby requests; it should ask clarifying questions when risk depends on context; and it should avoid turning every refusal into a lecture.

Safety examples therefore need paired contrastive coverage. For instance, a dataset might include a disallowed request for evading an access control system, an allowed request to explain access-control principles, and an allowed request to audit a user's own configuration. Without such near-neighbor examples, the model may learn surface keywords rather than policy boundaries. This is one reason SFT is usually followed by preference optimization and red-team evaluation rather than treated as the final safety layer [84, 107].

Refusal data also interacts with domain adaptation. In medicine, law, finance, education, and internal enterprise settings, a helpful response may require caution, uncertainty, and referral rather than a simple answer or refusal. A model trained only on generic safety data can sound polished while giving advice outside its evidence base. Chapter 11 returns to this problem for specialized domains.

9.6 Evaluation

SFT evaluation should answer three questions: whether the model follows instructions, whether it preserves base capabilities, and whether it behaves safely under distribution shift. A single benchmark score cannot answer all three.

Instruction following can be measured with held-out task suites, format validators, human pairwise comparisons, and model-graded rubrics. Format-heavy tasks should use deterministic checks whenever possible: a JSON parser is more reliable than a preference judge for JSON validity, and SQL execution is more reliable than surface matching when queries are semantically equivalent. Free-form tasks need human or

high-quality rubric-based review, but the review prompt must not leak the desired answer.

Capability preservation requires regression tests against the base model. SFT can improve chat behavior while degrading factual recall, translation, multilingual quality, code generation, or calibration. This is especially common when the SFT set is narrow, overly synthetic, or dominated by one language. A serious report includes before-and-after scores on tasks that were not optimized, not only a leaderboard where the tuned model looks good.

Safety evaluation must include both direct unsafe requests and benign requests that share vocabulary with unsafe requests. It should measure under-refusal and over-refusal separately. A model that refuses everything in a sensitive domain is not safe in a user-facing sense; it is unusable. A model that answers every request politely is not safe either. The operational target is a calibrated policy boundary with useful safe completions.

Finally, evaluation data needs quarantine. If public benchmark examples are used during SFT, then later benchmark scores are not evidence of generalization. Even when the exact examples are removed, paraphrases and synthetic variants can leak. The evaluation set should have documented provenance, a creation date, and a contamination check against the training mixture.

9.7 Limits of Imitation

SFT is imitation learning. It teaches the model to place probability mass on responses that look like demonstrations. It does not directly teach which of two plausible responses humans prefer, how to trade off helpfulness and safety, whether a retrieved passage is trustworthy, or how to recover after a tool fails. If the demonstrations contain confident hallucinations, the model learns confident hallucinations. If the demonstrations are verbose, the model becomes verbose. If the demonstrations always comply, the model learns compliance even where refusal is appropriate.

This limitation does not make SFT optional. SFT establishes the interface on which later methods operate. Preference learning needs candidate assistant responses to compare; tool-use training needs a call format; retrieval systems need answer synthesis behavior; safety tuning needs a refusal style. The right conclusion is that SFT should be narrow in its claims and disciplined in its data. It is the stage that turns a base model into a model that can participate in an assistant protocol, not the stage that proves the assistant is reliable.

9.8 Key Terms

Chat template A deterministic serialization of structured messages, roles, separators, and stop tokens into the token sequence consumed by the model.

Instruction tuning Supervised fine-tuning on examples that express tasks as natural-language requests and desired assistant responses.

Label mask A binary or ignored-label pattern specifying which positions contribute to the SFT loss.

Self-instruction A synthetic-data method in which a model generates new instructions and demonstrations that are filtered and used for fine-tuning.

Refusal data Demonstrations that teach the model to decline disallowed requests while preserving useful responses to safe neighboring requests.

Regression set A held-out evaluation suite used to detect capability loss after adaptation.

9.9 Exercises

1. Given a two-turn chat with a system message, two user turns, and two assistant turns, write a serialized template and mark every token span with $m_t = 0$ or $m_t = 1$ for Equation 9.2. State whether previous assistant turns are targets or context only.
2. Build a small audit table for 100 instruction examples. Include task type, language, source, target-token length, refusal category, and whether an automatic validator exists. What distributional gaps are visible?
3. Design a synthetic-data prompt for generating NL2SQL examples. Specify two automatic filters and one execution-based validator that would reject bad examples before training.
4. A model improves on an instruction-following leaderboard but gets worse at translation and arithmetic. List three plausible causes in the SFT mixture or training loop and one test for each cause.
5. Create three near-neighbor safety examples: one request that should be refused, one that should be answered, and one that requires clarification. Explain which surface features should not determine the policy decision.
6. Compare full fine-tuning and LoRA for the same SFT set. Report trainable parameters, checkpoint size, held-out instruction score, and one regression metric from the base model.

Chapter 10

Parameter-Efficient Adaptation

Abstract This chapter covers adapters, prefix tuning, prompt tuning, LoRA, QLoRA, target-module selection, rank choice, quantization, domain adaptation tradeoffs, and evaluation of adapted models.

10.1 What Is Being Made Efficient?

Full fine-tuning updates every parameter of a pretrained model. This is expressive, but it is expensive in three different ways. Training must store gradients and optimizer states for the full model; deployment may need a separate full checkpoint per task; and the adapted model may drift away from general capabilities learned during pretraining. Parameter-efficient fine-tuning, or PEFT, restricts the trainable variables while leaving most pretrained weights frozen.

Let θ_0 be the frozen base model and ϕ be the trainable adaptation parameters. PEFT solves

$$\min_{\phi} - \sum_{(x,y) \in \mathcal{D}} \log p_{\theta_0, \phi}(y | x), \quad |\phi| \ll |\theta_0|. \quad (10.1)$$

The inequality is the source of the savings, but it is also the source of the limitation. A PEFT method can only express changes reachable through the adapter subspace, prompt vectors, scaling vectors, or low-rank matrices it introduces. A small adapter may be enough to teach a format or a narrow domain. It may be insufficient to repair a weak tokenizer, add broad factual knowledge, or reverse a harmful pretraining bias.

Efficiency therefore has to be specified. Training memory depends on which parameters require gradients and optimizer states. Checkpoint storage depends on whether the base model is shared and only adapters are saved. Serving latency depends on whether the adapter adds extra computation, extra sequence length, or unmerged matrix products. Operational complexity depends on how many adapters must be versioned, loaded, combined, audited, and rolled back.

Table 10.1 Families of parameter-efficient adaptation.

Method	Trainable Object	Main Deployment Cost
Adapters	Bottleneck modules inside blocks	Extra projections and adapter routing
Prefix tuning	Per-layer continuous key/value prefixes	More attention states and prefix management
Prompt tuning	Input-level soft prompt embeddings	Consumes context budget; task-specific prompt loading
IA ³ -style scaling	Learned activation scaling vectors	Small extra elementwise operations
LoRA	Low-rank additive updates to selected matrices	Extra low-rank matmuls unless merged
QLoRA	LoRA on a quantized base	Quantization kernels and possible merge constraints

10.2 Adapter and Prompt Families

The original adapter idea inserts small trainable modules inside each Transformer block while freezing the pretrained weights [37]. A common bottleneck adapter maps a hidden vector $h \in \mathbb{R}^d$ down to a smaller dimension b , applies a nonlinearity, maps back to d , and adds the result residually:

$$\text{Adapter}(h) = h + W_{\text{up}} \sigma(W_{\text{down}} h), \quad W_{\text{down}} \in \mathbb{R}^{b \times d}, \quad W_{\text{up}} \in \mathbb{R}^{d \times b}. \quad (10.2)$$

When $b \ll d$, the adapter contributes far fewer parameters than the surrounding attention and feed-forward layers. The price is that every forward pass must execute extra projections unless the architecture provides a way to fold them into nearby operations, which is not generally true because of the nonlinearity.

Prefix tuning moves the trainable parameters from the residual stream into attention inputs. It learns continuous prefix vectors that act like task-specific key and value states available to the model at each layer [56]. The base model stays frozen, and generated tokens can attend to the learned prefix as if it were additional context. Prompt tuning is even simpler: it learns a small sequence of continuous input embeddings while leaving the model unchanged [51]. Prompt tuning is cheap to store and easy to swap, but it consumes effective context length and may be less expressive at smaller model scales.

Other PEFT methods use multiplicative gates or sparse updates. IA³, for example, learns vectors that rescale internal activations and can be highly parameter-efficient in few-shot settings [64]. The common principle is that the base model supplies a rich representation space and the adapter supplies a low-dimensional control signal.

10.3 LoRA

Low-Rank Adaptation, or LoRA, is the most widely used PEFT method for decoder large language models because it fits the linear algebra of Transformer blocks [38]. For a frozen weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, LoRA learns a low-rank update

$$W' = W_0 + \Delta W, \quad \Delta W = \frac{\alpha}{r} BA, \quad (10.3)$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$, $B \in \mathbb{R}^{d_{\text{out}} \times r}$, rank r is small, and α is a scaling hyperparameter. The trainable parameter count for this matrix is

$$|\phi_W| = r(d_{\text{in}} + d_{\text{out}}), \quad (10.4)$$

instead of $d_{\text{in}}d_{\text{out}}$. For square matrices with width d , the fraction is approximately $2r/d$.

The initialization is part of the method. A typical implementation initializes one low-rank factor randomly and the other to zero so that $\Delta W = 0$ at the start of training. The initial model is exactly the base model, and adaptation begins as a learned residual update. LoRA dropout can be applied on the adapter path, and the base weight remains frozen.

LoRA can target different matrices. Attention projections are common: query, key, value, and output projections each control different aspects of attention behavior. Many instruction-tuning recipes target query and value projections only for lower cost; stronger adaptation often also targets key, output, and feed-forward projections. There is no universal best target set. Code tasks, multilingual adaptation, and structured generation may benefit from broader targets; a small style adapter may not.

Merging is a practical advantage. At inference time one can compute $W_{\text{merged}} = W_0 + (\alpha/r)BA$ and serve the model as an ordinary dense model, provided the precision and deployment policy allow the merge. This removes adapter matmul overhead. Keeping adapters unmerged allows dynamic swapping, multi-tenant serving, and combining adapters, but it requires runtime adapter management. Multiple adapters can be added as sums of low-rank updates only when their target matrices, scaling, and licensing constraints are compatible; empirical interference still has to be tested.

10.4 QLoRA and Quantized Training

LoRA reduces trainable parameters but does not by itself remove the memory needed to store the frozen base model. QLoRA attacks that cost by keeping the base model in a 4-bit quantized representation while backpropagating through it into LoRA adapters [21]. The base weights are not updated; gradients update only the low-rank matrices. Activations, adapter weights, and some computation may still use higher precision, but the dominant model-weight storage is much smaller.

A rough memory comparison for a model with N parameters is

$$M_{\text{full}} \approx M(W) + M(\nabla W) + M(\text{Adam}_1) + M(\text{Adam}_2), \quad (10.5)$$

where the optimizer states can exceed the parameter memory. In QLoRA, the dominant terms become

$$M_{\text{QLoRA}} \approx M_{4\text{bit}}(W_0) + M(\phi) + M(\nabla \phi) + M(\text{optimizer}(\phi)) + M(\text{activations}). \quad (10.6)$$

The last term can still be large, especially with long sequences and large batches, so gradient checkpointing and sequence packing remain important.

Quantization is not free. A 4-bit base model introduces quantization error, kernel constraints, and sometimes slower wall-clock training if hardware kernels are inefficient. QLoRA uses NormalFloat 4-bit quantization, double quantization, and paged optimizers to make the recipe practical, but those techniques do not guarantee that every model and task will match full fine-tuning. Evaluation must compare task quality, regression behavior, and calibration, not just whether the run fits on one GPU.

10.5 Choosing Rank, Targets, and Hyperparameters

Rank r controls capacity. A rank that is too small underfits; a rank that is too large consumes more memory and may overfit a small dataset. The scaling parameter α controls the magnitude of the update relative to the base matrix. Practitioners often hold α/r or α in a conventional range, but the stable choice depends on optimizer, target modules, dataset size, and whether the base is quantized.

Target selection is usually more important than small changes in rank. If only the value projection is adapted, the model can alter what information is passed forward but has less control over how attention queries are formed. If feed-forward matrices are adapted, the adapter can change tokenwise transformations and often gains expressive power at a higher parameter cost. A defensible experiment grid varies one factor at a time: rank, target modules, learning rate, dropout, and number of training tokens.

Learning rates for adapters are often higher than full fine-tuning rates because the trainable matrices are newly initialized and small. This does not mean larger is always better. Too high a learning rate can produce brittle format imitation, loss spikes, or safety drift. A useful training log reports trainable parameter count, peak memory, tokens per second, gradient norm, training loss, validation loss, and task metrics.

The data collator remains critical. PEFT does not fix incorrect label masks, broken padding, or chat-template mismatches. A LoRA training implementation that freezes the base model, enables gradient checkpointing, disables the inference cache during training, pads to the longest sequence in the batch, and saves only trainable adapter weights captures the right systems shape. The textbook lesson is broader than the implementation: adapter training still inherits every preprocessing and evaluation obligation from SFT.

10.6 Systems Costs and Deployment

PEFT changes the deployment architecture. With full fine-tuning, each adapted model is a complete checkpoint. With LoRA or adapters, many tasks may share one base checkpoint plus small task-specific files. This is attractive for storage and governance: a base model can be approved once, while adapters are reviewed and versioned separately. It also creates runtime challenges.

If adapters are loaded dynamically, the server must batch requests that use different adapter ids. A batch containing many adapters may lose some efficiency because each request needs a different low-rank path or merged weight set. If adapters are premerged, the server regains dense inference speed but loses cheap swapping and may need many resident model copies. If the base model is quantized, merging can require dequantization or a specialized quantized-merge path.

Adapter composition is another operational risk. Two adapters trained separately may both improve their own tasks but degrade each other when summed or applied sequentially. A domain adapter that changes style can interfere with a safety adapter that depends on precise refusal wording. A multilingual adapter can improve target-language fluency while degrading English regression tests. Composition should be treated as a new model release, not as a bookkeeping operation.

Finally, PEFT does not remove the need for data security. An adapter can memorize sensitive examples even if it is small. Shipping only adapter weights is not a privacy guarantee. Membership inference, prompt extraction, and direct memorization tests are still relevant when the fine-tuning set contains private documents or user conversations.

10.7 Evaluation of Adapted Models

A PEFT report should include four classes of metrics. First, task metrics measure the reason for adaptation: exact match, execution accuracy, pass rate, retrieval-grounded answer quality, or human preference. Second, regression metrics measure what should not have changed: general chat quality, multilingual behavior, math, code, safety, and calibration. Third, systems metrics measure whether PEFT achieved the intended efficiency: trainable parameters, peak memory, training time, checkpoint size, tokens per second, and serving latency. Fourth, robustness metrics test sensitivity to prompt paraphrases, longer contexts, unseen schemas, and adversarial examples.

Comparisons must be fair. A LoRA model trained for 10,000 target tokens should not be compared to a full fine-tune trained for 10 million target tokens without accounting for data. A QLoRA run should not be declared equivalent to full fine-tuning because one benchmark matches. A prompt-tuned model should not be penalized for consuming context length unless the deployment actually has a tight context budget. The question is not which PEFT method is globally best; it is which method gives enough adaptation under the memory, latency, storage, and governance constraints of the application.

10.8 Key Terms

- Adapter** A small trainable module inserted into a frozen pretrained network, often using a bottleneck projection and residual connection.
- LoRA** A low-rank additive update to selected weight matrices, usually mergeable into the base weights for inference.
- QLoRA** A recipe that trains LoRA adapters while storing the frozen base model in a low-bit quantized format.
- Rank** The dimension r of the low-rank update in LoRA, controlling adapter capacity and parameter count.
- Target modules** The model matrices or layers to which adapters are applied, such as attention projections or feed-forward projections.
- Merge** The process of adding a trained LoRA update into the base weight matrix for ordinary dense inference.

10.9 Exercises

1. For a square projection matrix with width 4096, compute the LoRA parameter count for ranks 4, 8, 16, and 64. Compare each to the full matrix parameter count.
2. Implement Equation 10.3 for one linear layer. Initialize the adapter so that the adapted layer is initially identical to the base layer, then verify the equality numerically.
3. Fine-tune the same small model with LoRA on query/value projections only and on all attention projections. Report trainable parameters, validation loss, task metric, and one regression metric.
4. Design a QLoRA memory budget for a 7B-parameter model. List which terms come from quantized base weights, adapters, optimizer states, gradients, and activations.
5. Choose a domain task where prompt tuning might be preferable to LoRA and one where LoRA might be preferable to prompt tuning. Justify the choice using deployment constraints.
6. Suppose two adapters improve different tasks separately but fail when combined. Propose three diagnostic experiments to distinguish rank interference, conflicting data, and prompt-template mismatch.

Chapter 11

Domain and Language Adaptation

Abstract This chapter studies how LLMs are adapted to Chinese, domain-specific corpora, NL2SQL, medical-style retrieval tasks, and other specialized settings. It compares continual pretraining, supervised fine-tuning, retrieval, tokenizer extension, and evaluation under domain shift.

11.1 Adaptation Is a Diagnosis

Domain adaptation is not one method. It is a diagnosis of what the base model lacks. A model may fail in a new setting because the tokenizer fragments the language badly, because the relevant facts were absent or stale during pretraining, because the domain uses unfamiliar style and discourse conventions, because the task requires a structured output grammar, or because the answer must be grounded in private documents that should never be memorized into model weights. Each diagnosis points to a different intervention.

Table 11.1 gives a practical decision map. Continual pretraining, SFT, LoRA, tokenizer extension, and retrieval are complementary tools, not substitutes with the same risk profile. The cheapest intervention that addresses the actual failure should be tried before a more invasive one.

The most common error is to call every failure a fine-tuning problem. If the user asks about a private handbook updated last week, fine-tuning is the wrong default: retrieval gives freshness and access control. If the model cannot produce valid SQL for a known schema, supervised examples and execution validators may be appropriate. If the model uses too many tokens for a target language, tokenizer and continued pretraining choices matter before instruction style. If the application is medical, legal, or financial, the adaptation plan must include evidence boundaries and expert evaluation, not only a higher score.

Table 11.1 Diagnosing adaptation needs.

Observed Failure	Likely Cause	Candidate Intervention
Many tokens per word or character	Tokenizer mismatch	Vocabulary extension or a language-specialized tokenizer
Knows style but not facts	Missing or changing knowledge	Retrieval first; continual pretraining only for stable public corpora
Correct facts, wrong format	Interface or output contract missing	SFT or PEFT on demonstrations with validators
Fails on schemas or APIs	Poor grounding in structured context	Schema-aware prompting, SFT, execution feedback, or tool use
Good benchmark, bad deployment	Distribution shift or contamination	Private held-out evaluation and slice analysis
Unsafe advice in high-stakes domain	Policy and evidence boundaries unclear	Domain safety data, abstention rules, expert review, and monitoring

11.2 Language Adaptation

Language adaptation starts with tokenization. A subword tokenizer trained mostly on English may split Chinese, Thai, Arabic, or domain-specific symbols into long sequences. This increases context cost and makes long-range modeling harder. A simple diagnostic is token fertility:

$$F_L = \frac{1}{|S_L|} \sum_{s \in S_L} \frac{\text{tokens}(s)}{\text{units}(s)}, \quad (11.1)$$

where $\text{units}(s)$ may be characters, words, or language-appropriate segments. Fertility should be compared across languages and domains using representative text. High fertility does not prove poor model quality, but it is an early warning for cost and coverage.

There are three broad tokenizer strategies. The first is to keep the original tokenizer and adapt only weights. This is simple and preserves checkpoint compatibility, but it may waste context. The second is to extend the vocabulary with frequent target-language or domain tokens. New embeddings can be initialized from the average or composition of their old subpieces, then trained during continued pretraining. The third is to train a new tokenizer and convert or retrain the model. This can improve efficiency but is the most disruptive because embeddings, output heads, and compatibility with existing checkpoints change.

Chinese adaptation illustrates the tradeoff. Chinese LLaMA and Alpaca-style work added Chinese text encoding, secondary pretraining, and Chinese instruction tuning to make LLaMA-family models more useful for Chinese users [16]. The lesson is not that every language needs a separate model. The lesson is that language coverage is a full pipeline: tokenizer fertility, corpus selection, continued pretraining, instruction data, safety examples, and evaluation all matter.

Multilingual evaluation must be slice-based. A single averaged score can hide a model that improves simplified Chinese but degrades English, improves news text but

fails colloquial dialogue, or answers in the wrong language. Evaluation should include target-language perplexity, instruction following, translation or cross-lingual transfer where relevant, code-switching, culturally specific references, and safety examples written by native speakers.

11.3 Continual Pretraining

Continual pretraining, also called domain-adaptive pretraining when the corpus is domain-specific, continues the language-model objective on new unlabeled text. It is useful when the domain has stable terminology, discourse patterns, and public factual regularities. “Do not stop pretraining” showed that continuing pretraining on domain and task-related text can improve downstream performance for domain tasks [32]. The same idea scales to modern decoder models, but the risks scale too.

A common objective mixes domain data with replay data from the general distribution:

$$\mathcal{L}_{\text{CPT}}(\theta) = \lambda \mathbb{E}_{x \sim \mathcal{D}_{\text{domain}}} [-\log p_{\theta}(x)] + (1 - \lambda) \mathbb{E}_{x \sim \mathcal{D}_{\text{replay}}} [-\log p_{\theta}(x)]. \quad (11.2)$$

The replay term reduces forgetting. Without it, a narrow corpus can make the model better at local terminology while degrading general language quality, multilingual coverage, or safety behavior. The mixture coefficient λ , learning rate, number of tokens, and deduplication policy are central hyperparameters.

Continual pretraining should not be used to inject private or rapidly changing facts unless there is a strong reason to memorize them into weights. Weight updates are hard to audit, hard to revoke, and hard to access-control. Retrieval is usually better for internal policies, customer records, current prices, medical guidelines that update, or legal documents that differ by jurisdiction. Continual pretraining is better suited to stable public corpora such as scientific literature, code conventions, patent language, or target-language web text whose licensing has been reviewed.

Biomedical models illustrate both value and caution. BioGPT used large-scale biomedical literature to train a generative model specialized for biomedical text generation and mining [70]. Such specialization can improve terminology and task performance, but it does not by itself make the model clinically safe. Biomedical language competence is different from validated medical decision support.

11.4 Supervised Domain Tuning

SFT or PEFT is appropriate when the missing capability is a task interface rather than broad language modeling. A customer-support assistant may need to follow a ticket triage format. A writing assistant for a fictional world may need to continue in a specific voice. An enterprise assistant may need to output a strict JSON object. A database

assistant may need to map natural-language questions to SQL. These are supervised behaviors.

The training examples should separate three things: domain context, user request, and expected response. A useful schema for a writing task might contain a short memory summary, recent text, and the next passage. A useful schema for QA might contain question, evidence, and answer. A useful schema for NL2SQL contains the natural-language question, the database schema, optional foreign-key information, and the target SQL. Collapsing all fields into a raw string may work for a demo, but it hides the contract from preprocessing, validation, and evaluation.

PEFT is often the first adaptation method for supervised domain tuning because it is cheap to reverse and compare. LoRA can adapt a base model to a schema format or domain style while storing only adapter weights [38]. QLoRA can make the same experiment feasible under tighter hardware limits [21]. The danger is false confidence: a small adapter can overfit a small domain set and appear fluent while failing on unseen entities, new schemas, or adversarial wording.

Data splits must represent deployment. Randomly splitting examples in a domain dataset can leak entities, templates, or schemas. For an internal support assistant, split by customer or time. For a code assistant, split by repository. For fiction continuation, split by work or chapter. For NL2SQL, split by database when the deployment will see new schemas. The evaluation split should test the generalization claim the model is supposed to satisfy.

11.5 Structured Tasks: NL2SQL

Natural-language-to-SQL tasks expose the difference between fluent text and executable meaning. Given a question q and schema S , the model emits a SQL query y :

$$y \sim p_{\theta}(\cdot \mid \tau(q, S)). \quad (11.3)$$

The answer is correct only if the query is syntactically valid, refers to the right tables and columns, respects SQL dialect, and returns the intended result. Spider made cross-domain text-to-SQL evaluation important by splitting across databases and requiring generalization to unseen schemas [121].

Schema serialization is part of the model interface. A prompt that lists only table names is different from one that includes columns, types, primary keys, foreign keys, and example rows. Normalizing table names can help a model learn structural patterns instead of memorizing arbitrary identifiers, but it can also hide entity-linking difficulty. The preprocessing choice should match the intended evaluation. If deployment requires real table names, the model or postprocessor must recover them reliably.

Evaluation should include exact string match only as a diagnostic. SQL has many equivalent surface forms. Execution accuracy, where the predicted query is run against a database and compared to the expected result, is often more meaningful. It is still imperfect: two wrong queries can return the same result on a small database, and unsafe queries should not be executed outside a sandbox. A robust NL2SQL evaluation includes

parse validity, execution accuracy, schema-linking accuracy, latency, and manual review of a sample of failures.

For production, an NL2SQL model is usually part of a guarded system. The system can restrict SQL to read-only statements, require a query planner or parser check, ask for clarification when the question underspecifies a join, and show the generated query before execution. Fine-tuning the model is only one layer of safety.

11.6 Retrieval Versus Weight Updates

Retrieval-augmented generation and fine-tuning solve different problems. Retrieval supplies evidence at inference time; fine-tuning changes behavior stored in parameters. If the knowledge is fresh, private, user-specific, or legally sensitive, retrieval is usually the safer first step. If the behavior is a stable output pattern or the domain language is broadly missing, fine-tuning may be appropriate. If both are missing, combine them: retrieve evidence and fine-tune the model to use evidence faithfully.

A medical-style QA system is a useful example. The knowledge base can store prior cases, guidelines, or approved documents; a retriever finds relevant evidence; a reranker improves precision; and the generator drafts a response conditioned on the selected evidence. This architecture can update the knowledge base without retraining the model, and it can log which evidence was shown. It also introduces new failure modes: the retriever can miss relevant passages, the reranker can prefer superficially similar but clinically different cases, and the generator can overstate conclusions. Chapter 12 studies these retrieval failures in detail.

Fine-tuning on retrieved answers can improve style and evidence use, but it should not train the model to invent citations. The target answer should be traceable to evidence, and examples without adequate evidence should teach abstention or clarification. In high-stakes settings, a fluent answer without evidence is a failure even if it sounds domain-specific.

11.7 Safety and Governance in Domains

Domain adaptation often moves a model closer to consequential decisions. Medical, legal, financial, educational, and enterprise systems require stricter governance than general chat. The first question is scope: what is the model allowed to do, and what must remain with a qualified human or an external system? The second question is evidence: what sources can the model rely on, and how are they updated? The third question is accountability: what logs, audits, and rollback paths exist when the system fails?

Safety data should be domain-specific. A medical assistant needs examples that distinguish general health education from diagnosis, triage, emergency advice, dosage calculation, and clinician-facing summarization. A legal assistant needs jurisdiction, uncertainty, and non-attorney boundaries. A financial assistant needs suitability con-

straints and disclosure. Generic refusal data is not enough because domain risk often depends on subtle context.

Privacy is also central. A domain corpus may contain personal data, trade secrets, unreleased product plans, or copyrighted material. Deduplication and de-identification reduce risk but do not eliminate it. If private facts need to be revocable, retrieval with access controls is preferable to weight updates. If adaptation uses private examples, membership-inference and memorization checks should be part of release evaluation.

11.8 Evaluation Under Domain Shift

Domain adaptation should be evaluated by slices, not only by averages. A good report includes in-domain and out-of-domain metrics, pre-adaptation and post-adaptation comparisons, and failure analysis by entity type, language, length, source, and date. For language adaptation, report token fertility and target-language task quality. For structured tasks, report validity and execution. For retrieval-grounded tasks, report retrieval recall separately from answer quality. For high-stakes tasks, include expert review and abstention quality.

Contamination is a special concern. Domain benchmarks are often small and public. A model can memorize task templates or examples during continued pretraining or synthetic-data generation, especially when the same benchmark is used repeatedly to tune prompts and adapters. Evaluation sets should be held out before adaptation begins, and their overlap with training data should be checked by exact match, near-duplicate search, and task-specific leakage tests [115].

The final question is operational: does adaptation improve the deployed workflow? A model that raises an NL2SQL benchmark by two points but produces unsafe queries is worse. A model that sounds more medical but cites no evidence is worse. A model that improves Chinese chat but doubles context cost because of tokenization may be uneconomical. Adaptation is successful only when model quality, systems cost, safety, and governance improve together.

11.9 Key Terms

Continual pretraining Continuing the language-model objective from a pretrained checkpoint on additional corpora.

Domain-adaptive pretraining Continual pretraining focused on a target domain such as biomedical papers, legal documents, or code.

Token fertility The number of tokenizer tokens needed per language or domain unit, used as a diagnostic for tokenizer coverage.

Schema linking The process of connecting words in a natural-language query to database tables, columns, and values.

Execution accuracy The fraction of generated programs or SQL queries that return the expected result when executed in a controlled environment.

Replay data General-distribution data mixed into continual pretraining to reduce forgetting.

11.10 Exercises

1. Compute token fertility for 100 English sentences and 100 target-language sentences under the same tokenizer. Interpret the result as a cost and modeling diagnostic.
2. Design a continual-pretraining mixture for a legal-domain model. Specify domain sources, replay sources, deduplication rules, and three regression tests.
3. Build an NL2SQL evaluation plan for a new database. Include schema split, prompt serialization, parse validity, execution accuracy, and sandboxing rules.
4. Compare retrieval-only, LoRA-only, and retrieval-plus-LoRA baselines on a small domain QA set. Report retrieval recall separately from answer correctness.
5. Write three domain safety examples for a medical assistant: an allowed education question, a question requiring urgent referral, and a request the assistant should refuse or redirect.
6. A model improves in-domain perplexity after continual pretraining but gets worse at general instruction following. Propose two data-level causes and two training-level causes.

Part IV
Alignment, Applications, and Evaluation

Chapter 12

Retrieval, Tools, and Agents

Abstract This chapter covers retrieval-augmented generation, embeddings, vector databases, retrievers, rerankers, chunking, citation faithfulness, tool calls, ReAct-style reasoning-action loops, and agent workflow evaluation.

12.1 RAG as a Control System

Retrieval-augmented generation, or RAG, is often described as putting documents into the prompt. That description is too small. A production RAG system is a control system around an large language model. It decides what corpus is searchable, how documents are chunked, how queries are rewritten, which retriever and reranker are trusted, how much context is inserted, how the answer should cite evidence, when the model should abstain, and how to defend the system against hostile text inside retrieved documents.

The basic decomposition is

$$q \rightarrow \text{retrieve}(q) \rightarrow E_k \rightarrow \text{generate}(q, E_k) \rightarrow a, \quad (12.1)$$

where q is the user query, E_k is a selected evidence set, and a is the answer. RAG originally connected neural retrievers with sequence generators for knowledge-intensive NLP tasks [53]. Dense passage retrieval then made learned embedding search a practical alternative to purely sparse lexical search for open-domain question answering [42]. Retrieval-augmented pretraining systems such as RETRO showed that retrieval can also be integrated into model training and scaling, not only into prompting [9].

The central benefit is separation of concerns. Model weights store general linguistic and reasoning behavior; the retrieval layer stores mutable or private evidence. This separation improves freshness, auditability, and access control. It does not guarantee truth. If retrieval misses the relevant document, the generator may hallucinate. If retrieval returns misleading evidence, the generator may faithfully summarize the wrong source. If retrieved text contains instructions to ignore the user or reveal secrets, the generator may follow the wrong authority. The system must therefore evaluate retrieval, generation, and control logic separately.

12.2 Indexing and Retrieval

An index begins with documents, but retrieval usually operates over chunks. Chunking is a modeling decision. Short chunks improve lexical precision and citation granularity, but they can lose context. Long chunks preserve context, but they consume prompt budget and can bury the relevant sentence. Overlap helps when answers cross boundaries, but it increases storage and can produce duplicate evidence. A practical index stores each chunk with stable document id, source, timestamp, permissions, title, section path, and character offsets so that answers can cite and audit the original source.

Dense retrieval embeds queries and chunks into vectors. A typical scoring function is inner product or cosine similarity:

$$s_{\text{dense}}(q, d) = \langle f_q(q), f_d(d) \rangle, \quad D_k(q) = \text{topk}_{d \in C} s_{\text{dense}}(q, d). \quad (12.2)$$

Sparse retrieval, such as BM25-style lexical search, remains valuable because exact names, rare identifiers, error codes, and legal citations often matter. Hybrid retrieval combines dense and sparse signals:

$$s_{\text{hybrid}}(q, d) = \lambda \hat{s}_{\text{dense}}(q, d) + (1 - \lambda) \hat{s}_{\text{sparse}}(q, d), \quad (12.3)$$

where scores are normalized before combination. The weight λ is a tuning parameter, not a constant of nature.

Query rewriting can improve recall. The system may expand acronyms, generate paraphrases, add domain synonyms, or split a multi-part question into subqueries. In a medical-style retrieval system, for example, a symptom description may be rewritten into several clinically adjacent phrasings before vector search. Rewriting also adds risk: a rewrite can change the user’s intent. The original query and all rewrites should be logged, and evaluation should measure whether rewriting improves recall without increasing false positives.

12.3 Reranking and Context Construction

Retrievers are usually optimized for recall at a moderate k . The generator, however, needs a small set of highly relevant, non-duplicative chunks inside a finite context window. Reranking bridges this gap by scoring query-document pairs with a more expensive model after initial retrieval. A cross-encoder reranker can compare the query and chunk jointly, which is often more precise than independent embeddings, but it is slower and harder to run over a large corpus.

Context construction is not the same as taking the top five chunks. The system should remove near duplicates, respect source permissions, include enough neighboring context for interpretation, and order evidence deliberately. Long-context models do not always use middle-position evidence robustly; performance can be highest when relevant information appears near the beginning or end of the prompt [65]. Therefore, rank order, section grouping, and explicit citation markers can affect answer quality.

Table 12.1 Design choices in a retrieval-augmented generation pipeline.

Component	Choice	Failure Mode
Chunking	Size, overlap, metadata, boundaries	Relevant evidence split or buried
Embedding	Model, normalization, language coverage	Poor recall for domain terms or languages
Retriever	Dense, sparse, or hybrid search	Misses exact ids or semantic paraphrases
Reranker	Cross-encoder, LLM judge, or rules	High precision but high latency or bias
Context builder	Ordering, deduplication, compression	Prompt budget wasted or evidence distorted
Generator	Prompt, citation rules, abstention	Unsupported claims or citation laundering
Monitor	Logs, freshness, drift metrics	Silent index decay or stale answers

Table 12.1 lists design choices that should be documented in any serious RAG system.

12.4 Generation with Evidence

The generator should be instructed to answer from evidence, but instruction alone is insufficient. The prompt should separate system policy, user request, retrieved evidence, and tool outputs. Retrieved text is data, not authority. A document may contain malicious instructions, outdated statements, or user-provided content. The system prompt should state that retrieved documents are untrusted evidence and that higher-priority instructions come from the developer or system layer.

Grounded generation has three obligations. First, the answer should use the evidence that supports the claim. Second, the citation should point to the evidence actually used, not merely to a top-ranked chunk that seems related. Third, the model should abstain or ask for clarification when evidence is missing or conflicting. Recent RAG evaluation surveys emphasize that answer quality, context relevance, citation faithfulness, and robustness should be measured separately [26].

Citation faithfulness is harder than appending source ids. A model can cite a source that does not support the sentence, cite a source that supports only part of the sentence, or synthesize across sources without making the connection clear. A useful answer format ties each factual claim or paragraph to evidence ids and permits “not found in the provided sources” as a valid outcome. For high-stakes domains, unsupported claims should be treated as failures even if the natural language is plausible.

Compression is another risk. Systems often summarize retrieved chunks to save context. If the compressor removes uncertainty, dates, exceptions, or source qualifiers, the generator receives distorted evidence. Compression should be evaluated as its own component, and the system should preserve links to the original text for audit.

12.5 RAG Evaluation

RAG evaluation begins before generation. Retrieval recall asks whether the evidence needed to answer the question appears in the candidate set. If recall is low, no generator can reliably fix the system. Metrics include recall@k, mean reciprocal rank, and normalized discounted cumulative gain when relevance labels are graded. For enterprise systems, labels may come from human annotation, click logs, answer citations, or synthetic questions generated from documents and then audited.

After context construction, evaluate context precision: how much of the inserted context is relevant? A prompt filled with weakly related chunks can lower answer quality and increase cost. Then evaluate answer quality: correctness, completeness, citation support, abstention behavior, and style. These metrics should be reported by slice: document source, document age, language, query type, entity frequency, and permission group.

End-to-end judge scores are useful for triage but dangerous as the only metric. A judge model can reward fluent hallucinations, miss citation errors, or share the same blind spots as the generator. Deterministic checks should be used where possible: exact source id, quote span overlap, executable answer, policy compliance, or database result. Human review remains necessary for subtle factuality and high-stakes claims.

The right baseline matters. Compare against no retrieval, sparse retrieval, dense retrieval, hybrid retrieval, reranking, and an oracle context where the gold evidence is supplied. The gap between oracle-context generation and retrieved-context generation estimates how much quality is being lost in retrieval and context construction. The gap between oracle-context generation and the gold answer estimates generator limitations.

12.6 Prompt Injection and Trust Boundaries

RAG introduces untrusted text into the model's context. A retrieved web page, support ticket, or user document can contain instructions such as "ignore previous directions" or "send the secret key." The model sees these tokens in the same context window as legitimate instructions unless the system marks and handles them carefully. This is prompt injection.

The defense is architectural, not just a warning sentence. Retrieved documents should be delimited as data. Tool permissions should be enforced outside the model. Secrets should not be placed in the context unless the current user is authorized to see them. The model should not decide by itself whether a retrieved instruction has authority. Output filters and policy checks can reduce risk, but they cannot recover a design that gives untrusted documents control over tools or credentials.

Evaluation should include injection tests. Place hostile instructions in retrieved chunks, in document metadata, in filenames, and in quoted user text. Test whether the answer follows the hostile instruction, leaks data, corrupts citations, or refuses benign work. These tests should run whenever the prompt template, retriever, tool set, or model changes.

12.7 Tool Use

Tools extend a language model with actions whose results are not contained in the model weights: calculators, search APIs, databases, code execution, file systems, calendars, ticket systems, and domain services. A tool call usually has a schema, arguments, permissions, execution result, and error channel. Toolformer showed that models can be trained to decide when and how to call simple APIs from self-supervised traces [95]. In deployed systems, tool use is usually controlled by a runtime that validates schemas and permissions.

A minimal tool loop is

$$a_t \sim p_\theta(\cdot \mid h_t), \quad o_t = \text{Tool}(a_t), \quad h_{t+1} = h_t \oplus a_t \oplus o_t, \quad (12.4)$$

where h_t is the dialogue and state history, a_t may be a tool call or final answer, and o_t is the observation. The runtime, not the model, should validate that a_t is a permitted call. Tool observations should be treated like retrieved evidence: useful, but not automatically higher authority than system instructions.

Error recovery is part of tool competence. The model may call a function with invalid arguments, receive an empty result, exceed a rate limit, or encounter conflicting tool outputs. Training and evaluation should include these cases. A tool-using model that succeeds only when every call works on the first try is not robust.

12.8 Agents and Workflows

An agentic workflow repeatedly chooses actions, observes results, and updates state toward a task goal. ReAct made this pattern explicit by interleaving reasoning traces and actions so that a model can plan, call tools, and revise based on observations [120]. Modern agentic model reports emphasize coding, tool use, long-horizon tasks, and environment feedback, but the engineering problem remains the same: the system must manage state, tools, permissions, retries, and evaluation [43].

Planning can help when tasks require many steps, but plans are not guarantees. A plan may be impossible, stale after an observation, or optimized for a proxy goal. Memory can help preserve information across steps, but memory can also store wrong or sensitive content. Reflection can identify mistakes, but it can also rationalize them. Agent design should therefore prefer explicit state machines, typed tool calls, bounded retries, and verifiable checkpoints where the domain allows them.

Evaluation of agents should focus on task completion under constraints, not on whether the transcript looks thoughtful. Useful metrics include success rate, number of tool calls, wall-clock time, cost, rollback rate, permission violations, human interventions, and quality of final artifacts. For coding agents, run tests and inspect diffs. For data agents, verify queries and outputs. For workflow agents, check that the external system ended in the intended state.

12.9 Systems Costs

Retrieval, tools, and agents spend inference-time compute. A single user request may trigger embedding calls, vector search, reranking, prompt construction, a long prefill, multiple decode steps, tool calls, retries, and judge evaluations. The cost model is therefore wider than tokens generated by the final answer.

Latency has several components:

$$T_{\text{total}} = T_{\text{rewrite}} + T_{\text{retrieve}} + T_{\text{rerank}} + T_{\text{prefill}} + T_{\text{decode}} + T_{\text{tools}} + T_{\text{postcheck}}. \quad (12.5)$$

Optimizing one term can worsen another. Retrieving more chunks may improve recall but increase reranking and prefill time. Compressing chunks may reduce prefill but harm faithfulness. Calling tools can improve correctness but add network latency and failure modes. A production report should include latency percentiles, cost per request, cache hit rate, retrieval recall, answer quality, and safety failures together.

Freshness also has a systems cost. Indexes must be rebuilt or incrementally updated. Deleted documents must be removed from search results. Permission changes must take effect quickly. If the index is stale, a RAG answer may be worse than a base-model answer because it carries the appearance of grounded authority.

12.10 Key Terms

Chunk A retrievable unit derived from a document, usually with metadata and a pointer back to the source.

Dense retrieval Retrieval by learned vector representations and similarity search.

Hybrid retrieval Combining dense semantic scores with sparse lexical scores.

Reranker A second-stage model or rule that orders retrieved candidates more precisely before context insertion.

Citation faithfulness The degree to which cited sources actually support the claims attached to them.

Prompt injection An attack in which untrusted text inside context attempts to override higher-priority instructions or exfiltrate data.

Tool call A structured action request from the model to an external function, service, or environment.

Agentic workflow A bounded loop in which the model selects actions, receives observations, and updates state toward a goal.

12.11 Exercises

1. Build a small RAG benchmark with at least 30 questions and labeled supporting chunks. Report recall@5 before measuring answer quality.

2. Compare three chunking strategies for the same corpus: short chunks, long chunks, and overlapping chunks. Measure retrieval recall, context precision, and average prompt tokens.
3. Implement hybrid retrieval using Equation 12.3. Sweep λ and report which query types prefer dense versus sparse retrieval.
4. Create five prompt-injection documents and add them to a test index. Measure whether the system follows hostile instructions, leaks hidden text, or corrupts citations.
5. Design a tool schema for read-only SQL querying. Specify argument validation, permission checks, error messages, and how generated SQL will be sandboxed.
6. Evaluate an agentic workflow on a real task with a fixed budget. Report success rate, tool calls, cost, latency, and the most common failure mode.

Chapter 13

Preference Learning and Alignment

Abstract This chapter explains how language models are adapted with preference data after pretraining and supervised instruction tuning. It covers pairwise preference collection, reward modeling, RLHF with PPO, KL regularization, direct preference optimization, AI feedback, safety and refusal data, reward hacking, annotation protocols, systems costs, and the gap between preference learning and full alignment.

13.1 Alignment as Preference Modeling

Pretraining teaches a model to imitate text. Supervised instruction tuning teaches it a conversational interface. Preference learning then asks a harder question: among several plausible responses, which response should the deployed assistant prefer? The answer may depend on helpfulness, truthfulness, harmlessness, style, legal constraints, user intent, and institutional policy. The central engineering challenge is that these properties are not a single ground-truth label. They are observed through noisy comparisons, rubrics, red-team examples, and downstream outcomes.

Modern alignment pipelines therefore treat preference learning as a control layer around a capable base model. A typical pipeline starts from a supervised policy, samples multiple answers to user prompts, asks humans or AI judges to compare those answers, trains a reward or preference model, and then optimizes the policy toward the learned preference signal while constraining it not to drift too far from the reference model. This recipe descends from deep reinforcement learning from human preferences [15], language-model preference tuning [127, 103], and instruction-following RLHF [84]. The practical success of the recipe should not be confused with a proof of alignment. It is a way to optimize a proxy; the proxy can be incomplete, inconsistent, non-stationary, and vulnerable to gaming.

This chapter uses “alignment” in the operational sense common in LLM engineering: reducing systematic mismatches between model behavior and a specified preference or safety policy. It is not a claim that the model has internalized human values, that it will

generalize safely under all distribution shifts, or that all stakeholders share the same preferences.

13.2 Preference Data

13.2.1 From Labels to Comparisons

For many language tasks, direct labels are brittle. A summarization prompt may have many valid summaries; a medical information request may require both useful context and careful scope limits; a coding assistant may produce several correct patches with different maintainability tradeoffs. Preference learning replaces absolute labels with comparisons. For prompt x , annotators compare two completions y_a and y_b and produce a label such as

$$y_w \succ y_l \mid x,$$

where y_w is preferred to y_l under a rubric. Pairwise data is easier to collect than scalar reward because humans are usually better at choosing between concrete alternatives than assigning calibrated numerical scores.

The comparison label is still a measurement, not a fact about the universe. Annotators may disagree because one response is terse and another is explanatory, because a refusal is safer but less helpful, or because a prompt requires domain expertise. Good data collection therefore records more than the winning answer:

- the prompt source and sampling policy used to generate candidates;
- the rubric version, policy category, and intended user population;
- annotator qualifications, locale, and conflict-resolution process;
- ties, uncertainty, skipped examples, and known ambiguity;
- safety severity and whether the prompt is benign, dual-use, adversarial, or policy-violating.

This metadata is not administrative overhead. It determines whether a later reward model is learning user preference, annotator preference, policy preference, demographic majority preference, or artifacts of the candidate generator.

13.2.2 Sampling Candidate Responses

The distribution of candidate responses is part of the dataset. If both responses are produced by the same weak supervised model, comparisons teach the reward model about local quality differences near that model. If one response is produced by a stronger proprietary model and another by the current policy, the reward model may learn a teacher-model style. If candidates are generated with high temperature, the reward model sees diverse errors but may overemphasize fluency failures. If candidates

are generated after safety filtering, the reward model sees fewer dangerous negatives and may become weak at recognizing them.

Let $q(y | x)$ denote the candidate-generation distribution used for preference collection. A reward model trained on q is most reliable near q . After policy optimization, the new policy π_θ may generate text outside that region. This is the basic distribution-shift problem in RLHF: the optimizer searches for outputs that score highly under the reward model, including outputs unlike those shown to annotators.

13.2.3 Annotation Protocols

A preference rubric should decompose quality into observable criteria. A minimal assistant rubric often separates:

Instruction following. Does the response answer the actual user request, respect constraints, and use the requested format?

Truthfulness. Are factual claims correct, scoped, and supported when support is required?

Completeness. Does the response include the necessary steps, caveats, and assumptions without burying the answer?

Safety. Does the response avoid facilitating harm while remaining helpful for benign requests?

Style. Is the response clear, concise, respectful, and appropriate for the audience?

The rubric should say how to trade these criteria off. Without an explicit priority order, annotators will invent their own. Safety work especially needs boundary examples: requests that must be refused, requests that should be answered with safe alternatives, and requests that look sensitive but are legitimate.

13.3 Reward Models

13.3.1 The Bradley-Terry Objective

A reward model $r_\phi(x, y)$ maps a prompt-response pair to a scalar. Pairwise preference modeling commonly uses the Bradley-Terry likelihood:

$$P_\phi(y_w \succ y_l | x) = \sigma(r_\phi(x, y_w) - r_\phi(x, y_l)),$$

where σ is the logistic sigmoid. The negative log-likelihood over a comparison dataset $\mathcal{D}$ is

$$\mathcal{L}_{\text{RM}}(\phi) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} [\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))].$$

Only reward differences matter. Adding a constant to all rewards for the same prompt does not change the pairwise probability. This is why raw reward-model scores should

not be interpreted as calibrated utilities unless the training procedure includes explicit calibration.

The usual implementation takes a pretrained or instruction-tuned transformer, appends a scalar value head at the final token, and trains on packed pairs. For long responses, the reward may be read from the last non-padding token; for multi-turn dialogue, the entire conversation is serialized with the same chat template used by the policy. The reward model should see the same system-message and role-token conventions used at deployment, because formatting changes can move examples out of distribution.

13.3.2 Calibration and Uncertainty

Reward accuracy on a held-out preference set is necessary but insufficient. A reward model can rank easy pairs correctly while being overconfident on subtle pairs. Useful diagnostics include:

- accuracy by prompt category, language, length, and safety severity;
- calibration curves for predicted preference probability;
- accuracy as a function of reward margin $|r(y_w) - r(y_l)|$;
- disagreement between independent reward models trained on bootstrap samples;
- adversarial examples where verbosity, flattery, hedging, or refusal style changes the score without improving substance.

RewardBench-style evaluations are useful for comparing reward models across chat, reasoning, and safety cases [47], but a deployment team still needs in-domain preference data. A reward model that is strong on public chat comparisons may be poor at a company’s legal, medical, finance, or code-review policies.

13.3.3 Overoptimization

The reward model is a proxy. Optimizing against it too aggressively can reduce real user preference even while predicted reward rises. This is reward hacking. In language models, reward hacking often appears as excessive verbosity, confident but unsupported claims, policy boilerplate, false empathy, or responses that exploit annotation shortcuts. A model can learn that annotators prefer long answers, that safety raters reward visible disclaimers, or that a certain phrase pattern correlates with high scores.

The operational symptom is a divergence between offline reward and independent evaluation. As optimization steps increase, reward-model score continues upward, but human preference, factuality, or task success plateaus and then degrades. The fix is not simply a better optimizer. It requires better data coverage, stronger adversarial evaluation, regularization, early stopping, and repeated collection of comparisons from the current policy.

13.4 RLHF with PPO

13.4.1 Regularized Policy Optimization

In RLHF, the language model is a policy $\pi_\theta(y \mid x)$ over response sequences. The reward model provides a sequence-level score $r_\phi(x, y)$. Because unconstrained reward maximization can destroy language quality and exploit reward-model errors, the objective includes a penalty against a reference policy π_{ref} , usually the supervised instruction-tuned model:

$$\max_{\theta} \mathbb{E}_{x \sim \mathcal{P}, y \sim \pi_\theta(\cdot \mid x)} \left[r_\phi(x, y) - \beta \text{KL}(\pi_\theta(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x)) \right].$$

The coefficient β controls the alignment-strength versus drift tradeoff. A large β keeps the policy close to the reference and may underuse the reward signal. A small β permits larger improvements but increases the risk of reward hacking, style collapse, and safety regressions.

In autoregressive models, the KL term is usually estimated token by token during rollout:

$$\widehat{\text{KL}}(y) = \sum_{t=1}^T [\log \pi_\theta(y_t \mid x, y_{<t}) - \log \pi_{\text{ref}}(y_t \mid x, y_{<t})].$$

The scalar reward used by the RL algorithm can be written as a terminal reward plus per-token KL shaping. This turns a sparse sequence reward into a denser signal while preserving the regularized objective.

13.4.2 PPO Mechanics

Proximal Policy Optimization (PPO) is widely used because it limits the size of policy updates while using sampled rollouts [96]. For token-level actions, define the probability ratio

$$\rho_t(\theta) = \frac{\pi_\theta(y_t \mid x, y_{<t})}{\pi_{\theta_{\text{old}}}(y_t \mid x, y_{<t})}.$$

With an advantage estimate $\hat{A}_t$, the clipped PPO objective is

$$\mathcal{L}_{\text{PPO}}(\theta) = \mathbb{E}_t \left[\min \left(\rho_t(\theta) \hat{A}_t, \text{clip}(\rho_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right].$$

RLHF implementations also train a value head $V_\psi(x, y_{<t})$ to estimate expected future regularized reward, often using generalized advantage estimation. The value head reduces gradient variance but adds another model component that can become miscalibrated under distribution shift.

The InstructGPT pipeline used supervised fine-tuning, reward modeling, and PPO to produce models preferred by labelers over larger pretrained models on instruction-following tasks [84]. That result is historically important because it showed that post-

training data quality and preference optimization can dominate parameter count for user-facing behavior.

13.4.3 Systems Costs

RLHF is expensive because it runs several large models in the loop:

- the policy being optimized;
- the frozen reference model for KL computation;
- the reward model;
- a value model or value head;
- sometimes separate safety classifiers or filters.

For each rollout, the system performs autoregressive generation, reward scoring, KL scoring, advantage computation, and one or more policy-gradient updates. Memory pressure is high because the optimizer must store activations for the policy update and logits for KL or importance ratios. Many production pipelines therefore use smaller reward models, parameter-efficient adapters, sequence packing, aggressive batching, and periodic rather than continuous reward-model refreshes.

RLHF also has a data-systems cost. Prompts must be deduplicated, policy versions tracked, annotation decisions audited, unsafe content handled carefully, and privacy constraints enforced. If a dataset mixes user traffic, synthetic prompts, and red-team prompts, the provenance of each example should remain available during training and evaluation.

13.5 Direct Preference Optimization

13.5.1 The Implicit Reward View

Direct Preference Optimization (DPO) avoids online RL by deriving a supervised loss from the same KL-regularized preference objective [91]. Under the regularized optimum, an implicit reward can be written as

$$r(x, y) = \beta \log \frac{\pi_{\theta}(y | x)}{\pi_{\text{ref}}(y | x)} + C(x),$$

where $C(x)$ is a prompt-dependent constant that cancels in pairwise comparisons. Substituting this reward difference into the Bradley-Terry preference likelihood yields the DPO loss:

$$\mathcal{L}_{\text{DPO}}(\theta) = -\mathbb{E}_{(x, y_w, y_l)} \left[\log \sigma \left(\beta \left[\log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right] \right) \right].$$

DPO uses preference pairs directly, without training an explicit reward model or generating on-policy rollouts. This makes it simpler and more stable to implement than PPO, especially for smaller teams.

13.5.2 Tradeoffs

DPO is not free of assumptions. It is an offline method, so it learns from the candidate responses present in the preference dataset. If the dataset contains only weak negatives, DPO may not teach robust rejection of strong but unsafe completions. If preferred responses all share a teacher model’s style, DPO may distill style more than preference. The reference model still matters: the log ratio penalizes deviations from it, and changing the reference can change the effective reward.

Several related methods modify the DPO recipe. IPO, KTO, ORPO, and other preference objectives adjust the treatment of margins, binary desirable/undesirable labels, or reference-free training. These variants are useful engineering tools, but the deeper question is unchanged: what behavior does the data define, and where does that definition fail?

13.6 Safety, Refusal, and AI Feedback

13.6.1 Helpful-Harmless Tradeoffs

Assistant training usually optimizes both helpfulness and harmlessness. The two are not independent. A model that refuses every sensitive prompt is safe in a narrow sense but useless. A model that answers every request is helpful in a narrow sense but dangerous. Anthropic’s helpful-harmless work and Constitutional AI introduced explicit safety preference data and AI-generated critique/revision loops [6, 5]. RLAIIF scales this idea by using AI systems to generate some preference labels or direct rewards [48].

Safety preference data should include at least three classes:

- Comply. Benign requests that should be answered, including requests that mention sensitive topics in educational, defensive, journalistic, or fictional contexts.
- Refuse. Requests that directly facilitate harm, abuse, evasion, or illegal activity.
- Redirect. Requests where the assistant should avoid operational details but provide safe alternatives, high-level context, or help-seeking resources.

The hardest examples are near the boundaries. Overbroad refusal causes real harm when users are denied benign information about health, security, law, or history. Under-refusal causes harm when the model provides actionable misuse instructions.

13.6.2 Policy as a Training Object

Safety policy should be versioned like code. Each preference label is meaningful only relative to a policy version. If the policy changes, old labels may become inconsistent. This matters for reproducibility: a model trained under policy version p_1 and evaluated under policy version p_2 may appear misaligned even if it learned p_1 faithfully.

For high-risk domains, preference optimization should be paired with refusal evaluation, jailbreak testing, and human review. The policy should specify not only disallowed outputs but also expected safe completions. Users often need help reformulating a request, understanding risk, or finding legitimate resources. A refusal that merely says “I cannot help” may score well on a narrow safety metric while failing the broader user need.

13.7 Evaluation Cautions

13.7.1 Preference Is Not Ground Truth

Preference labels measure what annotators choose under a rubric and interface. They do not necessarily measure long-term user welfare, truth, fairness, legal compliance, or moral value. This limitation is central to open problems in RLHF [11]. Preference learning can make a model sound better without making it more reliable. It can also hide failures by teaching the model to hedge, flatter, or refuse in ways that satisfy raters.

13.7.2 Distribution Shift

Every stage introduces shift:

- pretraining data differ from instruction data;
- instruction prompts differ from preference prompts;
- preference candidates differ from optimized policy outputs;
- evaluation prompts differ from real deployment traffic;
- adversaries adapt after deployment.

A robust alignment process therefore repeats data collection after major model updates. Offline preference sets are useful for regression testing, but they cannot be the only source of evidence.

Failure mode	Symptom	Diagnostic response
Reward hacking	Reward score rises while human preference falls	Plot reward versus independent human eval across checkpoints; inspect high-reward samples
Verbosity bias	Longer answers win despite no added substance	Length-controlled comparisons; judge with strict concision criteria
Refusal overgeneralization	Benign sensitive prompts are refused	Boundary-set evaluation with comply/refuse/redirect labels
Safety undergeneralization	Harmful variants bypass refusal	Red-team paraphrases, multi-modal variants, and tool-use tests
Annotator drift	Label distributions change over time	Gold examples, rubric refreshes, annotator calibration meetings
Reward-model over-confidence	Large margins on ambiguous examples	Calibration curves, ensembles, and abstention thresholds

Table 13.1 Common preference-learning failure modes and diagnostics.

13.7.3 A Practical Diagnostic Table

13.8 Implementation Notes

A small but defensible preference-learning pipeline can be organized as follows:

1. Start from a supervised instruction-tuned policy with a frozen chat template.
2. Build a prompt mixture that includes real tasks, synthetic coverage prompts, and safety boundary cases.
3. Sample two or more candidate responses per prompt from known policy versions.
4. Collect pairwise labels with rubric metadata and allow ties or uncertainty.
5. Train a reward model or a direct preference objective; reserve a held-out set by prompt, not by pair.
6. Evaluate by category, length, language, and safety severity before optimizing the policy.
7. For PPO, monitor KL, reward, entropy, response length, refusal rate, and human preference across checkpoints.
8. For DPO, monitor chosen/rejected log-probability gaps, reference drift, and performance on out-of-domain prompts.
9. Run independent safety, factuality, and task-success evaluations before deployment.
10. Log deployed policy versions and sample real traffic for post-deployment regression tests, subject to privacy and consent constraints.

13.9 Key Terms

Preference data. Comparisons or ratings that express which model outputs are preferred under a rubric.

Reward model. A learned scalar scoring function used as a proxy for preference.

RLHF. Reinforcement learning from human feedback, usually involving a reward model and policy optimization.

KL regularization. A penalty that keeps the optimized policy close to a reference policy.

DPO. A direct offline preference objective that optimizes pairwise preferences without an explicit reward model.

RLAIF. Reinforcement learning from AI feedback, where AI systems generate some preference labels, critiques, or rewards.

Reward hacking. Improving the proxy reward while degrading the intended behavior.

Refusal boundary. The policy line between requests the model should answer, refuse, or redirect.

13.10 Exercises

1. Train a toy Bradley-Terry reward model on synthetic pairwise comparisons. Plot held-out accuracy and calibration as you increase label noise.
2. For a small instruction-tuned model, compute the token-level KL between sampled responses and a frozen reference model. Inspect examples with unusually high KL and explain what changed.
3. Implement the DPO loss for a batch of chosen/rejected responses. Verify that increasing β changes the strength of the preference update.
4. Design a safety boundary set with ten comply prompts, ten refuse prompts, and ten redirect prompts for a domain of your choice. Write the rubric before writing the prompts.
5. Take twenty model responses and label them under two different rubrics: one that prioritizes concision and one that prioritizes completeness. Measure how often the preferred response changes.
6. Write an evaluation memo for a preference-tuned assistant. Include the reward model's validation accuracy, independent human preference rate, refusal false-positive rate, and the main residual risks.

Chapter 14

Reasoning and Test-Time Compute

Abstract This chapter studies reasoning methods for LLMs as algorithms that allocate computation at inference time. It covers chain-of-thought prompting, self-consistency, decomposition, process supervision, verifiers, tree search, MCTS-style reasoning, reinforcement learning with verifiable rewards, GRPO-style training, DeepSeek-R1-style pipelines, budget forcing, frontier reasoning evaluations, systems costs, and the limits of spending more inference compute.

14.1 Reasoning as Budgeted Computation

Reasoning is not a single mechanism. In deployed LLM systems, it usually means that the system spends extra computation to improve the chance of a correct answer. The extra computation may appear as a longer generated scratchpad, multiple sampled solutions, a verifier that ranks candidates, a search tree over partial solutions, tool calls, or a policy trained to explore and check its own work.

This chapter treats reasoning methods as budgeted algorithms. A method should specify what is being computed, how the budget is allocated, how candidate answers are selected, what failure modes are expected, and how the method will be evaluated. Chain-of-thought prompting [112], self-consistency [109], process supervision [59], tree-of-thought search [119], and 2025 reasoning-RL systems such as DeepSeek-R1 [19] all fit this view. They differ less in their rhetoric than in their allocation of tokens, samples, supervision, and verification.

The central caution is that longer reasoning text is not the same as faithful reasoning. A visible trace can be useful working memory, a persuasive explanation, a post-hoc rationalization, or a mixture of all three. Evaluation must therefore separate answer correctness, trace usefulness, trace faithfulness, latency, cost, and robustness under distribution shift.

14.2 Prompted Reasoning

14.2.1 Chain of Thought

Chain-of-thought (CoT) prompting asks the model to produce intermediate steps before the final answer. In few-shot CoT, the prompt includes examples of step-by-step solutions. In zero-shot variants, the prompt asks for a reasoning process directly. CoT often helps arithmetic, symbolic, commonsense, and multi-hop tasks because it gives the autoregressive model more intermediate tokens on which to condition.

The simplest abstraction is:

$$z, a \sim \pi_{\theta}(\cdot \mid x),$$

where x is the problem, z is a generated reasoning trace, and a is the final answer. The trace is useful when it decomposes the task into intermediate states that make the next-token problem easier. It is harmful when it introduces unsupported assumptions, locks the model into an early mistake, or produces a convincing explanation for an incorrect answer.

CoT has three practical constraints:

- It consumes output tokens, increasing latency and cost.
- It may expose sensitive intermediate reasoning that a product should not reveal.
- It can be unfaithful: editing or hiding the trace need not preserve the model’s true internal computation [71].

For user-facing systems, it is often better to ask the model to use private scratch work and return a concise answer with checkable explanation, rather than expose every intermediate token.

14.2.2 Self-Consistency

Self-consistency improves over greedy CoT by sampling K reasoning traces and selecting the most common final answer:

$$\hat{a} = \arg \max_a \sum_{k=1}^K \mathbf{1}\{a_k = a\}, \quad z_k, a_k \sim \pi_{\theta}(\cdot \mid x).$$

The method works when independent samples make different mistakes but the correct answer has higher aggregate probability. It fails when many samples share the same misconception, when final answers are hard to normalize, or when the problem requires a rare insight that the model almost never samples.

Self-consistency is parallelizable. If one sample costs T generated tokens, K samples cost roughly KT tokens but can be run concurrently if serving capacity is available. This makes it attractive for offline evaluation and high-value tasks, but expensive for low-

latency products. Recent test-time scaling work reframes this as an allocation problem: spend extra samples, revisions, or search only when the expected improvement justifies the latency and cost [102, 75].

14.2.3 Decomposition and Programmed Reasoning

Reasoning prompts often decompose a problem into subproblems:

$$x \rightarrow (x_1, \dots, x_m) \rightarrow (a_1, \dots, a_m) \rightarrow a.$$

The decomposition can be natural language, a table, code, a set of equations, or calls to tools. Program-of-thought approaches ask the model to write executable code for arithmetic or symbolic manipulation, shifting part of the burden from language generation to an interpreter. Tool-using agents extend the same idea by calling search, calculators, theorem provers, databases, or code runners.

Decomposition is valuable only when subproblem boundaries are reliable. If the first step frames the problem incorrectly, later steps may make the wrong plan look rigorous. Systems should therefore include opportunities to revise decompositions, not merely solve them.

14.3 Search and Verification

14.3.1 Sample-and-Rank

A verifier $v_\psi(x, z, a)$ scores candidate solutions. The system samples K candidates and returns the answer from the highest-scoring candidate:

$$(\hat{z}, \hat{a}) = \arg \max_{(z_k, a_k)} v_\psi(x, z_k, a_k).$$

The verifier may be a trained reward model, a process reward model, a unit-test runner, a symbolic checker, or another LLM judge. Verifier-based selection is effective when the verifier is more reliable than the generator on the target error modes. It is dangerous when the verifier shares the generator’s blind spots or rewards superficial reasoning markers.

For code, mathematics, and structured outputs, verifiers can be strong because correctness is partially executable: tests pass, equations simplify, schemas validate, or proofs check. For open-ended analysis, the verifier often becomes another preference model and inherits the limitations from Chapter 13.

14.3.2 Outcome Versus Process Supervision

Outcome supervision labels only the final answer. Process supervision labels intermediate steps. In math, a process reward model (PRM) estimates whether each step is locally valid [59]. If a trace has steps $s_1, \dots, s_n$, a PRM provides scores

$$p_i = P(s_i \text{ is valid} \mid x, s_{<i}, s_i).$$

The trace score can then combine step scores, for example by summing log probabilities:

$$S(z) = \sum_{i=1}^n \log p_i.$$

Process supervision can catch early mistakes that lead to a lucky final answer, and it can guide search by pruning bad partial solutions. Its cost is annotation complexity. Step-level labels require expertise, clear definitions of validity, and careful handling of alternative solution paths.

14.3.3 Tree Search

Tree-of-thought methods generalize sampling by expanding partial reasoning states [119]. A node is a partial solution s . Expansion samples candidate next thoughts. Evaluation scores partial states. Search then chooses which nodes to expand next. Breadth-first search, beam search, depth-first search with backtracking, and Monte Carlo tree search (MCTS) are different policies for allocating the expansion budget.

A generic tree-search loop is:

1. Start with root state $s_0 = x$.
2. Select a frontier node according to a search policy.
3. Expand it by sampling b candidate next steps from the model.
4. Score candidates with a value function, verifier, or heuristic.
5. Stop when a candidate answer passes the acceptance criterion or the budget is exhausted.

The key design choice is not whether the structure is called a tree. It is whether the score used to guide expansion is valid. A weak heuristic can make tree search more expensive than self-consistency without improving correctness.

14.4 Reinforcement Learning with Verifiable Rewards

14.4.1 Verifiable Rewards

Reinforcement learning with verifiable rewards (RLVR) trains a policy on tasks where correctness can be checked automatically. Examples include math problems with exact answers, programming tasks with tests, theorem-proving goals with proof checkers, and structured extraction tasks with known labels. The reward can be as simple as

$$R(x, y) = \begin{cases} 1, & \text{if verifier accepts } y, \\ 0, & \text{otherwise.} \end{cases}$$

The attraction is scale: no human preference label is needed for each rollout. The risk is narrowness: the reward covers what can be verified, not necessarily what matters. A code solution can pass visible tests while being brittle; a math answer can be right for the wrong reason; a model can learn formatting tricks that exploit the verifier.

DeepSeek-R1 showed that large-scale RL on verifiable tasks can elicit strong reasoning behavior, including long traces, self-checking, and backtracking, without starting from human-written reasoning demonstrations in the R1-Zero stage [19]. Subsequent RLVR studies examine when such rewards incentivize correct reasoning rather than merely correct answers [113]. The answer depends on task design, verifier strength, exploration, and whether shortcuts are available.

14.4.2 Group Relative Policy Optimization

GRPO-style objectives replace a learned value function with group-relative baselines. For a prompt x , sample a group of G responses $\{y_i\}_{i=1}^G$ and compute rewards r_i . A normalized group advantage is

$$\hat{A}_i = \frac{r_i - \text{mean}(r_1, \dots, r_G)}{\text{std}(r_1, \dots, r_G) + \delta}.$$

The policy update then increases the likelihood of above-average samples and decreases the likelihood of below-average samples, usually with clipping and KL regularization against a reference policy. Removing the value model reduces memory and training complexity, but group sampling increases rollout cost. The method is most natural when rewards are cheap to compute and many samples can be generated per prompt.

A simplified clipped objective for a response $y_i = (a_{i,1}, \dots, a_{i,T_i})$ is

$$\begin{aligned} \mathcal{J}_{\text{GRPO}}(\theta) = & \frac{1}{G} \sum_{i=1}^G \frac{1}{T_i} \sum_{t=1}^{T_i} \min \left(\rho_{i,t}(\theta) \hat{A}_i, \text{clip}(\rho_{i,t}(\theta), 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}}) \hat{A}_i \right) \\ & - \beta D_{\text{KL}}(\pi_{\theta}(\cdot|x) \parallel \pi_{\text{ref}}(\cdot|x)). \end{aligned}$$

Here

$$\rho_{i,t}(\theta) = \frac{\pi_{\theta}(a_{i,t}|x, a_{i,<t})}{\pi_{\theta_{\text{old}}}(a_{i,t}|x, a_{i,<t})}.$$

Real implementations differ in token averaging, KL approximation, reward normalization, and whether the reward is attached only to the final answer or to intermediate verified steps. The important distinction from PPO is that the baseline is computed from the sampled group rather than from a learned critic. This makes memory use simpler, but it also makes optimization sensitive to group diversity: if all samples are equally bad, the relative advantages can still push the model toward the least bad local pattern.

GRPO and related methods also change the data distribution seen during training. The model learns from its own sampled attempts, not only from human-written solutions. This is powerful for exploration, but it amplifies verifier errors. If the verifier accepts a flawed shortcut, group-relative optimization can rapidly teach that shortcut.

14.5 Inference-Time Scaling

14.5.1 Compute Shapes

Test-time compute can be scaled in several shapes:

Sequential. Generate a longer trace, revise, reflect, or continue thinking.

Parallel. Generate multiple independent attempts and vote or rank them.

Tree-structured. Explore branches and allocate more budget to promising partial states.

Tool-augmented. Spend compute on retrieval, code execution, search, or external solvers.

If a prompt has prefill cost C_p , each generated token has decode cost C_d , and a strategy generates K samples of average length L , a crude serving cost is

$$C_{\text{serve}} \approx C_p + KLC_d + C_{\text{verify}} + C_{\text{tools}}.$$

This equation hides batching, KV-cache memory, parallelism, and hardware utilization, but it makes the product tradeoff explicit. A reasoning strategy that improves accuracy by two points may be unacceptable if it multiplies latency by ten for all requests. Adaptive policies attempt to spend more compute only on hard prompts.

14.5.2 Budget Forcing

Budget forcing controls the amount of reasoning by encouraging the model to continue or stop at chosen points. The s1 work demonstrated that simple test-time scaling and budget forcing can improve a distilled reasoning model on competition math tasks

[75]. The broader lesson is that inference-time behavior is a controllable interface: max tokens, stop conditions, prompts, verifier thresholds, and sampling parameters all shape the reasoning process.

Budget controls should be evaluated for both accuracy and failure mode. Forcing longer reasoning can help on multi-step math but hurt on perception-grounded multi-modal tasks, factual questions where the model lacks knowledge, or tasks where the first correct answer is later revised into an error. More thinking is useful only when the additional computation has access to a corrective signal.

14.5.3 Compute-Optimal Allocation

Test-time scaling work shows that the best allocation depends on the model and prompt [102, 13, 118]. Easy prompts may need one short answer. Medium prompts may benefit from a few samples. Hard prompts may require search, tools, or a larger model. Impossible prompts should trigger abstention rather than unbounded reasoning.

A practical router can estimate difficulty and choose among strategies:

$$\pi_{\text{route}}(x) \in \{\text{direct}, \text{CoT}, \text{self-consistency}, \text{tool}, \text{human}\}.$$

The router itself must be evaluated. If it sends too many prompts to expensive reasoning, cost explodes. If it misses hard prompts, accuracy suffers. If it routes safety-sensitive prompts to long free-form reasoning without strong policy constraints, risk increases.

14.6 Frontier Reasoning Systems

Frontier reasoning models released in 2024 and 2025 made test-time compute a first-class product feature. Public system cards for OpenAI reasoning models describe safety evaluations, tool use, and risk assessments for models that reason longer before answering [82]. Open-weight reports such as DeepSeek-R1 and Qwen3 document pipelines that combine supervised data, RL, verifiable rewards, distillation, and inference-time controls [19, 117].

These reports should be read as engineering evidence, not as final scientific explanations. They show that reasoning performance can be improved by post-training and inference-time computation. They do not prove that visible chains are faithful, that reward-trained reasoning generalizes outside verifiable domains, or that benchmark gains imply robust planning in the real world.

14.7 Evaluation Cautions

14.7.1 Answer Accuracy Is Not Enough

A reasoning evaluation should specify:

- answer metric: exact match, unit tests, proof check, human rubric, or preference;
- trace policy: hidden, visible, summarized, or unavailable;
- compute budget: max tokens, number of samples, tool budget, and wall-clock limit;
- selection method: greedy, majority vote, verifier rank, or human choice;
- contamination controls: held-out questions, transformed variants, and private tests;
- cost metric: tokens, latency, dollars, energy, or hardware occupancy.

Reporting only accuracy hides the central tradeoff. A model that solves 5% more tasks with 20 times more generated tokens may be valuable for theorem proving and unacceptable for customer support.

14.7.2 Trace Faithfulness

Visible reasoning traces are tempting evaluation artifacts. They let graders inspect steps, train process reward models, and debug failures. But a trace can be optimized for the grader. If the final answer is produced by internal activations and the visible trace is generated for presentation, then step-level natural-language evaluation may overstate reliability. Conversely, hiding all traces makes debugging and accountability harder. Many products therefore expose concise rationales or cited calculations, not raw chain-of-thought.

14.7.3 Reasoning Boundary Tests

Reasoning benchmarks should include boundary cases:

- problems with insufficient information, where abstention is correct;
- adversarially phrased easy problems, where overthinking causes errors;
- tasks requiring external knowledge not in the prompt;
- tasks with misleading but irrelevant details;
- tasks where a tool result contradicts the model’s prior belief;
- safety-sensitive tasks where the correct behavior is refusal or redirection.

These cases distinguish useful deliberation from verbose pattern completion.

14.8 Implementation Notes

A robust test-time reasoning stack usually contains five layers:

1. A prompt or policy that defines whether scratch work is private, visible, or summarized.
2. A budget controller for max tokens, number of samples, branch factor, and tool calls.
3. A verifier or acceptance test where the domain allows it.
4. A router that escalates hard, risky, or low-confidence requests.
5. Telemetry that logs budget, latency, selected strategy, verifier result, and user-visible answer.

For software engineering tasks, the verifier may be tests and static analysis. For math, it may be exact answer checking or symbolic verification. For factual tasks, retrieval and citation checks are often more valuable than longer unaided reasoning. For safety-sensitive tasks, policy checks should run outside the model’s own free-form scratchpad.

14.9 Key Terms

Chain of thought. Generated intermediate reasoning tokens used to support a final answer.

Self-consistency. Sampling multiple reasoning traces and aggregating final answers.

Verifier. A model, program, or human process that scores or checks candidate solutions.

Process reward model. A verifier that scores intermediate reasoning steps.

Tree search. A method that expands and evaluates partial reasoning states.

RLVR. Reinforcement learning with rewards from automatic verifiers.

GRPO. A group-relative policy optimization style that uses sampled group rewards as a baseline.

Test-time scaling. Improving performance by spending more computation during inference.

Budget forcing. Controlling the amount of reasoning computation by prompting or decoding constraints.

14.10 Exercises

1. Run a small math benchmark with greedy answers, CoT, and self-consistency with $K = 5$. Report accuracy, average output tokens, and latency.
2. Implement answer normalization for self-consistency on arithmetic problems. Measure how many apparent disagreements are only formatting differences.
3. Build a sample-and-rank loop using a unit-test verifier for simple programming tasks. Compare best-of- K selection to majority vote.

4. Create five problems where longer reasoning helps and five where it hurts. Explain the difference in terms of available corrective signals.
5. Derive the group-relative advantage for a batch of four sampled rewards. Show what happens when all rewards are equal.
6. Design an evaluation card for a reasoning model that reports accuracy, token budget, verifier type, hidden/visible trace policy, and contamination controls.

Chapter 15

Multimodal LLMs

Abstract This chapter introduces multimodal LLMs through contrastive representation learning, visual encoders, projection layers, query transformers, cross-attention adapters, multimodal instruction tuning, OCR and chart reasoning, image and video token budgets, multimodal evaluation, hallucination, privacy, and safety issues unique to systems that combine language with images, audio, video, and other continuous signals.

15.1 Modalities as Interfaces

A language model consumes tokens. Images, audio, video, tables, plots, documents, and sensor streams are not naturally token sequences in the same vocabulary. A multimodal LLM must therefore solve an interface problem: convert non-text signals into representations that a language model can use while preserving the information needed for the task.

The modern vision-language model (VLM) stack usually contains a modality encoder, a connector, and a language model. The encoder turns an image or video into patches, frames, or embeddings. The connector maps those representations into the language model’s hidden space. The language model then performs instruction following, reasoning, and generation. This design inherits strengths and weaknesses from both sides. It can use the language model’s broad knowledge and dialogue ability, but it can also hallucinate visually, over-rely on text priors, miss small spatial details, and obey malicious instructions embedded inside images.

This chapter focuses on LLM-centered multimodal understanding: models that take non-text input and produce language or structured text output. Image generation, diffusion models, and full robotics stacks are adjacent topics, but the core engineering ideas here are representation alignment, token budgeting, instruction data, evaluation, and safety.

15.2 Contrastive Pretraining

15.2.1 Image-Text Alignment

CLIP-style pretraining learns a shared embedding space for images and text using large collections of image-caption pairs [89]. Let $f_\theta(I_i)$ be the normalized embedding of image I_i , and $g_\psi(T_j)$ be the normalized embedding of text T_j . For a batch of N matched pairs, the image-to-text contrastive loss is

$$\mathcal{L}_{I \rightarrow T} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(f_\theta(I_i)^\top g_\psi(T_i)/\tau)}{\sum_{j=1}^N \exp(f_\theta(I_i)^\top g_\psi(T_j)/\tau)},$$

with a symmetric text-to-image term. The temperature τ controls the sharpness of the matching distribution. The model is not trained to generate captions. It is trained to place matched images and texts near each other and unmatched pairs farther apart.

This objective is powerful because web-scale captions provide broad supervision. It also has limits. Captions are incomplete, noisy, culturally biased, and often describe salient objects rather than fine-grained spatial relations. A CLIP encoder may know that an image is about a dog on a skateboard while missing whether the dog’s left paw touches the wheel. Downstream VLMs inherit these representation gaps unless later training addresses them.

15.2.2 Zero-Shot Classification

CLIP also demonstrates that language can define labels. To classify an image, one embeds prompts such as “a photo of a cat” and “a photo of a dog”, then chooses the text embedding with highest similarity to the image embedding. This is zero-shot transfer through natural-language class descriptions. The same idea appears in retrieval, image clustering, dataset filtering, and weak labeling.

For LLM-centered VLMs, contrastive encoders provide a strong visual backbone, but the assistant still needs a way to condition a generative language model on visual features. That connector is where many architecture differences arise.

15.3 Connecting Vision and Language

15.3.1 Frozen Encoders and Projection Layers

The simplest connector freezes a vision encoder and a pretrained LLM, then trains a projection module P_η :

$$H_{\text{vis}} = E_{\text{vis}}(I), \quad Z_{\text{vis}} = P_{\eta}(H_{\text{vis}}).$$

The projected visual tokens Z_{vis} are inserted into the LLM context alongside text tokens. LLaVA popularized this recipe for visual instruction tuning by connecting a CLIP vision encoder to a language model and training on image-grounded instruction data [63]. The appeal is simplicity: most parameters can remain frozen initially, and the connector learns the alignment.

Projection-only connectors are cheap, but they may pass too many image tokens or fail to select task-relevant details. If the vision encoder emits hundreds or thousands of patch features, each becomes part of the LLM context unless compressed. Long visual prefixes increase prefill cost and compete with text for context window space.

15.3.2 Query Transformers and Token Compression

BLIP-2 introduced a Querying Transformer (Q-Former) that learns a small set of query tokens to extract information from a frozen image encoder before passing it to a frozen LLM [55]. This design reduces the number of visual tokens and creates a trainable bottleneck:

$$Z_{\text{query}} = Q_{\eta}(Q_0, E_{\text{vis}}(I)).$$

The bottleneck can improve efficiency and force selection, but it can also discard details needed for OCR, diagrams, or counting. Many systems therefore vary token count by task or resolution: a low-resolution global view for general conversation, higher-resolution crops for documents, and frame sampling for video.

15.3.3 Cross-Attention and Interleaved Media

Flamingo-style models add cross-attention layers that let the language model attend to visual features while preserving a pretrained language backbone [2]. This design supports interleaved image-text sequences and few-shot multimodal prompting. The model can see examples such as image, question, answer, image, question, answer within a single context.

Cross-attention adapters are more complex than projection layers, but they avoid forcing all visual information into ordinary text-token positions. They also make it easier to handle multiple images or video frames. The tradeoff is implementation complexity, memory use, and the need to train or tune inserted layers across the language model depth.

Connector	Strengths	Typical risks
Linear or MLP projection	Simple, cheap, easy to retrofit to an LLM	Too many tokens; weak spatial grounding; connector bottleneck
Query transformer	Compresses visual features; keeps large backbones frozen	May discard small text or spatial details; extra pretraining stages
Cross-attention adapter	Handles interleaved media; flexible conditioning	More invasive architecture changes; higher memory cost
End-to-end multimodal training	Can optimize perception and language jointly	Expensive; greater risk of forgetting and data imbalance

Table 15.1 Common connector designs for LLM-centered multimodal models.

15.3.4 Architecture Tradeoffs

15.4 Multimodal Instruction Tuning

15.4.1 Training Stages

A practical VLM is rarely trained in one stage. A common sequence is:

1. Pretrain or choose a vision encoder using contrastive or supervised vision-language data.
2. Train a connector on image-caption pairs or image-text matching data.
3. Tune the combined model on image-grounded instruction data.
4. Add task-specific data for OCR, charts, documents, localization, visual math, or video.
5. Add safety and refusal data that includes both text-only and image-conditioned attacks.

The order matters. Early connector training teaches the language model to interpret visual embeddings as context. Later instruction tuning teaches conversational behavior and task formats. If the instruction data is too small or too synthetic, the model may learn a superficial response style without robust perception.

15.4.2 Data Sources

Multimodal instruction data can come from human annotation, synthetic conversations generated by stronger models, templated conversions of existing vision datasets, OCR/document datasets, chart QA datasets, and video QA datasets. Synthetic data is useful for coverage, but it can inject teacher-model hallucinations. Human data is expensive but essential for ambiguous tasks such as describing medical images, evaluating visual quality, or explaining charts for a specific audience.

The dataset should explicitly cover:

- object recognition, attributes, relations, and counting;

- OCR, handwriting, tables, charts, and diagrams;
- spatial reasoning and localization;
- multi-image comparison;
- visual math and science reasoning;
- uncertainty and abstention when the image is ambiguous or low resolution;
- safety cases where the image changes the policy decision.

Without such coverage, the VLM may answer from language priors. For example, if a prompt asks what color a stop sign is, a text-only prior may produce “red” even when the image is grayscale, occluded, or adversarially edited.

15.4.3 Chat Templates with Images

Image tokens must be represented in the chat template. A serialized example may look conceptually like:

```
<system> policy <user> <image> question <assistant> answer.
```

The literal tokens differ by implementation, but consistency is critical. Training and inference should use the same role markers, image markers, and ordering. Multi-image prompts need unambiguous references: “first image”, “second image”, bounding boxes, page numbers, or frame timestamps.

15.4.4 OCR, Charts, and Documents

Document understanding stresses VLMs differently from natural images. The model must read small text, preserve layout, associate labels with values, and reason over tables. Increasing image resolution helps but increases visual tokens. Some systems use specialized OCR tools and feed recognized text to the LLM; others rely on native visual reading. The best choice depends on the domain. OCR tools provide explicit text and coordinates, while native VLMs can combine layout and semantics but may hallucinate characters.

Charts require another layer: the model must map visual marks to quantities. It may need to infer axes, units, legends, and trends. Evaluation should check numerical accuracy, not merely plausible prose. A response that describes a bar chart fluently but swaps the largest and second-largest category is a failure.

15.5 Systems Costs

15.5.1 Visual Tokens and Prefill

For text-only LLMs, long prompts increase prefill cost. For VLMs, the image itself can contribute hundreds or thousands of tokens before the user question is considered. If an image is split into P patches and projected to M visual tokens, the LLM prefill length becomes approximately

$$L_{\text{prefill}} = L_{\text{text}} + M.$$

Attention cost during prefill grows roughly quadratically in total context length for standard attention, though optimized kernels and attention variants change constants. This makes high-resolution images, multi-page documents, and video expensive.

15.5.2 Video and Audio

Video adds a temporal dimension. A naive system that encodes every frame quickly exhausts context and compute. Practical systems sample frames, detect scene changes, compress frame features, or use a video encoder before the LLM. Video-LLaVA-style systems align image and video features before projection so that a language model can consume a unified visual representation [60]. Audio similarly requires chunking, spectrogram or codec features, speech recognition, speaker or prosody handling, and temporal alignment. AudioPaLM-style systems connect text and speech models so that speech recognition, speech translation, and speech generation can share a language-model backbone [94]. Omni models such as Qwen2.5-Omni make the deployment contract harder still: they must stream over text, image, audio, and video inputs while generating text or natural speech responses [87].

Temporal grounding is the core evaluation problem. A model may answer a global question about a video while failing to identify the frame, timestamp, or spoken segment that supports the answer. A publishable video or audio evaluation should separate event detection, temporal localization, transcription quality, speaker attribution, cross-modal coreference, and final reasoning. It should also report whether external tools are used. Tool-assisted OCR, ASR, chart extraction, object detection, or browser automation can make a system far stronger than a pure end-to-end model, but then the evaluation is of a pipeline, not only of a checkpoint.

System designers should report:

- maximum image resolution and resizing policy;
- number of visual tokens per image or frame;
- frame sampling rate for video;
- whether OCR, ASR, or other tools are used;
- latency split between encoder, connector, LLM prefill, and decode;
- memory impact on batching and KV cache.

Without these details, benchmark numbers are hard to interpret and deployment costs are hard to predict.

15.6 Evaluation

15.6.1 Benchmark Families

Multimodal evaluation must test perception and reasoning separately. MMMU evaluates multimodal understanding across expert-level disciplines [122]; MathVista focuses on mathematical reasoning in visual contexts [69]. POPE and HallusionBench probe visual hallucination and visual illusion failure modes [57, 31]. Recent frontier and open reports, including GPT-4o and Qwen2.5-VL, report broad multimodal capability and safety evaluations [80, 88, 33].

Benchmarks should be interpreted by task type. A model can be strong at OCR and weak at spatial reasoning, or strong at captioning and weak at chart arithmetic. A single leaderboard score is not a capability profile.

15.6.2 Evaluation Cautions

Key cautions include:

Language leakage. Questions may be answerable from text priors without inspecting the image.

Shortcut OCR. Datasets may contain visible text that gives away labels without visual reasoning.

Resolution sensitivity. A model may fail because preprocessing made the relevant detail unreadable.

Answer parsing. Free-form answers can hide wrong units, swapped labels, or approximate numbers.

Contamination. Public benchmark images and answers may appear in training data.

Tool ambiguity. A model using OCR or retrieval is not directly comparable to one using only pixels.

For serious deployments, create in-domain tests with private images, controlled perturbations, and human review of failures.

15.7 Safety and Governance

15.7.1 Visual Prompt Injection

Images can contain instructions. A screenshot may say “ignore previous instructions”, a document may include hidden white text, or a QR code may point to malicious content. If the VLM treats image text as user instruction rather than untrusted data, it can be prompt-injected. Visual jailbreaks such as typographic prompts demonstrate that safety training for text-only models does not automatically transfer to vision-language systems [28]. MM-SafetyBench-style evaluations show that harmful requests can be smuggled or amplified through images [66].

A robust system distinguishes instruction channels from content channels. Text extracted from an image should usually be treated as quoted content, not as a higher-priority instruction. Tool outputs, OCR text, and image captions should be wrapped with provenance markers before being passed to the language model.

15.7.2 Privacy and Sensitive Inference

Images often contain faces, addresses, medical information, documents, screens, location clues, and bystanders. A VLM can infer sensitive attributes even when the user did not ask for them. Policies should define which inferences are allowed, which require consent, and which should be refused. Logging and retention rules are especially important for multimodal systems because raw images are more identifying than many text prompts.

15.7.3 Hallucination and Grounding

Visual hallucination occurs when a model asserts objects, text, actions, or relations not supported by the image. It can come from language priors, weak visual features, low resolution, or overextended reasoning. Mitigations include higher-resolution crops, object detectors, OCR verification, uncertainty prompts, and training data with negative examples. The model should learn to say “I cannot tell from this image” when evidence is insufficient.

15.8 Implementation Notes

A minimal VLM implementation plan is:

1. Choose the visual domains: natural images, screenshots, documents, charts, video, or medical imagery.

2. Select a vision encoder and decide whether it is frozen, partially tuned, or trained end to end.
3. Choose a connector and visual-token budget.
4. Define the multimodal chat template and image-marker semantics.
5. Train the connector before broad instruction tuning unless using an already aligned VLM backbone.
6. Build evaluation slices for perception, OCR, spatial reasoning, chart reasoning, safety, and refusal.
7. Measure latency and memory by separating encoder, prefill, and decode costs.
8. Add prompt-injection defenses for image text and document content.

15.9 Key Terms

Vision encoder. A model that maps images or video frames into feature representations.

Contrastive learning. Training matched image-text pairs to be close in embedding space and mismatched pairs far apart.

Connector. A projection, query transformer, or adapter that maps modality features into the LLM interface.

Visual token. A representation of image content inserted into or attended by the language model.

Q-Former. A query transformer that compresses visual features into a small set of learned query outputs.

Visual hallucination. A claim about image content that is not supported by the image.

Visual prompt injection. Malicious or conflicting instructions embedded in image or document content.

Grounding. The degree to which a response is supported by specific parts of the input modality.

15.10 Exercises

1. Derive the symmetric CLIP loss for a batch of three image-text pairs and identify the positive and negative pairs.
2. Trace an image through a frozen vision encoder, projection layer, image marker, and language model context. List the tensor shapes at each step for a chosen patch size.
3. Compare two connector designs for a document QA system: an MLP projection with many visual tokens and a query transformer with compressed tokens. Discuss accuracy and cost tradeoffs.
4. Build a small evaluation set with five natural images, five screenshots, and five charts. Write separate scoring criteria for perception, OCR, and reasoning.
5. Create three visual prompt-injection examples in a benign test environment. Propose a serialization strategy that treats image text as untrusted content.

6. Measure how resizing an image affects OCR accuracy for a VLM or OCR tool. Report the smallest text size that remains reliable.

Chapter 16

Evaluation, Safety, and Governance

Abstract This chapter closes the book by treating evaluation, safety, and governance as part of LLM engineering rather than as afterthoughts. It covers benchmark design, benchmark contamination, human evaluation, LLM-as-judge methods, factuality, hallucination, robustness, red teaming, provenance, privacy, copyright, model cards, dataset documentation, system cards, frontier risk frameworks, operational monitoring, incident response, and deployment limits.

16.1 Evaluation as a Measurement System

Evaluation is the feedback loop that tells an engineering team whether a modeling decision helped. Safety and governance define whether the resulting system should be deployed, who is accountable, and what evidence must be collected before and after release. For LLMs, these concerns are inseparable. A model can improve on a public benchmark while becoming less reliable for the actual product. A safety filter can reduce obvious harmful completions while increasing refusal of benign requests. A benchmark can saturate, leak into training data, or reward the wrong behavior.

This chapter organizes evaluation as a measurement system. A measurement system has a construct, a dataset, a protocol, a metric, an uncertainty estimate, and a decision rule. “MMLU score” or “win rate” is not enough. The team must know what capability the metric is supposed to measure, where the examples came from, how the model was prompted, whether tools were allowed, how outputs were parsed, and what deployment decision follows from the result.

16.2 Evaluation Design

16.2.1 From Question to Decision

Every evaluation should begin with a decision. Examples:

- Can this model replace the previous production model for customer support?
- Does a new preference-tuning run improve helpfulness without increasing unsafe compliance?
- Is a reasoning model worth its added latency for code repair?
- Does a RAG system answer from retrieved evidence rather than parametric memory?
- Is the model acceptable for a regulated domain with human oversight?

The decision determines the metric. A customer-support assistant needs task completion, escalation accuracy, tone, and policy compliance. A code agent needs patch correctness, test pass rate, repository hygiene, and security review. A medical information assistant needs source faithfulness, abstention, and strict scope control.

16.2.2 A Measurement Template

A useful evaluation card should include:

Construct. The behavior being measured, such as factual precision, instruction following, refusal accuracy, or tool-use success.

Population. The users, prompts, languages, domains, and risk classes the evaluation represents.

Protocol. Prompt format, system message, sampling parameters, tool access, context budget, and time limit.

Metric. Exact match, pass@k, rubric score, pairwise win rate, calibration error, cost, or latency.

Uncertainty. Confidence intervals, bootstrap estimates, annotator agreement, and sample size.

Decision rule. The threshold or comparison that determines launch, rollback, escalation, or further testing.

This template prevents benchmark results from becoming detached from engineering decisions.

16.3 Benchmarks and Failure Modes

16.3.1 Capability Benchmarks

Public benchmarks give shared reference points. MMLU measures broad multiple-choice knowledge across academic subjects [34]. HELM argues for holistic evaluation across scenarios, metrics, and transparency artifacts [58]. GPQA targets graduate-level science questions designed to be difficult even with web access [93]. SWE-bench evaluates whether models can resolve real GitHub issues in real repositories [40]. Humanity’s Last Exam and BrowseComp reflect a 2025 shift toward harder, more multimodal, and agentic evaluations at the frontier [85, 81].

These benchmarks are useful, but each measures a specific construct. MMLU does not measure conversational helpfulness. GPQA does not measure safe deployment. SWE-bench depends on repository setup, tests, and agent harness design. BrowseComp measures web-browsing research behavior under a particular protocol, not general truthfulness. Benchmark literacy means knowing what a benchmark excludes.

16.3.2 pass@k and Selection Effects

Code and reasoning benchmarks often report pass@k: the probability that at least one of k generated samples passes. If a single sample has independent success probability p , then

$$\text{pass}@k = 1 - (1 - p)^k.$$

Independence rarely holds exactly, but the formula shows why pass@k is not the same as single-attempt reliability. A model with low pass@1 can look strong at pass@100 if sampling diversity is high and verification is cheap. For deployed agents, the relevant number may be pass@1 under a latency budget, pass@k with an automated verifier, or pass@k followed by human review.

16.3.3 Benchmark Contamination

LLMs are trained on large web corpora that may contain benchmark questions, answers, explanations, or near-duplicates. Contamination inflates scores and can make memorization look like reasoning. Surveys and detection methods document how difficult contamination analysis is for modern LLMs [100, 115]. Mitigations include private test sets, canary strings, deduplication against training corpora, transformed variants, time-split datasets, and post-release benchmark refreshes.

Contamination is not binary. A model may have seen the exact question, a paraphrase, a solution manual, a forum discussion, or many similar examples. Evaluation reports

should describe the level of contamination control, not simply assert that a benchmark is clean.

16.3.4 Benchmark Saturation and Gaming

When a benchmark becomes influential, model developers tune for it. Scores rise, but the benchmark may stop distinguishing real deployment quality. Saturated benchmarks are still useful as regression tests: a model that fails easy public tasks has a problem. They are poor launch gates for frontier capability. Harder benchmarks such as GPQA, HLE, and BrowseComp are partly responses to saturation, but they too can become targets.

16.4 Human Evaluation

16.4.1 Rubrics and Pairwise Preference

Human evaluation remains necessary when outputs are open-ended, values are contested, or task success is hard to automate. Pairwise preference is often more reliable than scalar scoring, but it needs a rubric. Annotators should know whether to prioritize factuality over style, safety over completeness, or concision over coverage. Ambiguous prompts should allow ties or “cannot judge” labels.

Report human evaluation with:

- number of prompts and comparisons;
- annotator qualifications and calibration process;
- rubric version;
- inter-annotator agreement;
- confidence intervals for win rates;
- examples of disagreements.

Without these details, a reported 55% win rate may be meaningful or noise.

16.4.2 LLM-as-Judge

LLM judges reduce cost and improve iteration speed. MT-Bench, Chatbot Arena, and related work show that strong models can approximate human preferences for many assistant comparisons [126, 14]. G-Eval and similar methods use structured prompts and rubrics to grade generated text [67]. These methods are useful for development, but they are not neutral measurement instruments.

Common judge biases include:

Position bias. Preferring the first or second answer because of ordering.
 Verbosity bias. Preferring longer answers.
 Self-preference. Preferring outputs from the judge’s own model family.
 Style bias. Rewarding polish over correctness.
 Policy bias. Applying the judge model’s safety policy rather than the target product policy.

Mitigations include answer-order randomization, length controls, judge ensembles, rubric-specific grading, calibration against human labels, and adversarial judge tests. For high-stakes launches, LLM judges should support human evaluation, not replace it.

16.5 Factuality, Hallucination, and Robustness

16.5.1 Truthfulness and Factual Precision

Truthfulness is not identical to confidence or fluency. TruthfulQA targets questions where imitation of common misconceptions leads to false answers [62]. FActScore decomposes long-form generations into atomic facts and measures support against a knowledge source [73]. HaluEval collects hallucination examples for detection and analysis [54]. RAG evaluation frameworks measure context relevance, answer faithfulness, and answer correctness for retrieval-augmented systems [23, 26].

For a generated answer A decomposed into atomic claims $c_1, \dots, c_n$, a simple factual precision estimate is

$$\text{FactPrecision}(A) = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{c_i \text{ is supported by evidence}\}.$$

This metric emphasizes precision, not completeness. A model can be precise by saying very little. Deployment evaluation should pair factual precision with coverage, abstention, and usefulness.

16.5.2 Calibration and Abstention

Many applications need calibrated uncertainty. A calibrated model’s stated confidence should match empirical correctness. If the model says it is 80% confident on many answers, about 80% should be correct. Calibration can be measured with reliability diagrams and expected calibration error, but LLM free-form confidence is unstable. Better signals often come from consistency across samples, verifier scores, retrieval support, or domain-specific checks.

Abstention is a behavior, not a failure, when the prompt lacks enough information. Evaluations should include unanswerable and underspecified prompts. Otherwise, models learn that every prompt must produce a confident answer.

16.5.3 Robustness

Robustness testing changes the input while preserving the intended task. Examples include paraphrases, typos, adversarial suffixes, distractor documents, misleading context, low-resource languages, unusual formatting, and distribution shifts over time. A robust model should not change a factual answer because the prompt uses a different dialect, and it should not follow malicious instructions in retrieved content.

For RAG and tool systems, robustness includes component failures. The retriever may return stale evidence, the tool may time out, a parser may fail, or an API may change. The evaluation harness should inject such failures and verify graceful degradation.

16.6 Safety Evaluation

16.6.1 Policy Taxonomy

Safety evaluation starts with a policy taxonomy. A typical taxonomy includes self-harm, violence, weapons, cyber abuse, fraud, privacy invasion, sexual content, hate and harassment, regulated advice, child safety, and illegal activity. Each category needs allowed, disallowed, and borderline examples. The taxonomy must also define how the assistant should respond: refuse, redirect, provide high-level educational context, ask a clarifying question, or escalate.

Safety metrics should separate:

- unsafe compliance rate on disallowed requests;
- false refusal rate on benign requests;
- quality of safe redirection;
- jailbreak robustness;
- policy consistency across languages and modalities;
- tool-use safety, such as whether the model calls an external tool for a disallowed purpose.

A single “safety score” hides the tradeoff between over-refusal and under-refusal.

16.6.2 Red Teaming

Red teaming searches for failures, not average-case performance. It should include manual experts, automated prompt mutation, multilingual attacks, role-play, prompt injection, tool-use attacks, long-context attacks, and multimodal attacks. Findings should be converted into regression tests, but the team should expect adversaries to adapt after deployment.

Frontier model system cards increasingly report safety evaluations, external red teaming, and risk-framework decisions. OpenAI’s 2025 Preparedness Framework up-

date and o3/o4-mini system card describe tracked risk categories and safeguards for reasoning models [83, 82]. Anthropic’s Claude system cards and Responsible Scaling Policy describe capability thresholds and AI Safety Level safeguards [3, 4]. Such reports are useful evidence, but they should be read with attention to scope, omitted details, and residual uncertainty.

16.7 Governance

16.7.1 Documentation

Documentation turns evaluation into an auditable artifact. Model cards describe intended use, limitations, evaluation results, and ethical considerations [74]. Datasheets and data statements document dataset motivation, collection, composition, preprocessing, recommended uses, and risks [27, 8]. For LLM systems, these ideas extend to system cards that cover model, tools, retrieval sources, filters, monitoring, and deployment constraints.

A useful LLM system card includes:

- model version, base model, and post-training summary;
- intended users and prohibited uses;
- data sources at a level compatible with privacy and security constraints;
- benchmark and in-domain evaluation results;
- safety policy version and refusal behavior;
- known limitations and failure modes;
- monitoring, logging, and incident response process.

16.7.2 Risk Management Frameworks

Governance frameworks provide structure for risk decisions. NIST AI RMF 1.0 organizes AI risk work around govern, map, measure, and manage functions [78]; the NIST Generative AI Profile adapts these ideas to generative AI risks such as hallucination, data privacy, misuse, and information integrity [79]. The European Union AI Act creates a risk-based legal framework with obligations for high-risk systems and general-purpose AI models [24]. These frameworks do not replace engineering judgment, but they force teams to document risk ownership, evaluation evidence, and post-deployment controls.

A concrete governance review should map every launch decision to an evaluation card. For example, a customer-support RAG assistant might require at least 95% retrieval recall on a private escalation set, factual precision above 98% on cited claims, unsafe-compliance rate below 0.5% on policy probes, false-refusal rate below 3% on benign sensitive prompts, and p95 latency below the product threshold. Each number needs an uncertainty interval. For a binary metric with k failures in n examples, a simple standard-error approximation is

$$SE = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}, \quad \hat{p} = k/n,$$

but low failure rates often require exact or bootstrap intervals rather than a normal approximation. A launch gate that ignores uncertainty can approve a model because the sample was too small to expose rare failures.

Operationally, governance also needs an owner and a rollback path. A release record should name the policy version, model version, retrieval index version, tool permissions, logging retention period, review queue, incident severity rubric, rollback trigger, and communication plan. Copyright and privacy review should happen before data enters training or retrieval: if a source cannot be redistributed, revoked, audited, or access-controlled, it should not be silently absorbed into a weight update.

16.7.3 Operational Monitoring

Pre-deployment evaluation is a snapshot. Deployed systems face changing users, prompts, tools, laws, and adversaries. Monitoring should track:

- traffic distribution and drift;
- refusal and escalation rates;
- user feedback and complaint categories;
- hallucination reports and citation failures;
- tool-call failures and suspicious tool use;
- latency, cost, and outage rates;
- safety incidents and near misses.

Monitoring must respect privacy. Logs should minimize sensitive data, enforce retention limits, and restrict access. For high-risk systems, human review workflows and incident response plans should exist before launch.

16.7.4 Deployment Limits

Some systems should not be deployed even if benchmark scores are high. Reasons include inadequate evaluation coverage, inability to monitor harms, high false-confidence risk, lack of domain oversight, legal restrictions, or severe misuse potential. A launch decision should weigh capability, uncertainty, reversibility, affected population, and available mitigations. “The model is better than the previous model” is not by itself a deployment argument.

Area	Question	Evidence
Task success	Does the system solve the user problem?	In-domain test set, human task review, tool success logs
Factuality	Are claims supported?	Atomic fact checks, retrieval faithfulness, abstention tests
Safety	Does it refuse the right requests?	Unsafe compliance and false refusal rates by category
Robustness	Does behavior survive perturbation?	Paraphrases, adversarial prompts, drift tests
Cost	Is the method deployable?	Latency, token use, hardware utilization, dollars per task
Governance	Can decisions be audited?	Model card, data documentation, policy version, incident process

Table 16.1 A compact checklist for LLM evaluation and deployment review.

16.8 Evaluation Checklist

16.9 Key Terms

Construct. The underlying capability or behavior an evaluation intends to measure.

Benchmark contamination. Exposure of benchmark items or solutions during training or tuning.

pass@k. Probability that at least one of k generated candidates passes an evaluation.

LLM-as-judge. Use of a language model to score, rank, or critique outputs.

Hallucination. Generated content that is unsupported or false in the relevant context.

Calibration. Agreement between stated confidence and empirical correctness.

Red teaming. Adversarial testing intended to discover failures.

Model card. Structured documentation of model use, evaluation, and limitations.

Risk management. The process of identifying, measuring, mitigating, monitoring, and governing risks.

16.10 Exercises

1. Write an evaluation card for a small LLM application. Include construct, population, protocol, metric, uncertainty, and launch threshold.
2. For a benchmark of your choice, list three ways contamination could occur and two mitigations that would still work after public release.
3. Run a pairwise human evaluation on twenty prompts with two model variants. Compute win rate and a bootstrap confidence interval.
4. Design an LLM-as-judge prompt for factual QA. Then list at least five judge biases and how you would test for them.
5. Create a safety evaluation matrix with allowed, disallowed, and redirect examples for three policy categories.

6. Draft a one-page incident response plan for a deployed chatbot that produces unsafe or defamatory content.

Appendix A

Appendix: Reproducibility and Notation Conventions

A.1 Notation

Vectors are written in bold lowercase, matrices in uppercase, and tensors with named shapes when ambiguity would otherwise obscure the implementation. Sequence length is denoted by T , hidden width by d_{model} , attention heads by h , and vocabulary size by V .

A.2 Experiment Records

Every executable experiment in the final manuscript should state the model, dataset, tokenizer, hardware, random seed policy, expected runtime, primary metric, and known failure modes.

A.3 Source Provenance

The manuscript prose is original and cites public primary sources for research claims. It does not include copied third-party code, slide text, dataset rows, model outputs, or figures. Exercises are written as specifications that readers can implement themselves; when an exercise is inspired by a public project or paper, the relevant source is cited in the chapter bibliography.

Glossary

Attention	A content-dependent mixing operation that forms query, key, and value representations and aggregates values according to query-key compatibility.
Benchmark contamination	Exposure of benchmark items, answers, or close variants during training, tuning, prompt development, or evaluation harness construction.
Context window	The maximum number of tokens a model can condition on during a forward pass.
Direct preference optimization	A preference-learning objective that fits a policy from chosen/rejected response pairs without training a separate online reward model.
Grouped-query attention	Attention in which several query heads share each key-value head, reducing KV-cache memory relative to full multi-head attention.
Instruction tuning	Fine-tuning a pretrained model on examples that pair user-like instructions with target responses.
KV cache	Stored key and value tensors from previous decoding steps, reused to avoid recomputing attention over the prefix.
Mixture of experts	A sparse architecture in which a router sends each token to a subset of expert networks, separating total parameters from active parameters.
Parameter-efficient adaptation	Fine-tuning methods that update a small set of parameters or adapters while leaving most base-model weights frozen.
Preference model	A model or implicit objective that represents which of two candidate responses should be preferred under a target policy or annotation protocol.
Retrieval-augmented generation	A generation pattern in which external documents are retrieved and inserted into the model context before or during answer generation.

RLHF	Reinforcement learning from human feedback, usually involving preference data, reward modeling, and policy optimization under a reference-model constraint.
RLVR	Reinforcement learning with verifiable rewards, where correctness can be checked by an answer key, execution, proof checker, or task-specific verifier.
RoPE	Rotary position embeddings, which inject position by rotating query and key coordinates before attention scores are computed.
State-space model	A sequence model family that updates a compact state across positions, often used to reduce long-context cost relative to full attention.
Synthetic data	Training data generated or transformed by models or programs, often filtered by validators, reward models, or human audits.
Test-time compute	Additional computation spent at inference time through longer reasoning traces, multiple samples, verification, search, or refinement.
Tool use	A system pattern in which a model selects and conditions on external operations such as retrieval, code execution, calculators, browsers, or database queries.

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